SEMESTER 3

BIOMEDICAL AND ROBOTIC ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	**GYMAT301**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	**Course Type**	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. **(Text 1: Relevant topics from sections 11.7, 11.8, 11.9)**	**9**
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w= z^2$, $w=e^z$, $w=\frac{1}{z}$, $w= \sin z$. **(Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)**	**9**
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected	**9**

	domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). **(Text 1: Relevant topics from sections 14.1, 14.2, 14.3)**	
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $cos\theta$ and $sin\theta$. **(Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)**	**9**

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	**K3**
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	**K3**
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	**K3**
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10th edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1st edition, 2011

SEMESTER S3

INTRODUCTION TO ROBOTICS

Course Code	PCBRT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 1: 0: 0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To help the students to get a basic idea about Robots.
2. Basic design of robots are introduced.
3. Trajectory planning, obstacle avoidance and kinematics of robots are introduced to students.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot., Trajectory planning, Tasks, Path planning, Trajectory Planning. Joint space trajectory planning- cubic polynomial, linear trajectory with parabolic blends, trajectory planning with via points; Cartesian space planning, Point to point vs continuous path planning. Obstacle avoidance methods- Artificial Potential field.	11
2	Robot Kinematics Direct Kinematics-Rotations-Fundamental and composite Rotations, Homogeneous co- ordinates, Translations and rotations, Composite homogeneous transformations, Screw transformations, Kinematic parameters,	

	The Denavit-Hartenberg (D-H) representation, The arm equation, direct kinematics problems (upto 3DOF) Inverse kinematics- general properties of solutions, Problems (upto 3DOF) Inverse kinematics of 3DOF manipulator with concurrent wrist (demo/assignment only) Tool configuration Jacobian, relation between joint and end effector velocities.	
3	Manipulator Dynamics Lagrange'sformulation – Kinetic Energy expression, velocity Jacobian and Potential Energy expression, Generalised force, Euler-Lagrange equation, Dynamic model of planar and spatial serial robots upto 2 DOF, modelling including motor and gearbox. Robot Control The control problem, Single axis PID control-its disadvantages, PD gravity control, computed torque control. Simulation of simple robot-control system-Matlab programming for control of robots (demonstration/assignment only)	**11**
4	Locomotion, Key issues for locomotion, Legged Mobile Robots, Wheeled Mobile Robots, Aerial Mobile Robots. Mobile Robot Kinematics (Differential Drive Robot), Mobile Robot Workspace- Degree of freedom, Holonomic and nonholonomic robots. Motion Control, Open Loop control, feedback control. Industrial application- Medical, material handling, pick and place robot. Robot selection- number of axes, work volume, capacity and speed, stroke and reach, repeatability , precision and accuracy, operating environment	**11**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarise with anatomy, specifications and types of Robots	**K1**
CO2	Obtain forward and inverse kinematic models of robotic manipulators	**K6**
CO3	Plan trajectories in joint space & Cartesian space and avoid obstacles while robots are in motion	**K3**
CO4	Familiarise with different types of mobile robots, kinematic models, motion control for mobile robots	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3
CO5	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of robotics–Analysis and control	Robert J. Schilling	Prentice Hall	
2	Introduction to Robotics (Mechanics and control)	John J. Craig	Pearson Education	
3	Introduction to Robotics	S K Saha	Mc Graw Hill Education	
4	Robotics and Control	R K Mittal and I J Nagrath	Tata McGrawHill	
5	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University press.	
6	Robotics Technology and Flexible Automation	S. R. Deb		2^{nd} Ed
7	Introduction to Autonomous Mobile Robots	Siegwart, Roland	Cambridge, Mass: MIT Press	2^{nd} ED

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Robotics	Sicilliano, Khatib	Springer	
2	Introduction to Robotics–Mechanics and Control	John J. Craig		
3	Modern Robotics Mechanics, Planning and Control	Kevin M. Lynch, Frank C. Park		
4	Robotics Modelling, Planning and Control	Bruno Siciliano, Lorenzo Sciavicco,		

		Luigi Villani, Giuseppe Oriolo		
5	Fundamentals of Robotics	D.K. Pratihar	Narosa Publishing House, New-Delhi	2017
6	Robotics	K.S. Fu, R.C. Gonzalez, C.S.G. Lee	McGraw-Hill Book Company	1987
7	Introduction to Robotics	J.J. Craig	Addison-Wesley Publishing Company	1986

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/107106090
2	https://onlinecourses.nptel.ac.in/noc19_me74/preview
3	https://www.classcentral.com/course/swayam-introduction-to-robotics-19912

SEMESTER S3

SENSORS AND ACTUATORS

Course Code	PCBRT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To provide an exposure for the student to learn about various sensors and transducers, enabling them to understand their principles and applications.
2. To facilitate the selection process by explaining the underlying theory followed by appropriate case studies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Requirement of sensors in robots used in industry, agriculture, medical field, transportation, military, space and undersea exploration, human robot interactions, robot control, robot navigation, tele-operational robot etc. **Sensor Characteristics**: Sensitivity, Linearity, Measurement/ Dynamic range, Response Time, Accuracy, Repeatability & Precision, Resolution & Threshold, Bandwidth. **Sensor classification**- touch, force, proximity, vision sensors **Proprioceptive or Internal sensors** Position sensors- encoders- linear, rotary, incremental linear encoder, absolute linear encoder, Incremental rotary encoder, absolute rotary encoder; potentiometers; LVDTs; velocity sensors- optical encoders, tacho generator, Hall effect sensor, acceleration sensors, Heading sensors- Compass, Gyroscope sensor, IMU, GPS, real time differential GPS; Force sensors-strain gauge based and Piezo electric based, Torque sensors; Block schematic representations; Interpreting typical manufacturer's data sheet of internal sensors.	**11**

	Note- Block schematic representations, Interpretation of typical manufacturer's datasheet and Numerical problems of the main sensors are to be covered.	
2	**Exteroceptive or External sensors-** contact type, noncontact type; Tactile, proximity-detection of physical contact or closeness, contact switches, bumpers , inductive proximity, capacitive proximity; semiconductor displacement sensor; Range sensors- IR, sonar, laser range finder, optical triangulation (1D), structured light(2D),performance comparison range sensors; motion/ speed sensors-speed relative to fixed or moving objects, Doppler radar, Doppler sound; Block schematic representations; Numerical problems; Block schematic representations; Interpreting typical manufacturer's data sheet of external sensors; Examples -use of Exteroceptive sensors in robots. **Vision based sensors-** Elements of vision sensor, image acquisition, image processing, edge detection, feature extraction, object recognition, pose estimation and visual servoing, hierarchy of a vision system, **Note- Block schematic representations, Interpretation of typical manufacturer's datasheet and Numerical problems of the main sensors are to be covered.**	**11**
3	**Vision based sensors-** CCD and CMOS Cameras, Monochrome, stereo vision, night vision cameras, still vs video cameras, Kinect sensor; Block schematic representations. Choosing sensor for different robotic applications and application of sensors inflexible manufacturing. **Introduction to transducers**; Requirement of transducers in robots, medicine etc; Differences between Sensors and transducers; Transducer performance characteristics based on static and dynamic properties; Classification of transducers based on physical effect, physical quantity and source of energy-Active vs Passive, Principle of transduction, Analog and Digital transducer, Primary and Secondary transducer; Transducer and Inverse Transducer	**11**
4	Position transducers, Displacement transducer– LVDT's, Captive Armatures, Unguided Armatures, Force-Extended Armatures; Velocity Transducers - LVT; Accelerometer- using potentiometer, Strain gage, Piezoelectric; Light transducers; Force transducers; Piezoelectric transducer;Temperature transducer–thermocouple, RTD- common errors, Thermistor, Integrated circuit; Pressure transducer-Bourdon tube, diaphragm, Capacitive Pressure transducer; Oscillator transducer; Flow transducer – Orifice Plate, venture,	**11**

	Defective type flow sensor, spin type flow sensor, Electromagnetic flow sensor; Level Transducer- Discrete Level, Level measurement by pressure sensor, differential pressure sensor, force sensor, capacitive level sensor; Inductive transducer; Ultrasonic transducer; LIDAR **Actuators for robots-** classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.	

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Analyse and select the most appropriate sensors and transducers for a robotic application	**K1**
CO2	Explain fundamental principle of working of sensors and transducers for robots	**K2**
CO3	Interpret typical manufacturer's datasheet of sensors and transducers and use them for selection in typical applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table(Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									3
CO2	3	2	2									3
CO3	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics Engineering: An Integrated Approach	Richard D. Klafter	Prentice Hall Inc.	
2	Sensors and Transducers	D. Patranabis	PHI Learning Private Limited	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Sensors and Actuators: Control System Instrumentation	Clarence W. de Silva	CRC Press	2007
2	Introduction to Robotics	SK Saha	Mc Graw Hill Education	
3	Mechatronics	W.Bolton	PearsonEducationLimited	
4	Automation, Production Systems and Computer Integrated Manufacturing	GrooverM.P,	PrenticeHallLtd.	1997
5	A first course on electric drives	PillaiS.K	WieleyEasternLtd,NewDelhi	
6	Journal of sensors, Special issue- Sensors for Robotics	Aiguo Song , GuangmingSong, DanielaConstantinescu,LeiWang, andQuanjunSong		Volume 2013
7	Mechatronics: Integrated mechanical electronic systems	K.P.Ramachandran,G.KVijayaraghavan	Wiley India	
8	Linear Electric Actuators	I.Boldea		
9	Piezoelectric Actuators (Electrical Engineering Developments)	Joshua E. Segel		2012
10	Sensors and Transducers Used in Robots", Basics of Robotics,	Morecki, Adam and Knapczyk	Springer	1999,pp 275—304
11	Robot sensors and transducers	Ruocco, S	Springer Science \&Business Media,	2013
12	Instrumentation, Measurement and Analysis	Nakra &Chaudhary		2016
13	HANDBOOKOFFORCETRANSDUCERS	HardcoverandStefanescu		
14	Transducer sand Instrumentation	DVSMurthy		

SEMESTER S3

ELECTRONIC CIRCUITS & LOGIC DESIGN

Course Code	PBBRT 304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104 Introduction to Electrical & Electronics Engineering	Course Type	Theory

Course Objectives:

1. Design analog signal processing circuits using diodes and first order RC circuit.
2. Analyse basic amplifiers using BJT.
3. Familiar with various number systems.
4. Design and implement combinational and sequential circuit.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Wave shaping circuits:** First order RC differentiating and integrating circuits, First order RC low pass and high pass filters. Diode Clipping circuits - Positive, negative and biased clipper. Diode Clamping circuits - Positive, negative and biased clamper. **Transistor biasing:** Need, operating point, concept of DC load line, fixed bias, self-bias, voltage divider bias, bias stabilization.	9
2	**BJT Amplifiers:** RC coupled amplifier (CE configuration) – need of various components and design, Concept of AC load lines, voltage gain and frequency response. **Feedback amplifiers:** Effect of positive and negative feedback on gain, frequency response and distortion.	9

	Oscillators: Classification, criterion for oscillation, RC phase shift oscillator, Wien bridge oscillator. (working principle and design equations of the circuits).	
3	**Number system:** Binary, Octal, Hexadecimal and conversions, Binary codes: BCD, XS-3 Grey code, Binary arithmetic, Basic logic operations, Universal gates. Boolean algebra: Operation, laws &Theorems, Demorgan's Theorem, SOP&POS Boolean expression & Truth table. Minterm and maxterm expansion, Incompletely specified function. Minimization technique: Karnaugh map, (up to 4 variable) realization using basic gates and universal gates.	**9**
4	**Combinational logic circuit & design**: Adders, Subtractors & types. Ripple carry and Carry lookahead adders, Multiplexers, Demultiplexers, Decoder and Encoder. **Sequential logic circuits and Design:** Latches–SR latch, Flip Flops-SR, JK, T, D Flipflops, level and edge triggered flipflops, master slave flip flops, conversion between flipflops. Shift registers: SISO, SIPO, PISO, PIPO shift registers, right and left shift register, bidirectional and universal shift registers. Counters: Asynchronous counter- up, down & up/down counter, mod -n counter. Shift register counter: Ring and Johnson counter.	**9**

Suggestion on Project Topics

- The objective of **Project based learning is** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the first class, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups.
- Each group should identify a project relevant to the subject of study.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Design analog signal processing circuits using diodes and first order RC circuit.	**K6**
CO2	Analyse basic amplifiers using BJT. Analyse basic feedback amplifiers using BJT	**K4**
CO3	Explain the elements of digital system abstractions such as digital representations of information, digital logic and Boolean algebra	**K2**
CO4	Design and implement combinational and sequential circuit.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1								
CO2	3	3	1	1								
CO3	3	3			3				3	3		
CO4	3	3			3				3	3		

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Electronic Devices and Circuit Theory"	Robert Boylestad and L Nashelsky,	Pearson	11/e, 2015
2	"Microelectronic Circuits"	Sedra A. S. and K. C. Smith	Oxford University Press	6/e, 2013
3	"Electronic Devices and Circuits"	David A Bell	Oxford University Press	2008
4	"Fundamentals of Microelectronics"	Razavi B.	Wiley	2015
5	FundamentalsofDigitalCircuits	A.Anandkumar	PHIlearning	2/e2010
6	DigitalFundamentals	ThomasLFloyd	Pearson	9thedition
7	Digital Design	Mano.M.M	PHI	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Electronic Circuits, Analysis and Design”	Neamen D.	TMH	3/e, 2007
2	“Microelectronic Circuits - Analysis and Design”	Rashid M. H.	Cengage Learning	2/e,2011
3	“Integrated Electronics”	Millman J. and C. Halkias	McGraw-Hill	2/e, 2010
4	“Digital Circuits and Systems”	D.V. Hall	Tata McGraw Hill	1989
5	FundamentalsofLogicdesign	Charles.H.Roth,Jr.	Thomsonbooks/cole	5th edition
6	“Modern digital Electronics”	R.P. Jain	Tata McGraw Hill	4th edition, 2009
7	“Digital Electronics – An introduction to theory and practice”	W.H. Gothmann	PHI	2nd edition, 2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/108/102/108102097/, https://www.youtube.com/watch?v=qXdBCResKvw
2	https://www.youtube.com/watch?v=Cqouhsl4Yx0, https://www.youtube.com/watch?v=1OKtejD225Y, https://www.youtube.com/watch?v=qmjk3jn7ask
3	https://www.youtube.com/watch?v=CeD2L6KbtVM, https://www.youtube.com/watch?v=BqP6sVYlrr0
4	https://www.youtube.com/watch?v=uv_RJ1Pv71s, https://www.youtube.com/watch?v=sUutDs7FFeA

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology
2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback
3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration
4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result
5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions
6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	**GNEST305**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to AI and Machine Learning**: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	**11**
2	**Mathematical Foundations of AI and Data science**: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	**11**

3	**Applied Probability and Statistics for AI and Data Science**: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	**11**
4	**Basics of Data Science**: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	**11**

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	**K3**
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	**K3**
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	**K3**
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	AurélienGéron	O'Reilly Media, Inc.	2nd edition,2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1st edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	**UCHUT346**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	**6**
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	**6**

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return -– Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	**15**	**12.5**	**12.5**	**50**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$.. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)
*****				

SEMESTER S3/S4
ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	**UCHUT347**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of ethics** - Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution**-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places**-accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies -** sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations**, Ethical values and practices in connection with**	6

	gender - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and	**6**

	regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. **Case Studies and Future Directions:** Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:
Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.

- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

MACHINE DRAWING AND SOLID MODELLING

Course Code	**PCBRL307**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST103	**Course Type**	Lab

Course Objectives:

1. Understand the basic principles of machine drawing as per standards and to get familiar with the different schemes of dimensioning, providing symbols with simple machine parts drawings
2. Understand and get familiar to specifying limits, fits, dimensional and geometric tolerances and surface roughness in machine drawings
3. Get familiar to assembly drawing practices and prepare assembly drawings of robotic components.
4. Get hands on using CAD software for preparing 2D drawings and 3D models of parts and to export them to various formats for different applications.
5. Get hands on preparing the assemblies of various machine parts using cad models and using them for various analysis purposes

Expt. No.	Experiments
1	Introduction Principles of drawing: - Free hand sketching, manual drawing, CAD drawing
2	Code of practice for Engineering Drawing: -BIS specifications – lines, types of lines, dimensioning, sectional views, Welding symbols, riveted joints, keys, fasteners –bolts, nuts, screws, keys..
3	Limits, Fits – Tolerances of individual dimensions – Specification of Fits – basic principles of geometric & dimensional tolerances. Surface roughness, indication of surface roughness, etc,
4	Preparation of production drawings and reading of part and assembly drawings, Exercises on Fasteners.
5	Couplings-Oldham’s coupling, flexible couplings, universal joints.
6	Introduction, input, output devices, introduction to drafting software like

	Auto CAD, basic commands and development of simple 2D and 3D drawings, Different file formats for3 Dmodelling (IGES, STLetc.)
7	Drawing, Editing, Dimensioning, Plotting Commands, Layering Concepts, Matching, Detailing, and Detailed drawings.
8	Exercises on 3D solid modelling:-Plummer lock, bearings, guideways, generating 2D from 3D
9	Assembly drawing of mechanical engineering parts
10	Exercises on valves, couplings, Gears and gear trains, belts, pulleys, modelling of robot grippers. Incorporating electronic components in CAD models.
11	Creating assembly models of Socket and spigot joint, Knuckle Joint, Rigid flange couplings, Bushed Pin flexible coupling.
12	Creating assembly models of Plummer block, Single plate clutch or Cone friction clutch.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic principles of machine drawing as per standards and to get familiar with the different schemes of dimensioning, providing symbols with simple machine parts drawings.	**K3**
CO2	Understand and get familiar to specifying limits, fits,dimensional and geometric tolerances and surface roughness in machine drawings	**K3**
CO3	Get familiar to assembly drawing practices and prepare assembly drawings of robotic components.	**K3**
CO4	Get hands on using CAD software for preparing 2D drawings and 3D models of parts and to export them to various formats for different applications.	**K3**
CO5	Get hands on preparing the assemblies of various machine parts using cad models and using them for various analysis purposes.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3				3							3
CO4	3	2			3							3
CO5	3	2			3							3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A text Book of Machine Drawing	Dr. R K DHAWAN	S CHAND	
2	Machine Drawing with AutoCAD	Goutam Rohith, Goutham Gosh	Pearson	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://iitg.ac.in/mech/academics/undergraduate/latest/sem-3/machine-drawing/
2	https://www.youtube.com/watch?v=ptJfomL1I7o

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

   - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

ELECTRONIC CIRCUIT AND LOGIC DESIGN LAB

Course Code	**PCBRL308**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PBBRT304	**Course Type**	Lab

Course Objectives:

1. Design and demonstrate the functioning of basic electronic circuits using discrete components.
2. Design and implement combinational circuit
3. Design and implement sequential circuit

Expt. No.	Experiments
1	RC integrating and differentiating circuits (Transient analysis with different inputs and frequency response)
2	Clipping and clamping circuits (Transients and transfer characteristics)
3	RC coupled CE amplifier - frequency response characteristics
4	Low frequency oscillators –RC phase shift or Wien bridge
5	Feedback amplifiers (current series, voltage series) - gain and frequency response
6	Low frequency oscillators –RC phase shift or Wien bridge
7	Study of basic logic gates and realization of logic gates using universal gates
8	Half adder, full adder, half substractor and full substractor using basic gates and universal gates
9	Design and implementation of code converters.
10	Flip-flop circuits (SR,JK,T,D &Master slave) and their IC's.
11	Asynchronous counters.
12	Johnson and ring counters

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Design and demonstrate the functioning of basic analog circuits using discrete components.	**K3**
CO2	Design and implement amplifier and oscillator circuits using BJT	**K3**
CO3	Designandimplementcombinationalcircuit	**K3**
CO4	Designandimplementsequentialcircuit	**K3**
CO5	Function effectively as an individual and in a team to accomplish the given task	**K1**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3												
CO4												
CO5												

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Electronic Devices and Circuit Theory”	Robert Boylestad and L Nashelsky,	Pearson	11/e, 2015
2	FundamentalsofDigitalCircuits	A.Anandkumar	PHIlearning	2/e2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Microelectronic Circuits”	Sedra A. S. and K. C. Smith	Oxford University Press	6/e, 2013
2	Digitalfundamentals	Thomas.LFloyd	Pearson	10/e,2011
3	“Fundamentals of Microelectronics”	Razavi B.	Wiley	2015
4	FundamentalsofLogicdesign	Charles.H.Roth,Jr.	Thomsonbooks/cole	5^{th}edition

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=2bprLH4cUSo
2	https://www.youtube.com/watch?v=YbcjMkaJsH8
3	https://www.youtube.com/watch?v=CeD2L6KbtVM, https://www.youtube.com/watch?v=sUutDs7FFeA
4	https://onlinecourses.nptel.ac.in/noc20_ee32/preview

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

BIOMEDICAL & ROBOTIC ENGINEERING

SEMESTER S4

MATHEMATICS FOR ELECTRICAL SCIENCE-4

Course Code	**GBMAT401**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	**Course Type**	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. **[Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]**	**9**
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. **[Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]**	**9**
3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and *t*-distribution), Hypotheses and Test	**9**

	Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. **[Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]**	
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. **[Text 2: Relevant topics from Chapter 6]**	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	**K3**
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	**K3**
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample t-test.	**K3**
CO4	Analyze random processes by classifying them, describing their properties, utilizing autocorrelation functions, and understanding their applications in areas like signal processing and communication systems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9^{th} edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3^{rd} edition, 2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1st edition, 2017

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER S4

IOT & BIOMEDICAL APPLICATIONS

Course Code	PCBRT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. AMQP protocol and how it works.
2. Different challenges of IoT.
3. Write a python code for connecting IoT printer to Raspberry Pi.
4. How IoT can be used in health monitoring.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Internet of Things –Definition and Characteristics of IoT, Physical Design of IoT IoT Protocols, Logical Design of IoT - IoT communication models, IoT Communication APIs, IoT enabled Technologies- Wireless Sensor Networks, Cloud Computing, Communication protocols, Embedded Systems. APIs Sensor Interfaces – ECG, EEG, PPG, Pulse Oximeter, Temperature Sensors, and Pressure Sensors Challenges of using IoT in Healthcare	11
2	IoT Protocols & Architecture: Protocols - Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFID Protocols – Issues with IoT Standardization – Unified Data Standards – IEEE802.15.4–BACNet Protocol– Modbus – KNX – Zigbee– Network layer – APS layer – Security Architecture - IoT Open source architecture (OIC)- OIC Architecture & Design principlesIoT Devices and deployment models- IoTivity : An Open source IoT stack - Overview- IoTivity stack architecture- Resource model and Abstraction	11

3	Wearable Health Monitoring System (WHMS) Basics of Wearable Health Monitoring System (WHMS), Need for Wearable Systems Applications of WHMS - Home Healthcare, Sports/Fitness, Disaster, Haptics, Design of Wearables Introduction, Architecture, and Attributes Design of Smart Textiles, Wearable Electronics, Dry Electrodes, and Textile Electrodes Issues of WHMS & Factors Inhibiting Growth	**11**
4	Building IoT with Arduino &Raspberry Pi: Installation, Interfaces (serial, SPI, I2C), Programming – Python program with Raspberry PI Controlling Hardware- Controlling AC Power devices with Relays, Controlling servo motor, DC Motor. Data Handling-Introduction, Bigdata, Types of data, Characteristics of Big data, Data handling Technologies, Flow of data, Data acquisition, Data Storage, Introduction to Hadoop.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze various protocols and architectures for IoT	Apply Analyse
CO2	Design a portable IoT using Raspberry Pi	Create
CO3	Analyze the components of Wearable Health Monitoring System (WHMS).	Apply Analyse
CO4	Deploy an IoT application and connect to the cloud	Evaluate

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	0	1	3					2		2
CO2	3	2	0	1	3					2		2
CO3	3	2	2	1	3					2		2
CO4	3	2	2	2	3					2		2
CO5	3	2	2	2	3					2		2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Internet of Things – A hands-on approach",	Arshdeep Bahga, Vijay Madisetti	Universities Press,	2015
2	"From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence",	Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle	Academic Press	1st Edition 2014
3	Getting started with raspberry PI. "	Richardson, Matt, and Shawn Wallace	O'Reilly Media	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“loT Fundamentals: Networking Technologies, Protocols and Use Cases for lnternet of Things	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry	Cisco Press	2017
2	“Recipes to Begin, Expand, and Enhance Your Projects”	Michael Margolis, Arduino Cookbook	OReilly Media	2nd Edition. 2011
3	Internet of Things, for Things, and by Things	Abhik Chaudhuri	CRC PRESS	2018
4	“lnternet of Things and Personalized Healthcare Systems”	Venkata Krishna, Sasikumar Gurumoorthy, Mohammad S. Obaidat	Forensic and Medical Bioinformatics,	2015

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/7iWriXyI2cE
2	https://youtu.be/EDjCUsJh01k
3	https://youtu.be/SwQUEYhffXA
4	https://youtu.be/mpk97ls0_yA

SEMESTER S4

ELECTRICAL AND ELECTRONIC INSTRUMENTATION

Course Code	PCBRT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0 :0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. Demonstrate basic concept of Measurement Systems and errors in Measurement.
2. Analyze basic electronic measuring Instruments and systems
3. Appraise various electrical and electronic circuits and systems used in biomedical instrumentation
4. Outline various electrical recording and measuring devices.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Electric and Magnetic units, SI units, conversion of units; Types of Errors- Gross erros, Systematic and random errors in measurement; propagation of errors; expression of uncertainty – accuracy, precision, sensitivity, resolution, loading effect. Electromechanical indicating instruments: suspension galvanometer, PMMC mechanism, DC ammeters and voltmeters, Ohmmeter, Multimeters; AC indicating instruments- Dynamometer, Watthour meter, power factor meter.	11
2	Bridges for measurement: Wheatstone, Kelvin bridges; AC bridges- Maxwell, Hay and Wein bridges; Measurement of R,L, C and Q meter. Motors (Basic Concepts only required): Motor Components; DC Motors- DC Shunt Motor, Separately Excited & Self Excited, DC Series Motor, DC Compound Motor; Permanent Magnet DC motor; AC Motors- Synchronous Motor, Three-phase Induction Motor.; Special Electrical Motors- Stepper Motor, Servo Motor, Brushless DC Motors, Hysteresis	11

	Motor, Reluctance Motor, Universal Motor, Electrostatic motor, Piezoelectric motor.	
3	Medical Instrumentation circuits: Analog Active Filters-Sallen and Key, Generalized Impedance Converter AFs for very low frequencies, two-Op Amp ECG amplifier, Instrumentation amplifier, Medical Isolation amplifiers, safety standards in medical electronic amplifiers, phase detectors, VCO, Phase Locked Loops. Digital Interfaces: Digital-to-Analog Converters, DAC Designs (R–2R ladder), Static and Dynamic Characteristics of DACs, Hold Circuits. Analog- to-Digital Converters (Tracking Converters, The Successive Approximation ADC, Flash Converters)	**11**
4	Devices for Measurement and Recording: Cathode ray oscilloscope (review), Digital storage oscilloscopes-DSO applications. Method of measuring voltage, current, phase, frequency and period using CRO, DSO. Strip chart recorder, X-Y recorder, Plotter, liquid crystal display (LCD), Waveform analysing instruments- Distortion meter, Spectrum analyser, Digital spectrum analyser.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Demonstrate basic concept of Measurement Systems and errors in Measurement	**K2**
CO2	Analyze basic electronic measuring Instruments and systems	**K4**
CO3	Appraise various electrical and electronic circuits and systems used in biomedical instrumentation	**K3**
CO4	Outline various electrical recording and measuring device	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2	2	2	3	1	1	-	-	-	-	-	-
CO2	1	2	3	2	2	2	2	1	1	-	1	2
CO3	1	2	3	2	2	2	2	1	1	-	-	1
CO4	1	2	3	2	2	1	-	-	-	-	-	1

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Modern electronic instrumentation and measurement techniques	Albert D. Helfrick and William D. Cooper Modern	Prentice Hall India	2008
2	Analysis and Application of Analog Electronic Circuits to Biomedical Instrumentation	Robert B. Northrop	CRC Press	2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A course in Electrical and Electronic Measurements and Instrumentation	A K.Sawhney	DhanpatRai Publishing	2012
2	Microelectronic Circuits Theory And Application	Sedra and Smith	Seventh Edition, Oxford University Press	2017
3	Electronic Instrumentation and Measurements	David A. Bell	Third edition, Oxford University Press	2013
4	Liquid Crystal Displays: Fundamental Physics and Technology	Robert H. Chen	Wiley	2012
5	Electric Motors and drivers: Fundamentals, Types and Applications	Austin Hughes and Bill Drury	Newnes	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/n1MinLtvnPY
2	https://youtu.be/0Owe848XA3k?si=9xRff7XYdCHbyfxm
3	https://youtu.be/6PhVUTRx3JA?si=j7mY8CAEKpiLCov5
4	https://youtu.be/0FVzYLEdSA8?si=AAg4Rrkt8CUrHu2q

SEMESTER S4

MICROCONTROLLERS & EMBEDDED SYSTEMS

Course Code	PBBRT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To understand the internal architecture of 8051 Microcontroller
2. To develop simple programs for 8051 using assembly language programming
3. To interface 8051 microcontroller with peripheral devices using ALP/Embedded C
4. To interpret the architecture and design concept of embedded systems
5. To design embedded systems based on Arduino

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Review-Basics of Computer Architecture **8051 microcontrollers:** Difference between microprocessor and microcontroller Architecture,8051 pin diagram**;** Architecture, I/O Port structure, Register organization - special function registers, Memory organization **8051 microcontrollers:** Instruction set, Addressing modes **Simple Assembly language programs:** Arithmetic (Addition, Subtraction, Multiplication& Division), Transfer a block of data from one internal memory location to another.	**9**
2	**8051 micro controller Programming and Peripherals**. Timers/Counters- Serial Communication **Interfacing of peripherals** –(ALP and embeddedC programming). LCD, ADC, DAC, Stepper Motor, key board interfacing,7 segment LED (embedded C programming).	**9**

3	**Introduction to Embedded Systems**: Definition, Features, Simple Example of Embedded Systems, Applications of embedded systems- Consumer electronics, Robotics, Automobiles **Embedded System Architecture**: HW - Processor, Controller, SoC, Memory, Peripherals; SW - Application, Middleware, OS, Device Drivers, Tool chain- Assembler, Interpreter, Compiler, Linker, Loader, Debugger Embedded system design process: Requirement Analysis, Specification Development, HW & SW Co-Design and Development, Module Integration, Testing	**9**
4	**Arduino Uno Board**: Board Study (Board level Block schematic) - Chip (Features only - Architecture not needed), GPIO, Memory, Programming Interface **Programming**: Arduino IDE, Sample Code (LED, Switch, DC motor), Temperature monitoring system using LM35 Temperature sensor &Seven Segment display	**9**

Suggestion on Project Topics

- The objective of **Project based learning is** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the first class, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups.
- Each group should identify a project relevant to the subject of study.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the internal architecture of 8051Microcontroller	**K2**
CO2	Develop simple programs for 8051 using assembly language programming	**K3**
CO3	Interface 8051 microcontroller with peripheral devices using ALP/Embedded C	**K3**
CO4	Interpret the architecture and design concept of embedded systems	**K4**
CO5	Design embedded systems based on Arduino	**K6**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				2					2	2
CO2	3	3	3			2					2	2
CO3	3	3	3			2					2	2
CO4	3	1									2	2
CO5	3	3	3			2					2	2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The 8051 Microcontroller and Embedded Systems: Using Assembly and C”	MuhammadAli Mazidi	Pearson	2nd Edition, 2007.
2	Embedded Systems: An Integrated Approach	Lyla B Das	Pearson	1e,2012
3	Beginning Arduino	Michael Mc Roberts	Apress	2008
4	The 8051 Microcontroller	Kenneth Ayala	Cengage Learning	3e,2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Embedded Systems Architecture, programming and Design	Raj Kamal,	Tata McGraw-Hill	3e, 2013
2	Embedded Systems Architecture, A Comprehensive Guide for Engineers and Programmers	Tammy Noergaard	Newnes – Elsevier	2e, 2012
3	Arduino: The complete guide to Arduino for beginners, including projects, tips, tricks and programming	James Arthur	Ingram	1e, 2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/108/105/108105102/
2	https://archive.nptel.ac.in/courses/108/105/108105102/
3	https://nptel.ac.in/courses/108102045
4	https://onlinecourses.swayam2.ac.in/aic20_sp04/preview

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

1. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

2. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

3. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving Final Result

4. **Final Presentation (5 Marks)**

    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions

5. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

PRINCIPLES OF MEDICAL IMAGING

Course Code	**PEBRT411**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. The subject deals with the study of basic imaging techniques.
2. Understand the techniques of all types of imaging techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Diagnostic Ultrasound Imaging:** Basic physics of ultrasound (Review Only): Wave nature, frequency, speed, wavelength, phase, pressure and intensity-acoustic pressure and intensities within the beam, power, acoustic impedance, reflection, diffuse reflection, scattering, refraction, absorption, interference, diffraction, Attenuation-dependence on frequency. Diagnostic Ultrasound Imaging: A-mode, B-mode and M-mode, Principle of B-Mode Image formation, B-mode formats, Transducers and Beam forming, B-Mode Instrumentation, B- mode image properties, limitations and artifacts, B-Mode Measurements, Doppler Ultrasound, Spectral Ultrasound, colour imaging, advanced techniques for flow and tissue motion, Three-Dimensional Ultrasound, 4D measurements, ultrasound contrast studies, Elastography, Quality assurance and safety of diagnostic ultrasound.	9
2	**Magnetic resonance imaging:** Physics of the Dynamics of the Spin (Review Only). MRI Principle-Magnetisation-Excitation-Relaxation-T1 and T2 phase relaxation and relaxation curves – Acquisition-Bloch Equations. MRI Instrumentation and Block Diagram - Main Magnet, Gradient System, Shim System, RF System, Console, MRI Suite,	9

	Magnet Quench system. Signal formation and imaging- Free induction decay, Echoes, Signal Coding- Slice Encoding, Phase Encoding, Frequency Encoding, slice thickness, 2D and 3D imaging- Spatial resolution and Field of View-K Space-MRI Fourier Reconstruction, Acceleration methods. Image Contrast- T1 Contrast, T2 Contrast, Proton Density Contrast. MR Sequence Parameters (TR,TE,FA,TI), Image Artifacts. MRI Pulse Sequences - Spin Echo Sequence, Multi Echo Sequence, Turbo Spin Echo Sequence, Fast Advanced Spin Echo Sequence, Gradient Echo Sequence, Inversion Recovery Sequence. Principles of Functional MRI, MR Angiography and Diffusion MRI.	
3	**Ray Computed Tomography:** Basic concepts of X-ray Generation, Components involved, Properties of X-rays, Different X-Ray Instruments, Impact of mA and KV. **Principle of Computed Tomography**- Geometries, First, Second, Third, Fourth generation Scanners- Multislice CT Scanners, CT Scanning in Spiral-Helical Geometry, Fifth- Generation Scanners, Sixth-Generation Scanners- The Dual-Source CT Scanner, Seventh- Generation Scanners- Flat-Panel CT Scanners. Slip-Ring Technology, System components- Gantry, Collimation, X-Ray tubes for CT applications - Pencil beam and Cone beam projections. CT Detector Technology, Detector Characteristics, Types, Design Innovations, Multirow/Multislice Detectors, Area Detectors, Image reconstruction algorithms, Image characteristics, Image matrix, CT numbers, Spatial resolution, System noise, Image Artifacts. Introduction to CT Angiogram and DSA.	**9**
4	**Radiology and Hybrid imaging:** Radio-isotopes in medical diagnosis, Interaction of nuclear particles with matter. Radio nuclide generators, nuclear radiation detectors, rectilinear scanner, Gamma camera, Single photon Emission Computed Tomography, Positron Emission Tomography, Biological effects. Hybrid Imaging instrumentation MR-PET Instrumentation-Mutual interference between MR and PET, MR Compatible PET detector technology, MR-PET system architecture. PET-CT, SPECT-CT. Radiation protection. **Infrared imaging and new imaging types:** Infrared Imaging - Physics of thermography – IR Detectors - photon & thermal detectors - thermal uncooled IR detectors: resistive micro bolometers, pyroelectric and	**9**

	ferroelectric detectors, thermoelectric detectors. Pyroelectric vidicon camera - camera characterization - thermographic image processing - clinical applications of thermography in rheumatology, neurology, oncology and physiotherapy. Introduction to Principles of Tactile Imaging and Photoacoustic Imaging. **Catheterization Laboratory-** Theory, Purpose, Types of Cath labs, CT angio vs Cat Lab Angio.	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the operating principle, instrumentation and imaging techniques of diagnostic Ultrasound.	**K2**
CO2	Discuss the operating principle, instrumentation and imaging techniques of Magnetic Resonance Imaging	**K6**
CO3	Analyse the operating principle, instrumentation and imaging techniques of X- Ray Computed Tomography, Cath Lab, and molecular imaging modalities.	**K4**
CO4	Explain the concept of Infrared, Tactile and Photoacoustic imaging modalities.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				2			2	2		2
CO2	3	3				2				2		2
CO3	3	3				2	3		2	2		2
CO4	3									2		2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Diagnostic ultrasound: Physics and equipment	Hoskins, Peter R., Kevin Martin, and Abigail Thrush	CRC Press	2019
2	Physics of Medical Imaging	M Flower, Webb	Taylor & Francis	2016
3	Hybrid MR-PET Imaging: Systems, Methods and Applications.	Shah, N. Jon,	Royal Society of Chemistry	2018
4	Computed Tomography e Book: Physical Principles, Clinical Applications, and Quality Control	Seeram, Euclid	. Elsevier Health Sciences	2015
5	The Physics of Medical Imaging	Webb	John Wiley & Sons	2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Computerized Tomographic Imaging	Avinash C Kak, Malcolm Slaney		2001
2	MRI Made easy…Well almost	Hans H Schild	Heenemann Gm BH	2012
3	Ultrasound imaging and therapy	Fenster, Aaron, and James C. Lacefield,	Taylor & Francis,	2015
4	, Nuclear Medicine Tex t book, Methodology and Clinical Applications	Duccio Volterrani Paola Anna Erba Ignasi Carrió H. William Strauss Giuliano Mariani	Springer	2019
5	Nuclear medicine and PET/CT: technology and techniques	Christian, Paul E., and Kristen M. Waterstram-Rich	Mosby/Elsevier	2007
6	Ultrasonic Bioinstrumentation	Douglas A Christensen	John Wiley, New York	1998

SEMESTER S4

BIOMATERIALS

Course Code	PEBRT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. The subject deals with the study of current biomaterials scene and analyses different types of biomaterials and application.
2. It helps to perform combination of materials that could be used as a tissue replacement implant.
3. It gives a good exposure about the concept of biocompatibility and methods for biomaterials testing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to biomaterials and definition & properties. Classification – Metals, polymers, ceramics, composites properties and application as biomaterial. Characterization of materials- Mechanical Properties, surface properties, electrical properties and visco elasticity. Metallic implant materials–Stainless steel, Co-based alloys, Ti and Ti- based alloys. Ceramic implant materials– Aluminium oxides, Glass ceramics, carbons. Bioactivity –Bio active glass & its applications	9
2	Polymeric implant materials – Polymerization – Polyolefin's, polyamides, acrylic polymers, fluorocarbon polymers, Rubbers, Thermoplastics, Physiochemical characteristics of biopolymers Strength and strengthening mechanisms of metals, ceramics and polymers. Tailor made composites, Bio-composites and nano bio - composites	9
3	Hard tissue implants: Orthopedic implants – wires, pins and screws, fracture plates, intra medullary and spinal fixation devices, Dental implants. Soft tissue replacement implants: bulk space filers, fluid	9

	transfer implants, maxillofacial implants Load carrying implants-Sutures, surgical tape and adhesives, wound dressings, Percutaneous and skin implants. Cardiovascular implants and Extracorporeal Devices Vascular implants and Heart valve implants	
4	Biocompatibility–Definition. Factors affecting biocompatibility–Carcinogenicity, mutagenecity, cytogenicity, toxicity. Blood Compatibility Factors affecting blood compatibility. Material response: Swelling and leaching, Corrosion and Dissolution, Reaction of Biological molecules with biomaterial surfaces. Testing of implants: Methods of test for biological performance- In vitro and In vivo implant test methods. Clinical testing of implant Materials, Design and selection of implant materials–Design Process	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To learn characteristics and classification of Biomaterials.	**K1**
CO2	To understand different metals and ceramics used as biomaterials	**K2**
CO3	To examine the relationship between structure and function of biomaterials	**K2**
CO4	To learn polymeric materials and combinations that could be used as a tissue replacement implants	**K2**
CO5	To know about artificial organ developed using these materials	**K1**
CO6	To give an idea of material testing	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		3	2	2	2	1	1	1	1		2
CO2	1		2	2	2	1	1	1	2	1		2
CO3	2		2	2	2	2	2	1	2	1		2
CO4	2		2	2	2	2	2	1	1	1		2
CO5	2		2	1	2	2	2	2	1	1		2
CO6	2		1	2	2	2	2	1	2	1		2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biomaterials: An Introduction	JB. Park, Roderic S. Lakes	Plenum Press, New York,	1992
2	Biological Performance of materials	Jonathan Black	Marcel Decker	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Encyclopaedic handbook of biomaterials and bioengineering(4vols.)	Donald L. Wise	Marcel Dekker, New York	1995
2	Biomaterials The intersection of biology and materials science	J.S Temenoff, AG Mikos	Pearson Education	First Edition
3	Polymeric Biomaterials	Piskin and AS Hoffmann	Martinus Nijhoff Publishers Dordrecht,	1986

SEMESTER S4

MEDICAL INFORMATICS

Course Code	**PEBRT413**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. State the definition & origin of health informatics.
2. Discuss on the basic theories & models involved in healthcare informatics.
3. List the key components of Electronic Health Records

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Overview of health informatics: healthcare data & analytics -** Definition of health informatics, Theories & models underlying health informatics-Systems Theory, Chaos's Theory, Complexity Theory, Information Theory, Diffusion of innovation theory, Learning Theory, Change Theory, System Life cycle models, Staggers & Nelson system Life cycle model, Evidence based practice & Informatics-Evidence based practice models, ACE star model of knowledge transformation, Knowledge ,Data mining & Practice based medicine	**9**
2	**Information system in healthcare delivery -** Electronic health Records– Definition, Electronic medical Record versus Electronic Health Record, EHR: components, Functions & Attributes, EHR applications used in clinical settings, benefits, Barriers to EHR adoption. Telehealth & Applications for delivering care at a distance–Technologies, Clinical Practice considerations for healthcare professionals, Operational & Organizational success factors & barriers, Telehealth challenges, Clinical Decision support system in Health care	**9**
3	**Health information exchange & health care analytics -** Health	**9**

	information exchange- Basic definitions, The Nationwide Health Information Network (NHIN) and, NHIN Exchange &NHIN Direct, Health Information Organizations (HIOs), Federal Gateway Overview, Web Services and Service Oriented Architecture, Healthcare Analytics-Introduction, Components of healthcare analytics, Tools: SQL (basics)-SQLite, Python features-Lists and tuples, Dictionaries, Sets, Python libraries- pandas and scikit-learn- Linear regression model, Logistic Regression	
4	**Healthcare security & Mobile computing in health care -** Patient medical Information-Privacy, Confidentiality, Security-Definition, Legal & historical context, Principles, laws & Regulations, the importance of information security, Current security Vulnerabilities- External, Internal Vulnerabilities, Use of ICT in healthcare. m-health-Introduction, Basic building block of m-Health, Old Episodic model of health care delivery, Benefits of m-health, Opportunities & obstacles in adoption of m-health.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the fundamentals of health information systems.	**K2**
CO2	Discuss the benefits and foundational role of electronic health records.	**K6**
CO3	Assess the role, benefits and challenges of health information exchanges and healthcare analytics.	**K2**
CO4	Appraise the security and privacy risks of healthcare data.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	3	3	3					2	3	2
CO2	3	2	3	3	3					2	3	2
CO3	3	2	3	3	3					2	3	2
CO4	3	2	3	3	3					2	3	2
CO5	3	2	3	3	3					2	3	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Health Informatics: An Inter professional Approach “	Ramona Nelson, Nancy Staggers	Elsevier	2nd Edition,2013
2	“Biomedical Informatics- Computer Applications In Health Care and Biomedicine”	Edward H. Shortliffe, James J. Cimino	Springer	Fourth edition, 2013
3	“An Introduction to Information Systems and Software in Medicine and Health “	David J. Lubliner	CRC Press, New Jersey Institute of Technology, Newark, USA	2015
4	“Health care informatics“	C Willam Hanson III MD	Medical Publishing Division McGraw-Hill Education / Medical;	1st edition, 2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Health Informatics	L.K. Schaper	IOS Press	2013
2	“Clinical Decision Support Systems Theory and Practice”	Eta S. Berner	Springer	2007
3	”Basics of Bioinformatics”	Rui Jiang • Xuegong Zhang • Michael Q. Zhan	Springer	2013

SEMESTER S4

BIOSTATISTICS

Course Code	**PEBRT 414**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. Explain the basic concepts of statistical analysis.
2. Analyse the concepts of probability and distributions
3. Examine the data and to identify the distribution forms relating to the variables
4. Analyse applications of correlation and regression in biostatistics.
5. Apply hypothesis testing via some statistical distributions in biological problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Descriptive statistics:** Measures of location, Arithmetic Mean, Median, Mode, Geometric mean and harmonic mean, measures of spread, Variance and standard deviation, coefficient of variation, properties of Arithmetic Mean and Variance, Grouped data, Graphic Methods.	9
2	**Probability:** Notations, Laws of probability, Bayes' rule and screening tests, Bayesian Inference, ROC Curves. Random variables. **Discrete Probability distributions:** Probability mass function, expected value, variance and cumulative distribution. Binomial distribution, Poisson distribution.	9
3	**Continuous Probability distribution:** Normal distribution, properties of standard normal distribution. **Estimation:** Relationship between Population and Sample, Random number trials, clinical trials, Estimation of Mean, Variance, Binomial distribution and Poisson distribution. One sided confidence Intervals.	9
4	Hypothesis testing: One sample inference: One sample test for the mean of a normal distribution, one- and two-sided alternatives. Categorical	9

	data: Fisher's Exact Test, Chi-Square Goodness-of-Fit Test. Nonparametric methods: The Sign Test, The Wilcoxon Signed-Rank Test Regression and Correlation Methods: Fitting regression lines- Method of least squares, Inference about parameters from regression lines, interval estimation for linear regression, Correlation coefficient, multiple regressions. Multisample Inference: Introduction to ANOVA	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of statistical analysis.	**K2**
CO2	Analyse the concepts of probability and distributions.	**K4**
CO3	Examine the data and to identify distribution forms relating to the variables.	**K4**
CO4	Analyse applications of correlation and regression in biostatistics.	**K4**
CO5	Apply hypothesis testing via some statistical distributions in biological problems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	3										2
CO3	3	3										2
CO4	3	3	3	2	2					1		2
CO5	3	3	3	2	2					1		2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of Biostatistics	Bernard Rosner	Duxbury Press	Fifth edition
2	Introduction to Biostatistics and Research Methods	PSS Sunder Rao and J Richard	PHI Learning	Fifth Edition

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of biostatistics	Khan, Irfan A., and Atiya Khanum	Ukaaz	2004
2	Introduction to the Practice of Statistics	Moore and McCabe	W.H. Freeman and Company	2009
3	Probability and Statistics”	N.G. Das	Atlantic Publishers	2016
4	Principles of Biostatistics	Marcello Pagano	Chapman and Hall/CRC	2^{nd} edition, 2018

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/kNfBcEHLiEk?si=Smx9w06aRFUARtBA
2	https://youtu.be/b7KWZNTR7Vo?si=Ey06xBPZ4p6nuqdt
3	https://youtu.be/o2LB3KQME3U?si=086ngD3nF1bPgEU7
4	https://youtu.be/14PQawp_rjk?si=ZrJf7uEJZcVIJIcNhttps://youtu.be/14PQawp_rjk

SEMESTER S4

DESIGN OF BIOMEDICAL DEVICES

Course Code	**PEBRT415**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. Documentation of medical device life cycle.
2. Describe process of developing specification for a medical device.
3. Discuss the common errors while designing a medical device.
4. Identify the steps in design transfer process

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Determining and Documenting device requirements**: Idea feasibility and Generating concept: Need identification, prototype development to prove the idea, proof of concept. Device requirements: Product specification, specification review, design specification Software and / or Hardware requirement specification, Software/Hardware design description, Device Records (Case study- Automatic BP monitor, ECG machine etc.) Medical device classification.	**9**
2	**Design Phase** – Risk analysis- hazards review, risk trace matrix, Failure Mode and Effects Analysis(FMEA): Design and system FMEA, Hardware design, Software design, design reviews, Design of experiments, safety margin, environmental protection, product misuse, Biocompatibility, sterilization requirements, human factors engineering, Bill of materials preparation (mechanical/electrical/software/system). (Case study: Example drug infusion system)	**9**
3	**Design Verification, Product Validation and Design Transfer**: Basic concepts, Design verification plan, protocols, design transfer process,	**9**

	software-hardware test plan for medical devices, system testing-subsystem and full device, Risk assessment of medical devices. Computer-Aided Design (CAD), and Computer-Aided Manufacturing (CAM) in biomedical device design.	
4	**Manufacturing supply chain & Medical Device Regulations**: Product manufacturing, Installation Qualification, Operational Qualification and Performance Qualification (IQ/OQ/PQ) protocols, Labelling-Instructions for Use (IFU), design and manufacture process, Quality assurance, audits, post market surveillance, Manufacturing supply chain-process optimization in manufacturing, Rapid prototyping-3D printing (Case study for manufacturing supply chain). Indian Certification for Medical Devices (ICMED)- ICMED 13485, Safety and essential performance of medical electrical equipment, Intellectual Property (IP)- protection for medical devices.	**9**

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	**15**	**10**	**10**	**40**

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. ***Problem Definition (K4 - 4 points)***
   a. *Clearly defines the real-world problem.*
   b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*
2. ***Problem Analysis (K4 - 4 points)***
   a. *Break-down and presents a well-reasoned solution approach.*
   b. *Compare and justify the proposed solutions with evidence and logical reasoning.*

3. ***Evaluate (K5 - 4 points)***
    a. *Thoroughly evaluate the proposed solutions.*
    b. *Compares trade-offs, advantages, and disadvantages.*
    c. *Considers feasibility, scalability, and practical implications.*
4. ***Implementation (K5 - 4 points)***
    a. *Select the most feasible solution by implementing the proposed solutions.*
    b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*
    c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*
5. ***Conclusion (K4- 2 points, K5 – 2 points)***
    a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***
    b. *Reflects critical thinking and informed decision-making.* ***(K5)***

<u>Scoring:</u>

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*
5.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Prepare documentation based on needs identification and specification preparation.	**K3**
CO2	Evaluate the design aspects of biomedical equipment and its safety.	**K5**
CO3	Identify steps in the verification and validation of the product design	**K4**
CO4	Assess the quality control and performance in manufacturing supply chain	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3		3				3		2
CO2	3	3	3	3		3		2		1		2
CO3	3	3	3	3		3				2		2
CO4	3	3	3	2	1	3				2		2
CO5	3	3	3	3		3		1				2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reliable design of medical devices	Fries, Richard C.	CRC Press	2012
2	"Medical Device Design Innovation from Concept to Market"	Peter Ogrodnik	Elsevier	2013
3	Biomedical devices: materials, design, and manufacturing	Lam, Raymond HW, and Weiqiang Chen	Springer	2019
4	"Design of Biomedical Device and Systems"	Paul H. King, Richard C. Fries, Arthur T. Johnson	CRC Press	Third Edition 2015
5	Introduction to product design and development for engineers	Jamnia, Ali	CRC Press CRC Press	2018

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical devices: regulations, standards and practices	Ramakrishna, Seeram, Lingling Tian, Charlene Wang, Susan Liao, and Wee Eong Teo	Woodhead Publishing	2015
2	Handbook of medical device design	Fries, Richard C	CRC Press	2019
3	"Handbook of Medical Device Regulatory Affairs in Asia"	Jack Wong,, Raymond Tong	Jenny Stanford Publishing	Second Edition, 2018
4	Medical device software verification, validation and compliance	Vogel, David A	Artech House	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/uoIxDeAyfdo
2	https://youtu.be/YIRiyCNlAsE
3	https://youtu.be/Hx6DXuildSc
4	https://youtu.be/I6dIlwkTEMs

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	**UCHUT346**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	**6**
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	**6**

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	**6**
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return -– Payback – Accounting Rate of Return.	**6**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	**50**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	**K2**
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	**K3**
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	**K2**
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7^{TH} Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: UCHUT346

Course Name: Economics for Engineers

Max. Marks: 50		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 8 marks

Module 1

9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

		Module 2		
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
		Module 3		
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNPfc from GDPmp.	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDPmp and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
		Module 4		
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

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SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	**UCHUT347**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Fundamentals of ethics -** Personal vs. professional ethics, Civic Virtue, Respect for others, **Profession and Professionalism,** Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, **Technology and digital revolution-**Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, **High technologies: connecting people and places-**accessibility and social impacts, **Managing conflict**, Collective bargaining, **Confidentiality**, Role of confidentiality in moral integrity, **Codes of Ethics**. **Basic concepts in Gender Studies -** sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, **Gender disparity and discrimination in education**, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations**, Ethical values and practices in connection with**	**6**

	gender - equity, diversity & gender justice, **Gender policy and women/transgender empowerment initiatives.**	
2	**Introduction to Environmental Ethics:** Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). **Sustainable Engineering Principles:** Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. **Ecosystems and Biodiversity:** Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. **Landscape and Urban Ecology:** Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	**6**
3	**Hydrology and Water Management:** Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. **Zero Waste Concepts and Practices:** Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. **Circular Economy and Degrowth:** Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. **Mobility and Sustainable Transportation:** Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	**6**
4	**Renewable Energy and Sustainable Technologies:** Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. **Climate Change and Engineering Solutions:** Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. **Environmental Policies and Regulations:** Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental	**6**

	policy-making. **Case Studies and Future Directions:** Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
		Total Marks		**50**

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis**: Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts**: Ability to apply course concepts to real-world problems and local contexts.
- **Creativity**: Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills**: Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	**K3**
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	**K4**
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	**K5**
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	**K4**
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges

and outcomes.

- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

BIOMEDICAL ELECTRONICS LAB

Course Code	**PCBRL407**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Lab

Course objectives:

1. To Familiarise with basic biomedical instruments
2. To Design and set up op-amp based and other electronic circuits used in biomedical equipment.
3. To Design and set up circuits using biomedical transducers
4. To Able to measure, test and troubleshoot electronic circuits

Expt. No.	Experiments
1	Bio amplifier
2	Phase detector
3	Active filters - Notch, LPF, HPF
4	VCO using 565
5	V to F, F to V converters
6	PLL using 4046
7	Flash ADC
8	Study of medical equipments 1. ECG machine 2. Sphygmomanometer 3. Patient monitor
9	Study of IC 4051 and its applications
10	Design of pacemaker circuits & characteristics
11	Thermistor characteristics
12	Skin contact impedance measurement

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarise with basic biomedical instruments	**K2**
CO2	Design and set up Op-Amp based and other electronic circuits used in biomedical equipment.	**K3**
CO3	Design and set up circuits using biomedical transducers	**K3**
CO4	Able to measure, test and troubleshoot electronic circuits	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	3	3	2	-	-	-	3	3	-	2
CO2	2	2	3	3	3	-	-	-	3	3		-
CO3	2	-	3	3	3	-	-	-	3	3	-	2
CO4	-	2	1	-	-	-	-	-	3	3	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.udemy.com/course/electronics-in-biomedical-engineering-theory-repair/?coup

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**
   - Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
   - Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.
2. **Conduct of Experiments (7 Marks)**
   - Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
   - Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
   - Teamwork: Collaboration and participation in group experiments.
3. **Lab Reports and Record Keeping (6 Marks)**
   - Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
   - Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

    - Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

    - Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
    - Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
    - Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
    - Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

    - Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

    - Accuracy of Results: Precision and correctness of the obtained results.
    - Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

    - Ability to explain the experiment, procedure results and answer related questions
    - Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

    - Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

MICROCONTROLLERS AND EMBEDDED SYSTEMS LAB

Course Code	PCBRL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course objectives:

1. To program and test a microcontroller system.
2. To Interface a microcontroller system to user controls and other electronic systems.
3. To Design embedded systems for the required applications

Expt. No.	Experiments
Assembly Language Program Using 8051 Trainer Kit	
1	Arithmetic Operations- addition, subtraction, Multiplication, Division
2	Sum of elements in an array
3	Sorting-Ascending and Descending Order
4	Matrix Addition
5	Largest and Smallest number in an array
6	Square and Cube of a number
Program using Embedded C	
7	Turn a Relay ON and OFF
8	Interfacing Seven Segment Display/Alphanumeric Display
9	Interfacing Stepper Motor
10	Control the speed of a DC Motor
11	Data transfer using different addressing modes
12	Square wave generation

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.***
- ***Endorsement by External Examiner:*** *The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Program and test a microcontroller system.	**K3**
CO2	Interface a microcontroller system to user controls and other electronic systems.	**K3**
CO3	Design embedded systems for the required applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3			2			2	2	2	3
CO2	3	3	3			2			2	2	2	3
CO3	3	3	3			2			2	2	2	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The 8051 Microcontroller and Embedded Systems: Using Assembly and C”	MuhammadAli Mazidi	Pearson	2nd Edition, 2007.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Embedded Systems Architecture, programming and Design	Raj Kamal,	Tata McGraw-Hill	3e, 2013

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students’ preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

BIOMEDICAL & ROBOTIC ENGINEERING

SEMESTER S5

MEDICAL ROBOTICS

Course Code	PCBRT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT404	Course Type	Theory

Course Objectives:

1. This course aims to make students able to understand the engineering aspects of Robots, analyse and design robotic structures and their applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Components of robotic system**: History of medical robots, Classification of medical robots, Present status and future trends. Basic components of robotic system. Basic terminology – Accuracy, Repeatability, Resolution, Degree of freedom. Mechanisms and transmission, End effectors, Grippers different methods of gripping, Mechanical grippers-Slider crank mechanism, Rotary actuators, Cam type gripper, Magnetic grippers, Vacuum grippers, Air operated grippers; Specifications of robot.; **Drive systems-** Pneumatic Drives-Hydraulic Drives-Mechanical Drives-Electrical Drives, D.C.Servo Motors, Stepper Motors, A.C.Servo Motors-Salient Features and Applications	**11**
2	**Sensors in robots-** Touch sensors, Tactile sensor, Proximity and range sensors, Robotic vision sensor, Force sensor, Light sensors, Pressure sensors. **Robot kinematics:** Forward Kinematics, Inverse Kinematics and Difference; Forward Kinematics and Reverse Kinematics of manipulators with Two, Three Degrees of Freedom (in 2 Dimension), Four Degrees of freedom (in 3 Dimension) Jacobians, Velocity and Forces-Manipulator Dynamics, Trajectory Generator.	**11**

3	**Robot programming Languages**-VAL Programming-Motion Commands, Sensor Commands, End Effect or commands and simple Programs **Robots in health care**: Tele operated Robot-Assisted Minimally Invasive Surgery, Robot design concepts Robot assisted Image-Guided Interventions: CT, MR, Ultrasound, Other applications: Robotic catheters for heart electrophysiology, Humanoids, Micro/nano robots for biomedicine-Delivery, surgery, sensing, and detoxification	**11**
4	**Case Studies**: Cardiac, abdominal, and urologic procedures with tele-operated robots (3 procedures each), Neurosurgery and Orthopedic surgery with robots. Other types of healthcare robots: Physically assistive robots, socially assistive robots, Rehabilitation robotics, Design methodologies-Technological choices - Security. Professional and medical ethics; Robo ethics- social and ethical implications	**11**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of robotics and its components and also understand drive mechanisms in robotics	**K2**
CO2	Understand sensors in robotic sand also Illustrate the Kinematics and Dynamics of robotics	**K2**
CO3	Analyse applications of robots in healthcare	**K3**
CO4	Develop programs for controlling robots.	**K6**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1			1				1		2
CO2	2	2	1			2				1		1
CO3	2	2	1			1				1		1
CO4	3	2	2			1				1		1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Robot Modeling and Control”	Mark W. Spong, Seth Hutchinson, and M. Vidyasagar	Wiley Publishers	2006.
2	"Medical robotics -Minimally Invasive surgery",	Paul a Gomes	Wood head	2012
3	Industrial Robotics– Technology, Programming and Applications,	Mikell and Groover	McGraw Hill,2/e	2012
5	Introduction to Robotics. Analysis and control, applications-	Saeed B. Niku	Wiley student edition	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Introduction to Robotics Mechanics and Control”,	Craig J.J	Pearson Education,	2008
2	“Robotics Technology and Flexible Automation”	Deb S. R	Tata McGraw Hill Book Co.	1994
3	“Medical Robotics”,	Achim Schweikard, Floris Ernst	Springer	2015
4	“Medical Robots”,	Daniel Faust,	Rosen Publishers	2016.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/a6_fgnuuYfE?si=lXClvqDUReU_xMmQ
2	https://youtu.be/L_de-Xlh3Sg?si=8eBFtVJrRH9AmwD5
3	https://youtu.be/ZBMIGbKLi30?si=ZLCXWIpQTXinlpmT
4	https://youtu.be/topzzV2oKRw?si=nHfB34FYaGFgnbbh

SEMESTER S5

ELECTRICAL DRIVES AND CONTROL SYSTEMS

Course Code	**PCBRT502**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	THEORY

Course Objectives:

1. Recognize the different power semiconductor device and their working principles
2. Demonstrate design of various speed control techniques of DC motors
3. Introduction To Control systems

SYLLABUS

Module No.	**Syllabus Description**	**Contact Hours**
1	**Introduction to electric motors used for robotic applications.** **DC Motor** -Construction– principle of operation –Back emf–Torque - characteristics of shunt, series and compound motors - necessity of starters-starting methods of dc motors **Introduction to Special Electrical Machines-** DC servo motor, AC servo motors, stepper motor- variable reluctance - permanent magnet- hybrid, BLDC	**11**
2	**DC-DC converters** – step down and step up choppers – control methods-two-quadrant & four quadrant chopper. **DC Motor drives-** Solid state speed control of DC motors-Armature control and field control, Single phase fully controlled converter drives (Rectifier and inverter mode). Chopper controlled DC drives**.**	**11**

3	**Introduction To Control systems-** Components of a Control System, Open-Loop Control Systems and Closed-Loop Control Systems **Mathematical modelling of control systems**: Mechanical systems. Time response of first and second order systems to standard test signals (Step and Ramp)	**11**
4	**Stability of linear control systems** Stability of linear control systems: concept of BIBO stability, Routh's Hurwitz Criterion **Root Locus Techniques**-Introduction, properties and its construction	**11**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the working of different types of motors commonly used in robotics and the need for Electric drives	**K1**
CO2	Speed control of DC motors	**K2**
CO3	Analyse electromechanical systems by mathematical modelling and derive their transfer functions	**K3**
CO4	Determine absolute stability and relative stability of a system	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									2
CO2	3	2										2
CO3	3	3	2									2
CO4	3	3	3									2
CO5	3	2	1									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Power electronics converters applications and design"	Ned Mohan, Tore m Undeland, William P Robbins	John Wiley and Sons.	
2	. "Power semiconductor control drives"	Dubey G. K.	Prentice Hall, Englewood Cliffs, New Jersey	1989
3	Control Systems Engineering	I.J. Nagarath, M.Gopal	New Age International Pub. Co	2007
4	Automatic Control Systems	Farid Golnaraghi, Benjamin C. Kuo	Wiley India	9/e

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Electric drives (concepts and applications)”	VEDAM SUBRAMANIAM	Tata McGraw-Hill	2001
2	Permanent magnet synchronous and Brushless DC motor Drives’	R. Krishnan	CRC Press	2001
3	Control System Engineering	Norman S. Nise	Wiley India.	5/e
4	Modern Control Systems	Richard C Dorf and Robert H. Bishop	Pearson Education,	
5	Modern Control Engineering	Ogata K.	Pearson Education,	2002

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/ndseqKIlR9c
2	https://youtu.be/h7cwYHhBBPA
3	https://youtu.be/EtMRJkCqJN0
4	https://youtu.be/HSzm5QXrjR8

SEMESTER S5

POWER ELECTRONICS & APPLICATIONS

Course Code	PCBRT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To describe various power electronic devices so that the learner will be able to meet technical demands concerning power electronics in biomedical applications
2. To introduce students to the basic theory of power semiconductor devices and passive compo nents, their practical applications in power electronics.
3. To familiarize students to the principle of operation, design and synthesis of different power c onversion circuits and their applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Power electronic devices: Power diodes, Power transistors, MOSFETS and IGBTs. Structure, fabrication, Regions of operations, switching characteristics turn ON and turn OFF switching waveforms, Safe operating area, Paralleling of transistors. Thyristor family: SCRs, DIAC, TRIAC and GTOs. V I characteristics of SCR. Application of power devices in renewable energy systems and Electric vehicles	11
2	AC-DC converters: Single phase-controlled rectifiers – half wave, full-wave, half-controlled and fully controlled - typical waveforms with R, RL, RL with diode and RL with voltage source - voltage and current equation for half-wave-controlled rectifier. DC-DC converters: Basic chopper operation using semiconductor switches, Buck, Boost and Buck-Boost converters Three	11

	phase half-wave and full-wave controlled rectifier with R load, waveforms.	
3	Selection of devices, Electrical and thermal stress, Isolated converters, Half bridge and full bridge converters, Understanding governing equations and switching waveforms DC-AC Inverters, Square wave operations, half bridge and full bridge inverters, Carrier based unipolar and bipolar PWM inverters, Effect of harmonics and its elimination, Sinusoidal PWM inverters	**11**
4	Power electronic systems: Solar power plants, ON grid and OFF grid types, Traction control in EVs, Forward motoring and generating modes, PWM control techniques: voltage mode, current mode and gated oscillator controls, Closed loop responses and stabilization. Gated river circuits, Isolated and non-isolated.	**11**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe basic operation and compare performance of various power semiconductor devices.	**K2**
CO2	Analyze the performance power converter circuits.	**K4**
CO3	Design and implement chopper circuits suitable for various applications.	**K3**
CO4	Describe the working of various types of inverters including PWM inverter.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1							2
CO2	3	3	2		1							2
CO3	3	3	3		1							2
CO4	3	3	1		1							2
CO5	3	3	3		1		2					2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Power Electronics- Essentials and applications	L Umanad	Wiley	2009
2	Power Electronics Circuits, Devices and Applications	Muhammad H. Rashid	Pearson Education	4^{th} edition,2013
3	Power Electronics	M. D. Singh, K. B. Khanchandani	Tata Mcgraw Hill,	3^{rd} edition,2008
4	Power Electronics	Dr. P. S. Bimbhra	Khanna Publishers	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Power electronics: converters, applications, and design	Ned Mohan, Tore M. Undeland	John Wiley & Sons.	1989
2	Power Electronics,	Daniel. W. Hart	Pearson Education	2010

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/pwjrtIjkGak?si=CxSC3bln54Av3LKY
2	https://youtu.be/1Auay7ja2oY?si=6bcpTV0Y75rY2ol0
3	https://youtu.be/Dg5AIy0bY1A?si=KpqrF0fRxaPpVDa6
4	https://youtu.be/o9IOsdnp7w8?si=HSeiPlDmt_ShXDgc

SEMESTER S5

BIOMEDICAL SIGNAL PROCESSING

Course Code	**PBBRT504**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3-0-0-1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Pre-requisites	GBMAT201	**Course Type**	PC-PBL

Course Objectives:

1. Interpret, represent and process discrete/digital signals and systems
2. Thorough understanding of frequency domain analysis of discrete time signals.
3. To design & analyse DSP systems like FIR and IIR Filter etc.
4. Understand filters for Noise Removal from Biomedical signals.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Signals and Signal Processing**-Characterization and Classification of Signals, Typical signal Processing Operations, Examples of Biomedical Signals and Analog-to-Digital Conversion (block diagram only) **Discrete-Time Signals -**Typical Sequences and Sequence. Representation, Discrete-Time Systems-Classification of Discrete-Time Systems	9
2	**Discrete-Time Fourier Transform (DTFT)-**Definition, Properties, Discrete Fourier Transform-Definition, Properties, FFT-Decimation-in-Time, Decimation-in-Frequency. **The z- Transform**-Properties of the z-Transform, Poles and Zeros, Inversion of the z-Transform- Using partial fraction expansion, Analysis of Linear Time-Invariant Systems in the z-Domain- Transfer function, Frequency response	9

3	**LTI systems as frequency selective filters**-Ideal filter characteristics-Simple Digital Filter Design**-**Preliminary Considerations, Design of linear-phase FIR Filter using windows (Bartlett, hamming, Hanning, Kaiser) and frequency sampling method-Applications of FIR filters in Biomedical systems - pre-processing of bio signals. **IIR Filter Design**-Approximation of Derivatives, Impulse invariance, IIR digital filters for ECG analysis	**9**
4	**Periodogram**-Averaged periodogram, Blackman-Tukey Spectral estimation, QRS detection in ECG, Analysis of Heart Rate Variability using periodogram **Adaptive Filters-**Principle of Wiener Filters**,** Principle of adaptive filtering, Principle of adaptive noise cancellation, Cancellation of 50-Hz interference in ECG	**9**

Suggestion on Project Topics

- The objective of **Project based learning is** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the first class, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups.
- Each group should identify a project relevant to the subject of study.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the fundamentals of discrete-time signals and systems	**K4**
CO2	Apply domain transformation techniques in discrete-time signals	**K3**
CO3	Design digital filters and apply the concepts in biomedical scenarios.	**K6**
CO4	Apply different power spectrum estimation and noise cancellation techniques for biomedical applications.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biomedical signal processing: principles and techniques.	Reddy, D. C.	McGraw-Hill	2005
2	Digital Signal Processing Principles,Algorithms and Applications	John G Proakis& Dimitris G Manolakis	Prentice Hall of India	2005
3	Biomedical signal processing	Akay, Metin.	Academic press	2012
4	"Biomedical digital signal processing."	Tompkins, Willis J.	Editorial Prentice Hall	1993
5	"Digital signal processing (Book)." Research supported by the Massachusetts Institute of Technology, Bell Telephone Laboratories, and Guggenheim Foundation. Englewood Cliffs, N. J.	Oppenheim, Alan V., and Ronald W. Schafer	Prentice-Hall,Inc.	1975. 598 p

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1								
CO2	3	3	1	1								
CO3	3	3			3				3	3		
CO4	3	3			3				3	3		

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital signal processing: a computer-basedapproach. Vol. 2. New York	Mitra, Sanjit Kumar, and Yonghong Kuo.	McGraw-Hill	2006
2	Signals and systems for bioengineers: a MATLAB-based introduction	Semmlow, John.	Academic Press	2011
3	Biomedical Signal Analysis	Rangaraj M Rangayyan	John Wiley	2002

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_ee28/preview, https://nptel.ac.in/courses/117105134
2	https://onlinecourses.nptel.ac.in/noc22_ee28/preview, https://onlinecourses.nptel.ac.in/noc20_ee15/preview, https://archive.nptel.ac.in/courses/108/106/108106151/
3	https://archive.nptel.ac.in/courses/108/106/108106151/, https://onlinecourses.nptel.ac.in/noc22_ee19/preview, https://archive.nptel.ac.in/courses/108/105/108105055/, https://archive.nptel.ac.in/courses/108/101/108101174/
4	https://nptel.ac.in/courses/117105075, https://onlinecourses.nptel.ac.in/noc20_ee41/preview, https://www.youtube.com/watch?v=tcMY6FUwYSI

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
    - Clarity and feasibility of the project plan
    - Research and background understanding
    - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
    - Individual contribution to the presentation
    - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**
    - Active participation and individual contribution
    - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**
    - Adherence to the project timeline and milestones
    - Application of theoretical knowledge and problem-solving
    - Final Result

5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S5

PLC AND DISTRIBUTED CONTROL SYSTEMS

Course Code	PEBRT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To learn PLC hardware and its general architecture.
2. To design logic circuits to perform industrial control functions of medium complexity and realise the same using ladder logic.
3. To apply the industrial protocols used with PLC in real applications.
4. To familiarize the DCS architecture and its functions.
5. To determine hardware and communication requirements of DCS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to PLC -Construction of relay logic circuits with different control elements-Need for PLC –Evolution of PLC. PROGRAMMABLE LOGIC CONTROLLERS: Architecture of PLC -Types of PLC –PLC modules, Input and Output modules –Digital and Analog Input/Output- examples of Digital and Analog Inputs/Outputs in PLC-AC-DC power supplies in PLC- isolators- PLC Configuration -Scan cycle -Capabilities of PLC-Selection criteria for PLC –PLC Communication with PC and software-PLC Wiring-Installation of PLC and its modules. PLC programming languages: Ladder Logic, Functional Block Diagram FBD -Sequential Flow Chart SFC - Structured Text - Instruction List	9

2	PROGRAMMING OF PLC: – Ladder Logic Programming – Realization of simple logic circuits programming on-off inputs/ outputs. Auxiliary commands and functions: PLC Basic Functions: Register basics- timer functions- counter functions. Program control instructions- math instructions- sequencers- PLC based traffic light system, stepper motor & servo motor control using PLC, Analog sensor interfacing with PLC APPLICATIONS OF PLC: Case studies of manufacturing automation and process automation.	**9**
3	PLC programming tools as per IEC 61131-Developing programs using Sequential Function Chart and FBD Networking PLC: usingRS232, RS485, Protocols-ethernet- Modbus, CANOpen, OPC-Open Platform Communication- OPC UA Industrial Automation: General Block diagrams of DDC- Supervisory control- Data Acquisition system	**9**
4	Distributed Control System- DCS - Architectures, Comparison, Local control unit, Process interfacing issues, Communication facilities- Distributed Control System Basics: DCS introduction- Various function Blocks- DCS components/block diagram- DCS Architecture of different makes- comparison of these architectures with automation pyramid Interfaces In DCS: Operator interfaces- Low level and high-level operator interfaces - Operator displays- Engineering interfaces- Low level and high-level engineering interfaces-	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Learn PLC hardware and its general architecture.	**K2**
CO2	Design logic circuits to perform industrial control functions of medium complexity and realise the same using ladder logic.	**K3**
CO3	Apply the industrial protocols used with PLC in real applications.	**K3**
CO4	Familiarize the DCS architecture and its functions.	**K1**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3	2									
CO3	3	3	2									
CO4	3	3										

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programmable logic controllers,	Frank D Petruzella	McGraw-Hill	2011
2	Distributed Control Systems’	Michael P. Lukas	Distributed Control Systems	1986
3	Industrial Process Automation Systems: Design and Implementation	B.R. Mehta and Y. Jagan mohan Reddy	ELSEVIER	1st Edition November 26, 2014

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Programmable Controllers, Hardware, Software & Applications	George L. Batten Jr	Mc Graw Hill	2^{nd} Edition, 1994.
2	Programmable logic controllers	W. Bolton	Elsevier Ltd	2015
3	Programmable Logic Controllers: Programming Methods and Applications	John R Hackworth and Fredrick D Hackworth Jr.	Pearson Education	2006

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/102/112102011/
2	https://archive.nptel.ac.in/courses/112/102/112102011/
3	https://archive.nptel.ac.in/courses/108/105/108105062/
4	https://archive.nptel.ac.in/courses/108/105/108105062/

SEMESTER S5

MOBILE ROBOTICS

Course Code	**PEBRT522**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. With the increase in automation, robots are used in all walks of life.
2. It was observed by many researches that the ability of a robot can be increased several folds if it was capable of movement.
3. To introduce the basic issues in bringing mobility to robots and how those issues are resolved through its various modules.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Mobile Robots:** Fundamental problems – Path Planning, Localization, Sensing, Mapping, Simultaneous Localization and Mapping **Locomotion:** Introduction to stepper Motor and Servo Motor Control, Wheeled Mobile Robots -Differential Drive, Synchronous Drive, Steered wheels, Tricycle Drive, Car drive - Limbed Locomotion – Vehicle Stability, Number of legs, Limb design and Control, forward and inverse kinematics, Gait and Body Control, Dynamic Gaits **Off-Board Communication:** Tethered, Untethered, Radio Modems	9
2	**Non Visual Sensors :** Basic concepts – Sensors: Bumbers, Accelerometers, Gyroscopes, infrared sensors, Sonar – Transducer Model, Data interpretation, Laser Range finders - Data Fusion: Kalman Filter **Visual Sensors :**Perspective Camera, Planar Homography, Camera Calibration, - Object Appearance and Shading–Signals and Sampling–Image features and Their Combination– Measuring Depth – Active Vision	9

3	**Representation and Reasoning about Space:** Representing Space: Spatial Decomposition, Geometric Representations, Topological Representations– Representing the Robot: Configuration Space, Simplification of C-space– Path Planning for Mobile Robots: Constructing a discrete search space, Retraction Methods - Searching a Discrete State Space: Graph search, Depth-first Search, Breadth-first Search, Dijkstra's Algorithm – Searching a Continuous State Space: Vector Field Algorithm, Bug algorithm – Spatial Uncertainty – Dynamic Environments , Probabilistic path planning	9
4	**System Control:** Horizontal Decomposition – Vertical Decomposition – Hybrid Control Architectures–Middleware–High Level Control–Alternative Control Formalisms–The human Robot interface. **Pose Maintenance and Localization:** Simple landmark measurement - Servo Control – Recursive Filtering – Non-Geometric methods: Perceptual Structure – Correlation based localization – Global Localization **Mapping:** Sensorial Maps–Geometric Maps–Topological Maps.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the fundamental computational issues involved in mobile robotics and issues related to locomotion	**K2**
CO2	Translate the working principle of different visual and non-visual sensors to select the appropriate ones for a particular application	**K3**
CO3	Explain the techniques used for representing and reasoning about space	**K2**
CO4	Classify the different software architecture in the development of robotic applications	**K1**
CO5	State the techniques used for pose maintenance and localization techniques used in robotics	**K1**

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2		2									2
CO3	3		2		2							2
CO4	3	2	3	2	2							2
CO5	3	2	3		2							2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computational Principles of Mobile Robotics	Gregory Dudek and Michael Jenkin	Cambridge University Press	
2	Introduction to Autonomous Mobile Robots	Scaramuzza	MITPress,USA	2011.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier, USA	2014
2	Sensors for mobile robot	H R Everett	CRCPress	

SEMESTER S5

KINEMATICS AND DESIGN OF MACHINE ELEMENTS

Course Code	**PEBRT 523**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. To Understand the kinematic details of machines
2. To apply the concepts of stress analysis, theories of failure and material science to analyse and design commonly used machine components
3. Provides an in-depth understanding on the design of gear, belt drives which are critical components of automation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Basics of mechanisms:** Links, kinematic pairs, kinematic chain, mechanism and machine, schematic diagrams and description of common mechanisms like linkages, cams- follower mechanisms, gear trains, belt and chain drives- no derivations, degrees of freedom (DoF), Kutzbach's formula, determination of DoF of planar linkages and mechanisms with cam-follower pairs. Definition of position analysis problem, derivation of solutions for simple mechanisms like fourbar, slider-crank, and multi DoF closed and open loop mechanisms, multiple branches of solution, exposure to graphical approach.	**9**
2	**Velocity analysis:** Definition of the velocity analysis problem, angular velocity of a rigid link and relative velocity of points, analytical method of velocity analysis, derivation of equations and numerical solutions, forward and inverse velocity analysis of open loop 3R mechanism. **Acceleration analysis:** Definition of the acceleration analysis problem, angular	**9**

	acceleration of a rigid link and relative acceleration of points, analytical method of acceleration analysis and derivation of equations, numerical solution for simple mechanisms.	
3	Introduction to Design- Definition, steps in design process, preferred numbers, standards and codes in design. Springs- classification, spring materials, stresses and deflection of helical springs, Selection of springs, concentric springs, end constructions Shafts, design based on strength, rigidity and critical speed, design for static and fatigue loads, repeated loading, reversed bending; Design of Coupling- selection, classification, rigid and flexible coupling, design of keys and pins	**9**
4	Gears- classification, Gear nomenclature, Tooth profiles, Materials of gears, virtual or formative number of teeth, gear tooth failures, Lewis equation, Buckingham's equation for dynamic load, wear load, endurance strength of tooth, surface durability, heat dissipation – lubrication of gears, Merits and demerits of each type of gears. Design of spur gear. Basic idea of flat belt-materials for belts Selection of V-belt drives, Advantages and limitations of V-belt drive	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the kinematic details of machines, kinematic pairs and degrees of freedom and determining the loop closure equations of various linkages and find known and unknown coordinates	**K2**
CO2	Determine the velocity and acceleration of a point in open loop planar mechanisms.	**K2**
CO3	To provide knowledge in the design of common machine elements such as shafts, springs and couplings.	**K1**
CO4	To provide basic design methods for gear , belt drives and bearings	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1									3
CO2	3	3	3									3
CO3	2	1										3
CO4	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Kinematics and Dynamics of Machinery	Norton R L	McGraw-Hill	
2	Theory of Machines and Mechanisms	John J. Uicker. Jr, Gordon R. Pennock, Joseph E. Shigley	Oxford HED	4th Edition
3	Machine Design	Jalaludeen	Anuradha Publications	2016
4	Design of Machine elements	V. B. Bhandari	McGraw Hill	2016
5	Machine Design – An Integrated Approach	R. L. Norton	Pearson Education	2001

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Kinematics, Dynamics and Design of Machinery	Kenneth J. Waldron, Gary L. Kinzel, Sunil K. Agrawal	Willey	3rd Edition, 2016
2	Fundamentals of Kinematics and Dynamics of Machines and Mechanisms	Oleg Vinogradov	CRC press	
3	Theory of Machines	Rattan S S	Tata McGraw-Hill	
4	Fundamentals of Machine Component Design	Juvinall R.C &Marshek K.M	John Wiley	2011
5	Design of Machine Elements	M. F. Spotts, T. E. Shoup	Pearson Education	2006
6	Machine Design	Rajendra Karwa	Laxmi Publications (P) LTD, New Delhi	2006
7	Mechanical Design of Machines	Siegel, Maleev & Hartman	International Book Company	1983

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/MJeRFzs4oRU?si=cw8vLLusOqOzFoKD
2	https://youtu.be/40RU9lWdfTA?si=5s9lZfWYuVRbXg0L
3	https://youtu.be/T4IgtIkBnOo?si=DabloxjpyKORzSVn
4	https://youtu.be/Fb4weO0HLxk?si=PbEdIojStbpM4NA7

SEMESTER S5

DESIGN FOR MANUFACTURING AND ASSEMBLY

Course Code	**PEBRT524**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. The main objective of this course is to understand the basic design rules for manufacturing and material selection and applying the production process for ease of manufacturing. Apply the concepts of design for manufacturing and assembly for product manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Design philosophy, steps in design process, general design rules for manufacture ability, basic principles of designing for economical production, creativity in design; materials: Selection of materials for design, developments in material technology, criteria for material selection, and material selection interrelationship with process selection **Design For Machining Process:** Overview of various machining processes, general design rules for machining, dimensional tolerance and surface roughness, design for machining ease.	**9**
2	**Design for Metal Casting:** Appraisal of various casting processes, selection of casting process, general design considerations for casting - casting tolerances - use of solidification simulation in casting design - product design rules for sand casting **Design for Metal Joining:** Appraisal of various welding processes, Factors	**9**

	in design of weldments - general design guidelines - pre and post treatment of welds - effects of thermal stresses in weld joints - design of brazed joints	
3	**Forging:** Design factors for forging, closed die forging, design parting lines of dies, drop forging die design, General design recommendations **Extrusion:** Sheet metal work and plastics, Design guide lines for extruded sections, Design principles for punching, blanking, bending, deep drawing, Keeler -Goodman formability diagram, (forming limit diagram) Component design for blanking	**9**
4	**Design Of Manual Assembly:** Design for assembly fits in the design process, general design guidelines for manual assembly, development of the systematic DFA methodology, assembly efficiency, classification system for manual handling, classification system for manual insertion and fastening, effect of part symmetry on handling time, effect of part thickness and size on handling time, effect of weight on handling time, parts requiring two hands for manipulation, effects of combinations of factors.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Remember the basic principles of designing for economical production and understand the principles of selection of materials for product development	**K1**
CO2	Enumerate the general design considerations for casting, casting tolerances andRemember the factors in design of weldments.	**K2**
CO3	Analyze the effects of thermal stresses in welded joints and Understand the variousadvantages and limitations of joining techniques	**K4**
CO4	Remember Keeler-Goodman formability diagram and its concept and Applydesign guidelines to assembly	**K1**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	1						1	1
CO2	3	2	1	1	1						1	1
CO3	3	3	1	2	1						1	1
CO4	3	2	1	1	1						1	1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Assembly Automation and Product Design"	Geoffrey Boothroyd,	Marcel and Dekken	2002
2	Design - Material & Processing Approach",	George E. Deiter, "Engineering	Mc GrawHill Intl. 2nd Edition	2010
3	" Hand Book of Product Design",	Geoffrey Boothroyd	Marcel and Dekken,	2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Hand Book of Product Design"	Geoffrey Boothroyd	Marcel and Dekken, 1stEdition,	2013

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
2	https://youtu.be/mzWMdZZaHwI?si=tHIBi7_GlsjGG-dl
3	https://youtu.be/ZLlwfXSXEVc?si=QOjqJ5GR78wxThr8
4	https://youtu.be/vEPpKjIdpt0?si=oiuKP0FXnFtQe1T7

SEMESTER S5

ARTIFICIAL INTELLIGENCE FOR ROBOTICS

Course Code	**PEBRT525**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	THEORY

Course Objectives:

1. Compare and contrast different types of search strategies
2. Ontology in Al
3. Expert systems in AI

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction and Problem solving:** History, state of the art, Need for AI in Robotics. Thinking and acting humanly, Intelligent agents, structure of agents. Solving problems by searching –Informed search and exploration– Constraint satisfaction problems– Adversarial search	**9**
2	**Knowledge representation and Planning:** Knowledge representation, first order logic. Planning with forward and backward State space search – Partial order planning – Planning graphs– Planning with propositional logic – Planning and acting in real world.	**9**
3	**Reasoning and Decision making:** Uncertainty – Probabilistic reasoning, Dynamic Bayesian Networks; Basis of utility theory, decision theory, sequential decision problems;	**9**
4	Forms of learning – Knowledge in learning – Statistical learning methods –reinforcement learning, communication, perceiving and acting,	**9**

	Probabilistic language processing, and perception. **AI In Robotics and Applications:** Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics	

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	**15**	**10**	**10**	**40**

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

<u>Criteria for evaluation:</u>

1. ***Problem Definition (K4 - 4 points)***
    a. *Clearly defines the real-world problem.*
    b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*
2. ***Problem Analysis (K4 - 4 points)***
    a. *Break-down and presents a well-reasoned solution approach.*
    b. *Compare and justify the proposed solutions with evidence and logical reasoning.*
3. ***Evaluate (K5 - 4 points)***
    a. *Thoroughly evaluate the proposed solutions.*
    b. *Compares trade-offs, advantages, and disadvantages.*
    c. *Considers feasibility, scalability, and practical implications.*
4. ***Implementation (K5 - 4 points)***
    a. *Select the most feasible solution by implementing the proposed solutions.*
    b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*
    c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*

5. ***Conclusion (K4- 2 points, K5 – 2 points)***
    a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***
    b. *Reflects critical thinking and informed decision-making.* ***(K5)***

Scoring:

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Identify and solve problems using appropriate AI methods	**K1/K2**
CO2	Formalize a given problem in the language/framework of different AI methods	**K3**
CO3	Describe the learning methods adopted in AI	**K4**
CO4	Interpret various applications of AI in Robotic Applications	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3
CO5	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Artificial Intelligence: A modern approach"	Stuart Russel, Peter Norvig	Pearson Education, India	2016
2	Machine Learning; A Probabilistic Perspective	Kevin Murphy	MIT Press	2012
3	"Artificial Intelligence: A guide to Intelligence Systems"	Negnevitsky, M, , Harlow	AddisonWesley	2002
4	Introduction to Robotics	S K Saha	McGraw Hill Education	
5	Introduction to Autonomous Mobile Robots	Siegwart, Roland,	Cambridge, Mass. : MIT Press	2nd ed

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Introduction to AI Robotics"	Robin Murphy, Robin R. Murphy, Ronald C. Arkin	MIT Press	2000
2	"Artificial Intelligence: Robotics and Machine Evolution"	David Jefferis	Crabtree Publishing Company	1992
3	"Artificial Intelligence for Robotics"	Fransis X. Govers	Packt Publishing	2018
4	"Handbook of Robotics"	Sicilliano, Khatib	Springer	
5	Introduction to Robotics – Mechanics and Control	John J. Craig		

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/XCPZBD9lbVo
2	https://youtu.be/ARywou8HLQk
3	https://youtu.be/GqxOwxwH1C8
4	https://youtu.be/yCXm5cgG0UA
5	https://youtu.be/bRsjfbqHj8k

SEMESTER S5

ROBOTIC PROCESS LAB

Course Code	PCBRL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	LAB

Course Objectives:

1. Test forward, inverse kinematic modelling and path planning of robotic manipulators
2. Test basic control algorithms in mobile robots to move to a point, to follow a line, to follow a path and for obstacle avoidance. Calibrate sensors used in robots.
3. Familiarise about localisation of mobile robots.
4. Design and develop sensor-based systems in robots.

Details of Experiment

Expt. No	Experiment
1	Joint space and Cartesian space trajectory planning for a pick and place task
2	Obtain forward and inverse kinematic models (check end effector and joint positions with theoretical and actual values)
3	Point to point control and continuous path control
4	Control of mobile robot for moving to a point(xg, yg) , following a line (ax+by+c=0), moving to a specific target orientation (θ_g) (Closed loop control considering kinematicmodels)
5	V Obstacle avoidance of a mobile robot while moving to a point.
6	Localization of a mobile robot using LIDAR
7	Calibration of sensors-sonar, IR sensors and obtain the calibration curve
8	Object detection using any one standard algorithm
9	Object tracking and visual serving
10	Following a moving target/ Object tracking from a moving vehicle
11	Design and develop a servo controlled robotic manipulator (1 DOF) with visual feedbackfor pick and place task
12	Design and develop a mobile robot capable of obstacle avoidance and localisation
13	Assemble a quadcopter drone kit and make it hover.

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Test forward, inverse kinematic modelling and path planning of robotic Manipulators	**K3**
CO2	Test basic control algorithms in mobile robots to move to a point, to follow a line, to follow a path and for obstacle avoidance.	**K3**
CO3	Familiarise about localisation of mobile robots	**K3**
CO4	Calibrate sensors used in robots. Design and develop sensor-based systems in robots.	**K3**

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2		2			2	2		3
CO2	3	2	2	2		2			2	2		3
CO3	3	2	2	2		2			2	2		3
CO4	3	2	2	2		2			2	2		3

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics (Mechanics and control)	John. J. Craig	Pearson Education Asia	2002
2	Introduction to Robotics	S K Saha	Mc Graw Hill Education	
3	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill, New Delhi	2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University press.	
2	Robotics Technology and Flexible Automation	S. R. Deb		2^{nd} Ed
3	Introduction to Autonomous Mobile Robots	Siegwart, Roland	Cambridge, Mass: MIT Press	2^{nd} Ed
4	Robotics, Vision and Control: Fundamental Algorithms in MATLAB	Peter Corke,	Springer	

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.

- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

BIOMEDICAL SIGNAL PROCESSING LAB

Course Code	**PCBRL508**	**CIE Marks**	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	**ESE Marks**	50
Credits	2	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PBBRT504	**Course Type**	Theory

Course Objectives:

1. To design and implement techniques to process and analyse signals
2. Apply suitable signal processing algorithms for biomedical signal analysis.
3. To design solutions to address the issues during bio-signal acquisition and processing.

Details of Experiment

Expt.	Experiment
1	Acquisition and basic operations of biosignals using standard signal acquisition systems
2	Implementation of I/D sampling rate converters.
3	Impulse response of first order and second order systems.
4	Design of 50 Hz notch filter for ECG signal
5	Detection of events and heart rate measurement from ECG
6	Generation of IBI and analysis of HRV
7	Spectral modeling and analysis of ECG signals
8	DCT/IDCT Computation
9	Time domain Filtering to remove baseline drift & high frequency artefacts in ECG signal.
10	Powerline interference rejection from ECG signals
11	Arrhythmia analysis
12	Detection of Premature Ventricular Contraction in ECG

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Implement different operations on signals and systems.	**K3**
CO2	Apply suitable signal processing algorithms for biomedical signal analysis.	**K3**
CO3	Design solutions to address the issues during bio-signal acquisition and processing.	**K6**
CO4	Implement FIR and IIR filters	**K3**
CO5	Function effectively as an individual and in a team to accomplish the given task	**K6**

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3		2								2
CO3		2		2								2
CO4		2		2								2
CO5		2		2								2

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biomedical Signal Analysis	Rangaraj M Rangayyan	John Wiley	2nd Edition, 2015.
2	Signals & Systems in Biomedical Engineering	Suresh R Devasahayam	Springer	2^{nd} Edition, 2013
3	Biomedical signal processing: principles and techniques.	Reddy, D. C.	McGraw-Hill	2005
4	Digital Signal Processing Principles, Algorithms and Applications	John G Proakis& Dimitris G Manolakis	Prentice Hall of India	2005
5	Practical Guide for Biomedical Signals Analysis Using Machine Learning Techniques	AbdulhamitSubasi,	Academic Press	1st Edition, 2019
6	Biomedical signal processing	Akay, Metin.	Academic press	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Biomedical digital signal processing."	Tompkins, Willis J.	Editorial Prentice Hall	1993
2	"Digital signal processing (Book)." Research supported by the Massachusetts Institute of Technology, Bell Telephone Laboratories, and Guggenheim Foundation. Englewood Cliffs, N. J.	Oppenheim, Alan V., and Ronald W. Schafer	Prentice-Hall,Inc.	1975. 598 p
3	Digital signal processing: a computer-basedapproach. Vol. 2. New York	Mitra, Sanjit Kumar, and Yonghong Kuo.	McGraw-Hill	2006
4	Signals and systems for bioengineers: a MATLAB-based introduction	Semmlow, John.	Academic Press	2011

<u>Continuous Assessment (25 Marks)</u>

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.

- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

BIOMEDICAL & ROBOTIC ENGINEERING

SEMESTER S6

MECHATRONICS

Course Code	**PCBRT601**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104, PCBRT502	**Course Type**	Theory

Course Objectives:

1. To get a basic idea about mechatronics – the working of differently powered systems as one unit
2. To familiarize the mechanical actuators / mechanism and its electrical and electronic supporting systems and along with the application of mechatronic systems in industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Mechatronics Design System and Interfacing**. Introduction to Mechatronics. Historical development and definitions of mechatronic systems. Key elements of Mechatronic Systems. Functions of mechatronic systems. Design approach -Properties of Conventional and Mechatronic Design Systems. Ways of integration. Concurrent Design Procedure for Mechatronic Systems. Examples of mechatronic systems. **Mechatronic System Interfacing**: Input Signals and Output signals of a Mechatronic System. **System modelling** -Basic building blocks of general mechanical, electrical, fluid and thermal systems	**11**
2	**Actuation Systems**: Pneumatic & Hydraulic Systems: Process Control Valves, Directional and Pressure Control valves, Linear and Rotary	**11**

	actuators. **Mechanical Actuation Systems**: Translational and Rotational motions, Kinematic Chains, Cams, Gear Trains, Ratchet and Pawl, Belt and Chain drives, Bearings. Electrical Actuation Systems, Performance characteristics, and System modelling. **Electrical Actuation Systems**: Mechanical and Solid State Relays, Solenoids, DC & AC motors, Servo & Stepper motors	
3	**Sensors and Transducers-** Static and dynamic performance characteristics, Internal sensors, External sensors and microsensors. **Mechatronic System Controllers**, Mechatronic System Controllers (basics only): ON/OFF, P, I, D, PI and PID Controllers, Digital controllers, Intelligent Controllers in Mechatronics. Fundamentals of numerical control - advantages of NC systems- classification of NC systems - point to point and contouring systems - NC and CNC - incremental and absolute systems -open loop and closed loop systems	**11**
4	Basics of Programmable Logic Controller and Industrial Robotics. **Programmable Logic Controller**- Basic PLC structure, I/O processing. Industrial robotics -basic concepts-robot anatomy-robotics and automation - specification of robots - resolution - repeatability and accuracy of manipulator - classification of robots.	**11**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Analyze various mechatronic instruments and tools	**K4**
CO2	Understand actuators and its electrical and electronic supporting systems	**K2**
CO3	Apply the principle of various controllers and numerical control machines for system design.	**K3**
CO4	Design and implement basic robotic solutions and analyse its specifications	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2							2		3
CO2	3	3	2							2		3
CO3	3	3	2							2		3
CO4	3	3	2							2		3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Mechatronics, Electronic Control systems in Mechanical and Electrical Engineering	Bolton. N	Pearson Education	4^{rth} edition, 2008
2	Mechatronics-An Introduction	Robert H Bishop	C R C press	1^{st}edition,2005
3	Introduction to Mechatronics	K. K. Appukuttan	Oxford University Press	1^{st}edition,2007.
4	Mechatronics-Principles, Concepts and Applications	Nitaigour P Premchand	TMH	11^{th}edition,2011.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Mechatronics-Integrated Technologies for Intelligent Machines	A. Smaili, F. Mrad	Oxford	1st edition,2009
2	Introduction to Mechatronics and Measurement Systems	David G Alciatore, Micheal	TMH	3rd edition,2007
3	Mechatronics – Electronics in Products & Processes.	Dradly. D.A, Dawson. D, Burd N. CandLoader A. J,	Springer	1st edition, 1993
4	Electromechanics- Principles concept and Devices	James Harter	Prentice-Hall	2nd edition, 2002
5	Numerical Control of Machine Tools	Yoram Koren & Ben Yuri	Khanna Publishers	1st edition,1984

SEMESTER S6

PRINCIPLES OF MEDICAL IMAGE PROCESSING

Course Code	PCBRT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Summarize the types of CT used. Explain their need and principle
2. Mention the types of image reconstruction techniques used in MRI**.**
3. The applications of transforms in biomedical engineering**.** The different spatial domain smoothing techniques
4. an image observation model and explains the reasons for degradations in the observed image

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**X- Ray Imaging**: Basic principles of image formation, Block diagram of an X-Ray machine, Digital radiography - Basic principles **X-Ray Computed Tomography**: Basic principles, Contrast scale, Different generations of CT scanners, Basic principles of image reconstruction. **Ultrasonic Imaging**: Physical principles, Transducer parameters, Different modes - A-mode, M-mode (echocardiograph), B-mode, Principles of Doppler ultrasonic imaging. **Magnetic Resonance Imaging**: Principles of MRI, T1 weighted images, T2 weighted images, Proton density weighted images, Applications of MRI **Nuclear Medicine Imaging Modalities**: Emission Computed Tomography- SPECT & PET	**9**
2	**Image Perception:** MTF of the visual system, Monochrome vision model, Color vision model - HSI model, Image sampling and quantization. **Image Transforms:** Two dimensional orthogonal and unitary transforms,	**9**

	Properties of unitary transforms- DCT, KL Transform	
3	**Image Enhancement in Spatial Domain**: Grey level transformations, Histogram processing - Histogram equalization and modification, Smoothing and sharpening spatial filters, Zooming and interpolation. **Image Enhancement in Frequency Domain:** Filtering- low pass, high pass, band pass and band stop filters, Homomorphic filter	**9**
4	**Image Restoration**: Inverse filtering, Wiener filtering. **Image Segmentation:** Detection of discontinuities- Point, line, edge, canny edge detector. Edge linking - Hough Transform. Segmentation based on thresholding - Basic global thresholding, variable thresholding. Region based segmentation- region growing, region splitting and merging	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Identify major processes involved in formation of medical images.	**K3**
CO2	Understand transforms used in image processing.	**K2**
CO3	Understand the different methods of image processing both in the time and transform domains.	**K2**
CO4	Describe fundamental methods for image enhancement and segmentation	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2							1		1
CO2	3	2	3							1		1
CO3	3	3	2							1		1
CO4	3	2	2							1		1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Webb's The Physics of Medical Imaging	Flower M. A	CRC Press	2012
2	Digital Image Processing	Gonzalez Rafel C, Woods Richard E,	Taylor & Francis	2018
3	Fundamentals of Digital Image Processing	Jain Anil K	Prentice Hall of India	1989

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical Imaging Analysis	Atam P Dhawan	Wiley Inter science publication	2003
2	Ultrasound Physics & Instrumentation	D L Hykes, W R Hedrick & D E Starchman	Churchill Livingstone, Melbourne	1985
3	Ultrasonic Bioinstrumentation	Douglas A Christensen	John Wiley, New York	1988
4	Digital Image Processing	Pratt William K	John Wiley and Sons	2001
5	Biomedical Image Processing	Thomas M. Deserno	Springer-Verlag Berlin Heidelberg	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/117/105/117105135/
2	https://onlinecourses.nptel.ac.in/noc22_ee116/preview

SEMESTER S6

INDUSTRIAL MANIPULATORS

Course Code	PEBRT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT502	Course Type	THEORY

Course Objectives:

1. Inverse kinematic transformation of position**.**
2. To understand the basics of industrial manipulator, manipulator kinematics and mechanics of robot motion, forward and inverse kinematic transformation of position.
3. To understand forward and inverse kinematic transformation of velocity, end effectors force transformations.
4. To understand about manipulator dynamics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Robot Subsystems -Classification of Robots -Industrial Applications Robotic configurations- robot motion- joint notation schemes-End effectors-types- mechanical, vacuum, magnetic grippers- power transmission systems-modelling and control of single joint robot. Industrial robot- manipulator structures Kinematics-: Position definitions- Coordinate frames - Different orientation descriptions - Free vectors- Translations, rotations and relative motion - Position and Orientation of a Rigid Body- Rotation Matrix- Elementary Rotations- Representation and rotation of a Vector - Composition of Rotation Matrices-Euler Angles- YZY angles- Roll-Pitch-Yaw Angles-Angle and axis	9

2	Introduction to manipulator kinematics- position representation -forward and reverse transformations of the 2-DOF arm—adding dimensions – Homogeneous transformation and robot kinematics. Direct Kinematics: Kinematic Chains - Denavit Hartenberg Representation- Kinematics of manipulator structures- Three link planar arm- parallelogram arm- spherical arm - anthropomorphic arm- spherical arm – Stanford manipulator. Manipulator Kinematics-2 Inverse Kinematics- The General Inverse Kinematics Problem- kinematic decoupling - inverse position – inverse orientation- joint space and operational space -workspace.	**9**
3	Differential Kinematics and Statics:- Geometric Jacobian- Derivative of a Rotation Matrix - Link Velocity -Jacobian Computation.-Jacobian of Typical Manipulator Structures- Three- link Planar Arm - Anthropomorphic Arm- Stanford Manipulator - Kinematic Singularities- Singularity Decoupling - Wrist Singularities- Arm Singularities -Analysis of Redundancy - Differential Kinematics Inversion -Redundant Manipulators - Kinematic Singularities -	**9**
4	Manipulator Dynamics-1 Lagrange Formulation -Properties of Dynamic Model -Dynamic Model of Simple Manipulator Structures - Two-link Cartesian Arm- Two-link Planar Arm.- Parallelogram Arm - Dynamic Parameter Identification-Newton-Euler Formulation- Link Acceleration Manipulator Dynamics-2 Trajectory Planning - Path and Trajectory -Joint Space Trajectories-Point-to-point Motion Path Motion- Operational Space Trajectories-Path Primitives - position -Orientation - Dynamic Scaling of Trajectories - Motion Control: Joint Space Control- Independent Joint Control- Feedback Control- Decentralized Feedforward Compensation	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the basics of industrial manipulator, manipulator kinematics and mechanics of robot motion, forward and inverse kinematic transformation of position	**K2**
CO2	Understand forward and inverse kinematic transformation of velocity, end effectors force transformations	**K2**
CO3	Understand about manipulator dynamics	**K2**
CO4	Understand robot control schemes	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	2	2										
CO3	2	2										
CO4	2	2										
CO5	3	3	1									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics	S K Saha	MC GRAW HILL INDIA	
2	Industrial Robotics	Groover Mikell P., M. Weiss, R.N. Nagel, N.G. Odrey	McGrawHill	1986, ISBN-13: 978-0070249899 ISBN-10: 007024989X
3	Modelling & Control of Robot Manipulators	Sciavicco, L., B. Siciliano	Springer Verlag	2nd Edition, 2000
4	Robot Dynamics and Control	Mark W. Spong & M. Vidyasagar	John Wiley & Sons	1989, ISBN: 978-0-471-61243-8

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Calibration of Industrial Robot Manipulators	Patrick Maurine	Wiley	ISTE, 2015, ISBN-10: 1848212542 ISBN-13: 978-1848212541
2	Advanced Robotics & Intelligent machines,	Gray J.O., D.G. Caldwell(Ed)	The Institution of Electrical Engineers, UK	1996, ISBN-13: 978-0852968536 ISBN-10: 0852968531
3	Introduction to Robotics: Mechanics & Control	Craig, John J.	Pearson Education	2nd Edition, 1989, ISBN 0131236296
4	Chaouki T.Abdallah Robot Manipulator Control Theory and Practice	Frank L.Lewis, Darren M.Dawson	Marcel Dekker Inc	2006, ISBN: 0-8247-4072-6

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=B9jDCj581Os, https://archive.nptel.ac.in/courses/112/104/112104308/, https://www.youtube.com/watch?v=X7iBT5l599c
2	https://www.youtube.com/watch?v=nAwVfwSHAP0, https://www.youtube.com/watch?v=f5GMjGRsKi4, https://www.youtube.com/watch?v=KHJdSfcck78
3	http://acl.digimat.in/nptel/courses/video/112106304/L19.html
4	https://www.youtube.com/watch?v=QN-Awth50aA

SEMESTER S6

ROBOT MOTION PLANNING

Course Code	**PEBRT632**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT302	**Course Type**	Theory

Course Objectives:

1. To explain the concepts of topology and configuration space in motion planning
2. To describe the classical motion planning techniques
3. To describe the working of sampling-based motion planners
4. To apply sensor information to motion planning and obstacle avoidance
5. To explain dynamic motion planning techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Overview of robot motion planning:** Application areas of robot motion planning - Personal Transport Vehicles, Planetary Exploration, Demining, Fixed-base Robot Arms in Industry, Search and Rescue Robots, Surgical Robots; Concepts in Motion Planning – Task, Properties of the Robot, Properties of the Algorithm **Basic Topological Concepts:** Basic definitions of Topological Spaces, Homeomorphism, Diffeomorphism, Differentiable Manifolds, Paths and Connectivity – Paths, connected vs. path connected **Configuration space:** Specifying a Robot's Configuration, Obstacles and the Configuration Space - Circular Mobile Robot, Two-Joint Planar Arm, C obstacles, Topology of the Configuration Space	9

2	**Roadmaps:** Definition, Accessibility, Departability, Connectivity, Types of roadmaps: visibility maps – visibility graphs, deformation retracts – Generalized Voronoi Diagram GVD,Sensor-Based Construction of the GVD, Polygonal Spaces, Grid Configuration Spaces - The Brushfire Method, Silhouette Methods - Canny's Roadmap Algorithm **Cell decomposition:** Exact cell decomposition, Approximate cell decomposition, Trapezoidal Decomposition, Morse Cell Decomposition, Boustrophedon Decomposition **Potential Field Planning**: Potential Field Method, Attractive, Repulsive potential	**9**
3	**Sampling-Based Algorithms:** Characteristics of Sampling-Based Planners, Probabilistic Roadmaps: Basic PRM – Roadmap Construction, Query Phase, PRM Sampling Strategies - Sampling near the obstacles, Sampling inside narrow passages, Visibility-Based Sampling, Manipulability-Based Sampling, Single-Query Sampling-Based Planners - Expansive-Spaces Trees EST, Rapidly-Exploring Random Trees RRTs	**9**
4	**Sensor-Based Motion Planning Algorithms**: Bug1, Bug2, Tangent Bug algorithms, Dynamic window approaches, A*, D* Algorithms **Multiple Robot Motion Planning:** Problem Formulation, Composite Configuration Space, Centralized Planning, Decoupled planning - Prioritized planning - Fixed-path coordination, Fixed-roadmap coordination **Dynamic Motion Planning**: Moving Obstacles, Configuration Time space, Planning without velocity bound - Exact Cell Decomposition, Approximate Cell Decomposition, Planning with velocity bound - Asteroid Avoidance Problem, Approximate Cell Decomposition, Velocity Tuning	**9**

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the concepts of topology and configuration space in motion planning	**K2**
CO2	Describe the classical motion planning techniques	**K1**
CO3	Describe the working of sampling-based motion planners	**K1**
CO4	Apply sensor information to motion planning and obstacle avoidance	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Principles of Robot Motion - Theory, Algorithms, and Implementation	Choset, Lynch, Hutchinson, Kantor, Burgard, Kavraki, Thrun	MIT Press	2007
2	Planning Algorithms	Steven M. LaValle,	Cambridge University Press	1999
3	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh,	MIT Press	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning,Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	2019
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/112/104/112104308/
2	https://archive.nptel.ac.in/courses/112/104/112104308/
3	https://archive.nptel.ac.in/courses/112/104/112104308/
4	https://archive.nptel.ac.in/courses/112/104/112104308/

SEMESTER S6

INDUSTRY 5.0

Course Code	PEBRT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To create a manufacturing system whereby machines in factories are augmented with wireless connectivity and sensors to monitor and visualize an entire production process and make autonomous decisions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Industry 5.0, Comparison of Industry 4.0 Factory and today's Factory - The 10 most important things that will change with Industry 4.0, Difference between conventional automation and Industry 4.0, Internet of Things (IoT) - Industrial Internet of Things (IIoT) - Internet of Services, Big Data - Cyber-Physical Systems - Digital Twins - Cloud Computing - Cloud Manufacturing - Security issues within Industry 4.0 networks	9
2	Introduction to Cyber-Physical Systems (CPS), Core elements of Cyber-Physical Systems - CPPS - Control theory and real time requirements, Communication in CPS - Design Methods for Cyber-physical Systems – Modelling - Programming - Model-Integrated Development, Applications for cyber-physical systems - Examples - Manufacturing – Traffic - Medical technology.	9

3	The intelligent work piece as basic functionality in implementing Industry 4.0, Work piece tagging - QR codes and RFID - Communication between work piece and environment, Multi-agent systems in production, Applications for smart work pieces (examples of existing or future applications in the field of manufacturing). The connected worker within the industry 4.0,	**9**
4	Basic concepts of Digital Twins, Benefits - Impact – Challenges, Features and Implementation of Digital Twins, Types of Digital Twins, Applications for digital twins in production. Diversity - driven workplaces (barrier free workplaces, accessibility in production), Human and task-centred assistance systems - Technical tools - Ambient Assisted Working (AAW) - Mobile information technologies - Shop floor information systems, Production line support systems - Manipulator systems and intelligent chairs - Human work support by using exoskeletons	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of Industry 5.0 and Industry 4.0	**K2**
CO2	Describe cyber physical system and the emerging applications	**K2**
CO3	Summarize the basic concepts of smart work piece	**K1**
CO4	Understand the fundamental Digital Twins in production	**K1**
CO5	Comprehend the fundamentals of Assistance systems for production	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	2	-	2	-	-	-	-	2
CO2	3	2	-	-	2	-	2	-	-	-	-	1
CO3	2	-	-	-	1	-	2	-	-	-	-	1
CO4	2	-	-	-	1	-	2	-	-	-	-	1
CO5	3	2	-	-	2	-	1	-	-	-	-	1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Industry 4.0	Jean-Claude André	Wiley- ISTE	4^{th} Edition, , 2019
2	Industry 4.0 Technologies for Education: Transformative Technologies and Applications,	P. Kaliraj, T. Devi,	United States: CRC Press	1^{st} Edition, 2023,
3	Industrial Internet of Things: Cyber manufacturing Systems	Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat,	Germany: Springer International Publishing	2^{nd} Edition, , 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Concept Industry 4.0: An Empirical Analysis of Technologies and Applications in Production Logistics, ,	Bartodziej, C. J.	Germany: Springer Fachmedien Wiesbaden	2^{nd} Edition,
2	Introduction to Industrial Internet of Things and Industry 4.0,	Mukherjee, A., Roy, C., Misra, S.,	United States: CRC Press	4^{th} Edition, 2021
3	Industry 4.0: The Industrial Internet of Things,	Gilchrist, A.,	United States: A press,	3^{rd} Edition, 2016
4	Industry 4.0: Current Status and Future Trends,		United Kingdom: Intech Open	3^{rd} Edition,

SEMESTER S6

DIGITAL CONTROL SYSTEMS

Course Code	**PEBRT634**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	THEORY

Course Objectives:

1. Data reconstruction is done in zero order hold and first order hold
2. The mapping of the following locus from s-plane to z-plane
    a. Constant damping loci
    b. Constant frequency loci
    c. Constant damping ratio loci
3. The state space representation of the following transfer function
4. The concept of pole placement by state feedback

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Basic Elements of discrete data control systems, advantages of discrete data control systems, examples. Signal conversion & processing: Digital signals & coding, data conversion & quantization, sample and hold devices, Mathematical modelling of the sampling process; Data reconstruction and filtering of sampled signals: Zero order hold, first order Hold and polygonal hold.	9
2	Digital control systems- Pulse transfer function. z transform analysis of closed loop open loop systems-Modified z- transfer function Difference equation. Solution by recursion and z transform. Steady state error analysis- Examples on static error coefficients. Bilinear transformation- mapping from s-plane to z-plane.	9

3	Stability of linear digital control systems- Routh Hurwitz criteria, Jury's test. Root loci of digital control systems – rules for construction of root locus. Frequency domain analysis - Bode Plots-Gain margin and Phase margin	**9**
4	Review of state space techniques to continuous data systems, state space representation of discrete time systems- Transfer function from state space model-various canonical forms- conversion of transfer function model to state space model-characteristics equation- solution to discrete state equations. Controllability and Observability - Response between sampling instants using state variable approach.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Analyse discrete data control system	**K1**
CO2	Understand Z-transform, its properties and do steady state error analysis of digital control system	**K2**
CO3	Understand and gain knowledge in stability analysis of digital control systems	**K3**
CO4	Test controllability and observability of linear systems also design discrete control systems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2								
CO3	3	2		2								
CO4	3	2		2								
CO5	3	2		2								

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Digital control systems"	B. C. Kuo	Oxford University Press	**2007**
2	"Discrete Time control systems	K. Ogatta	PHI	**1995**
3	"Digital Control systems and state variable methods"	M. Gopal	Tata McGraw Hill	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	“Continuous & Discrete Control Systems “	John Dorsey	MGH	
2	“Continuous & Discrete Control Systems “,	R John Dorsey	MGH	
3	Control System Engineering	R Nagrath & Gopal	Wiley Eastern	
4	“Digital control of Dynamic Systems”	R F. Franklin, J.D. Powell, and M.L. Workman	Addison - Wesley Longman, Inc., Menlo Park, CA	1998

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/14cMhrp5wlk
2	https://youtu.be/NCLo4Kqm3P8
3	https://youtu.be/3rzR3Zay1VA
4	https://youtu.be/RcuGxWc0HyQ
5	https://youtu.be/P_EL2iHFZkg

SEMESTER S6

MACHINE VISION

Course Code	PEBRT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. This course will enable the students to learn fundamental digital image processing and machine vision concepts and their application to the fields of robotics and automation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Image Acquisition, Lenses and Cameras: Pinhole camera, Gaussian Optics, Depth Field, Telemetric lenses, Lens aberrations Cameras: CCD cameras, CMOS Camera, Colour cameras: Single chip cameras, Three chip cameras, Camera performance parameters: noises, dynamic range, Camera-Computer interfaces, Digital video signals: camera link, IEEE1394, USB2.0, USB 3 vision	9
2	Image Fundamentals: Images, regions, piecewise contours, Image Enhancement: contrast enhancement, contrast normalization, Image Smoothing: Temporal Averaging, Mean Filter, Noise Suppression by Linear Filters, Median and Rank Filters, 2-D Fourier Transform: Continuous Fourier Transform, Discrete Fourier Transform. Geometric Transformations: Affine Transformations, Projective Transformations	9
3	Image Transformations: Nearest-Neighbour Interpolation, Bilinear Interpolation, Bicubic Interpolation, Smoothing to Avoid Aliasing, Projective Image Transformations, Transformations, Image Segmentation Thresholding: Global Thresholding, Automatic Threshold Selection,	9

	Dynamic Thresholding, Variation Model, morphological operations	
4	Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object detection Pattern Analysis: Clustering: K-Means, Mixture of Gaussians, Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised; Classifiers: Bayes, KNN, ANN models; Dimensionality Reduction: PCA, LDA, ICA	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. ***Problem Definition (K4 - 4 points)***
   a. *Clearly defines the real-world problem.*
   b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*
2. ***Problem Analysis (K4 - 4 points)***
   a. *Break-down and presents a well-reasoned solution approach.*
   b. *Compare and justify the proposed solutions with evidence and logical reasoning.*
3. ***Evaluate (K5 - 4 points)***
   a. *Thoroughly evaluate the proposed solutions.*
   b. *Compares trade-offs, advantages, and disadvantages.*
   c. *Considers feasibility, scalability, and practical implications.*
4. ***Implementation (K5 - 4 points)***

a. *Select the most feasible solution by implementing the proposed solutions.*

b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*

c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*

5. ***Conclusion (K4- 2 points, K5 – 2 points)***

   a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***

   b. *Reflects critical thinking and informed decision-making.* ***(K5)***

Scoring:

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the vision capturing systems and its industry standards	**K2**
CO2	Acquire images and standardize the images by applying standard techniques like smoothing and filtering.	**K3**
CO3	Apply various transform tools like frequency domain and affine transform.	**K3**
CO4	Apply various segmentation algorithms. Apply state-of-the-art pattern analysis techniques like clustering, classifying and dimensionality reduction	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3				2						2
CO2	3	2										
CO3	3	2										
CO4	3	3	1									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Machine Vision Algorithms and Applications	Carsten Steger, Markus Ulrich, Christian Wiedemann	WILEY-VCH, Weinheim	2008
2	Algorithms and Applications,	Richard Szeliski, Computer Vision	Springer-Verlag London Limited	2011
3	Computer Vision: A Modern Approach	D. A. Forsyth, J. Ponce	Pearson Education,	2003

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Digital Image Processing,	Rafael C. Gonzalez and Richard E.woods	Addition –Wesley Publishing Company, New Delhi	2007
2	High-Level Vision: Object recognition and Visual Cognition,	Shimon Ullman	ABradford Book, USA	2000
3	A Do-It-Yourself Guide to Robot OperatingSystem - Volume I,	R.Patrick Goebel	A Pi Robot Production	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/uvXTZxSzdMk?si=VO8rnpQ2ymfrrMM1
2	https://youtu.be/VYgFAdvDpzM?si=Xwn72-QGN6JDxTVx
3	https://youtu.be/RCfjuSDKuTY?si=rGVstg_QYcRgGR82
4	https://youtu.be/3qJej6wgezA?si=ZWNDEMVALDI2HAGj
5	https://youtu.be/Z66_2_VG_k8?si=ZHLLKO9GuHQIu5YN

SEMESTER S6

SOFT COMPUTING AND COMPUTER VISION FOR ROBOTS

Course Code	**PBBRT604**	**CIE Marks**	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	**ESE Marks**	40
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	-	**Course Type**	Theory

Course Objectives:

1. Apply Neural Networks - architecture, functions and various algorithms in relevant problem areas.
2. Develop Fuzzy logic systems for typical computational applications.
3. To review image processing techniques for computer vision
4. To understand shape and region analysis

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to soft computing techniques: Neural networks, fuzzy logic, genetic algorithms. Artificial neural networks: Biological neural networks, model of an artificial neuron, architecture, activation functions, learning methods, McCulloch & Pitts model, Linear separability, Hebb net.	9
2	Types of networks: Perceptron networks, ADALINE, MADALINE, Feedback Networks: Discrete Hopfield nets, Feed Forward Networks: Back Propagation Networks - architecture, training algorithm, Problems. Competitive net: Maxent.	9
3	Review of image processing techniques: Digital filters, linear filters-Homomorphic filtering, Point operators- Histogram, neighbourhood operators, thresholding	9

4	Mathematical morphology, Binary shape analysis, Binary shape analysis, Erosion, Dilation, Opening and Closing, Hit-or-Miss Transform, connectedness, object labelling and counting, Boundary descriptors – Chain codes. Properties of Binary Regions, Geometric Features, Statistical Shape Properties.	9

Suggestion on Project Topics

- The objective of **Project based learning is** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the first class, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups.
- Each group should identify a project relevant to the subject of study.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks **(8x2 =16 marks)**	• 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. **(4x6 = 24 marks)**	**40**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply Neural Networks - architecture, functions and various algorithms in relevant problem areas.	**K3**
CO2	Develop Fuzzy logic systems for typical computational applications.	**K4**
CO3	Understand digital filtering operations for CV applications.	**K2**
CO4	Apply basic morphological and boundary operators for Computer vision applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	3					2		2
CO2	3	3	3	2	3					2		2
CO3	3	3	2		2						2	3
CO4	3	3	2		2						2	3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of neural networks: architectures, algorithms andapplications	Fausett, Laurene V	Pearson Education India	2006
2	Principles of soft computing	Deepa, S. N., and S. N. Sivanandam.	Wiley	3rd edition 2018
3	Computer and Machine Vision -Theory Algorithm and Practicalities,	E. R. Davies,	Academic Press,	2012
4	Computer Vision: Algorithms and Applications,	Richard Szeliski,	Springer	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Neural Networks and Deep learning	Charu C. Aggarwal	Springer International Publishing	2018
2	Neural Networks, Fuzzy Logic and GeneticAlgorithm, Synthesis and Applications	S. Rajasekaran, G.A. Vijayalakshmi Pai	PHI Learning Pvt. Ltd	2017
3	Digital Image Processing and Computer Vision,	R. J. Schalkoff,	John Wiley,	2004
4	Computer Vision: Models, Learning, and Inference,	Simon J D Prince,	Inference, Cambridge University Press,	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/106105173
2	https://nptel.ac.in/courses/106105173
3	https://onlinecourses.nptel.ac.in/noc19_cs58/preview
4	https://onlinecourses.nptel.ac.in/noc19_cs58/preview

PBL Course Elements

L: Lecture (3 Hrs.)	**R: Project (1 Hr.), 2 Faculty Members**		
	Tutorial	**Practical**	**Presentation**
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		**30**

1. **Project Planning and Proposal (5 Marks)**
   - Clarity and feasibility of the project plan
   - Research and background understanding
   - Defined objectives and methodology

2. **Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
   - Individual contribution to the presentation
   - Effectiveness in answering questions and handling feedback

3. **Involvement in the Project Work and Team Work (3 Marks)**
   - Active participation and individual contribution
   - Teamwork and collaboration

4. **Execution and Implementation (10 Marks)**
   - Adherence to the project timeline and milestones
   - Application of theoretical knowledge and problem-solving
   - Final Result

5. **Final Presentation (5 Marks)**
    - Quality and clarity of the overall presentation
    - Individual contribution to the presentation
    - Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**
    - Overall quality and technical excellence of the project
    - Innovation and originality in the project
    - Creativity in solutions and approaches

SEMESTER S6

BIOMEDICAL INSTRUMENTATION

Course Code	**OEBRT611**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. To understand inverse kinematic transformation of position**.**
2. To understand the basics of industrial manipulator, manipulator kinematics and mechanics of robot motion, forward and inverse kinematic transformation of position.
3. To understand forward and inverse kinematic transformation of velocity, end effectors force transformations.
4. To understand about manipulator dynamics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to human physiological system:** Physiological systems of the body (brief discussion on Heart and cardio vascular system, Anatomy of nervous system, Physiology of respiratory systems) problems encountered in biomedical measurements. Sources of bioelectric potentials – resting and action potentials -propagation of action potentials – bio electric potentials example (ECG, EEG, EMG, ERG, EOG, EGG etc.) **Bio potential electrodes and ECG** Bio potential electrodes – theory – microelectrodes – skin surface electrodes – needle electrodes –biochemical transducers –transducers for biomedical applications. Electro conduction system of the heart. Electro cardiograph – electrodes and leads – Einthoven triangle, ECG read out devices, ECG machine – block diagram.	9

2	**Measurement of blood pressure, blood flow and heart sound** Measurement of blood pressure – direct and indirect measurement– oscillometric measurement – ultrasonic method, measurement of blood flow and cardiac output, plethysmography –photo electricand impedance plethysmographs. Measurement of heart sounds –phonocardiography.	9
3	**Measurement of EEG, EMG and Respiratory Parameters** Electro encephalogram –neuronal communication – EEG measurement, recording and analysis. Muscle response– Electromyogram (EMG) – Nerve Conduction velocity measurements- Electromyogram Measurements. Respiratory parameters – Spiro meter, pneumography **Therapeutic Aid** Cardiac pacemakers – internal and external pacemakers, defibrillators. Ventilators,	9
4	**Advances in Radiological Imaging** X-rays- principles of generation, uses of X-rays- diagnostic still picture, fluoroscopy, angiography, endoscopy, and diathermy. Basic principle of computed tomography, magnetic resonance imaging system and nuclear medicine system – radiation therapy. Ultrasonic imaging system – introduction and basic principle. **Electrical safety** Electrical safety– physiological effects of electric current –shock hazards from electrical equipment –method of accident prevention, introduction to tele-medicine	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the human anatomy and physiological signal Measurements. Illustrate various techniques used for measurement of Blood flow, blood pressure, and respiration rate and body temperature.	**K2**
CO2	Measurement of blood pressure, blood flow and heart sound	**K2**
CO3	Measurement of EEG, EMG and Respiratory Parameters and therapeutic aid	**K2**
CO4	Advances in Radiological Imaging and Electrical safety	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	3	3										
CO4	3	3										

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical Instrumentation, Application and Design,	J. G. Webster	John Wiley and Sons	
2	Biomedical Instrumentation Measurements	L. Cromwell, F. J. Weibell and L. A. Pfeiffer	Pearson education, Delhi	1990
3	Handbook of Biomedical Instrumentation	R. S. Khandpur	Tata Mc Graw Hill	
4	Introduction to Biomedical Equipment Technology	J. J. Carr and J. M. Brown	Pearson Education	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Joseph Bronzino, Introduction to Biomedical Engg.	John Enderele , Susan Blanchard	Academic Press	
2	Biomedical Instruments, Theory and Design	Welkovitz	Elselvier	
3	Medical Imaging Signals & Systems	Jerry L Prince, Jonathan M Links	Pearson Education	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=DEsTkXTtOLA, https://www.youtube.com/watch?v=mK6sPBbChqc
2	https://www.youtube.com/watch?v=DQxiUYXi4UE, https://www.youtube.com/watch?v=3EFq5WbvJTQ
3	https://www.digimat.in/nptel/courses/video/108108125/L45.html, https://www.youtube.com/watch?v=q57ybN7m6V0
4	https://www.digimat.in/nptel/courses/video/102105090/L07.html, https://www.youtube.com/watch?v=TpPXxJ7fPDs

SEMESTER S6

FUNDAMENTALS OF ROBOTICS

Course Code	OEBRT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To help the students to get a basic idea about Robots.
2. Basic design of robots is introduced.
3. Trajectory planning, obstacle avoidance and kinematics of robots are introduced to students.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Trajectory planning Tasks, Path planning, Trajectory Planning. Joint space trajectory planning- cubic polynomial, linear trajectory with parabolic blends, trajectory planning with via points; Cartesian space planning, Point to point vs continuous path planning. Obstacle avoidance methods- Artificial Potential field.	9

2	Robot Kinematics Direct Kinematics- Rotations-Fundamental and composite Rotations, Homogeneous co- ordinates, Translations and rotations, Composite homogeneous transformations, Screw transformations, Kinematic parameters, The Denavit-Hartenberg (D-H) representation, The arm equation, direct kinematics problems (upto 3DOF) Inverse kinematics- general properties of solutions, Problems (upto 3DOF) Inverse kinematics of 3DOF manipulator with concurrent wrist (demo/assignment only) Tool configuration Jacobian, relation between joint and end effector velocities.	**9**
3	Manipulator Dynamics Lagrange'sformulation – Kinetic Energy expression, velocity Jacobian and Potential Energy expression, Generalised force, Euler-Lagrange equation, Dynamic model of planar and spatial serial robots upto 2 DOF, modelling including motor and gearbox. Robot Control The control problem, Single axis PID control-its disadvantages, PD gravity control, computed torque control. Simulation of simple robot-control system-Matlab programming for control of robots (demonstration/assignment only)	**9**
4	Locomotion, Key issues for locomotion, Legged Mobile Robots, Wheeled Mobile Robots, Aerial Mobile Robots. Mobile Robot Kinematics (Differential Drive Robot), Mobile Robot Workspace- Degree of freedom, Holonomic and nonholonomic robots. Motion Control, Open Loop control, feedback control. Industrial application- Medical, material handling, pick and place robot. Robot selection- number of axes, work volume, capacity and speed, stroke and reach, repeatability , precision and accuracy, operating environment	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Familiarise with anatomy, specifications and types of Robots	**K1**
CO2	Obtain forward and inverse kinematic models of robotic manipulators	**K6**
CO3	Plan trajectories in joint space & Cartesian space and avoid obstacles while robots are in motion	**K3**
CO4	Familiarise with different types of mobile robots, kinematic models, motion control for mobile robots	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3
CO5	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall	
2	Introduction to Robotics (Mechanics and control)	John. J. Craig	Pearson Education	
3	Introduction to Robotics	S K Saha	Mc Graw Hill Education	
4	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	
5	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University press.	
6	Robotics Technology and Flexible Automation	S. R. Deb		2^{nd} Ed
7	Introduction to Autonomous Mobile Robots	Siegwart, Roland	Cambridge, Mass: MIT Press	2^{nd} ED

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of Robotics	Sicilliano, Khatib	Springer	
2	Introduction to Robotics – Mechanics and Control	John J. Craig		
3	Modern Robotics Mechanics, Planning and Control	Kevin M. Lynch, Frank C. Park		
4	Robotics Modelling, Planning and Control	Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, Giuseppe Oriolo		
5	Fundamentals of Robotics	D.K. Pratihar	Narosa Publishing House, New-Delhi	2017
6	Robotics	K.S. Fu, R.C. Gonzalez, C.S.G. Lee	McGraw-Hill Book Company	1987
7	Introduction to Robotics	J.J. Craig	Addison-Wesley Publishing Company	1986

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/107106090
2	https://onlinecourses.nptel.ac.in/noc19_me74/preview
3	https://www.classcentral.com/course/swayam-introduction-to-robotics-19912

SEMESTER S6

AI IN ROBOTICS

Course Code	**OEBRT613**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	THEORY

Course Objectives:

1. Compare and contrast different types of search strategies
2. Ontology in Al
3. Expert systems in AI

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction and Problem solving:** History, state of the art, Need for AI in Robotics. Thinking and acting humanly, Intelligent agents, structure of agents. Solving problems by searching –Informed search and exploration– Constraint satisfaction problems– Adversarial search,	**9**
2	**Knowledge representation and Planning:** Knowledge representation, first order logic. Planning with forward and backward State space search – Partial order planning – Planning graphs– Planning with propositional logic – Planning and acting in real world.	**9**
3	**Reasoning and Decision making:** Uncertainty – Probabilistic reasoning, Dynamic Bayesian Networks; Basis of utility theory, decision theory, sequential decision problems;	**9**

4	Forms of learning – Knowledge in learning – Statistical learning methods –reinforcement learning, communication, perceiving and acting, Probabilistic language processing, and perception. **AI In Robotics and Applications:** Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Identify and solve problems using appropriate AI methods	**K1/K2**
CO2	Formalize a given problem in the language/framework of different AI methods	**K3**
CO3	Describe the learning methods adopted in AI	**K4**
CO4	Interpret various applications of AI in Robotic Applications	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3
CO5	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Artificial Intelligence: A modern approach"	Stuart Russel, Peter Norvig	Pearson Education, India	2016
2	Machine Learning; A Probabilistic Perspective	Kevin Murphy	MIT Press	2012
3	"Artificial Intelligence: A guide to Intelligence Systems"	Negnevitsky, M, , Harlow	AddisonWesley	2002
4	Introduction to Robotics	S K Saha	McGraw Hill Education	
5	Introduction to Autonomous Mobile Robots	Siegwart, Roland,	Cambridge, Mass. : MIT Press	2nd ed

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Introduction to AI Robotics"	Robin Murphy, Robin R. Murphy, Ronald C. Arkin	MIT Press	2000
2	"Artificial Intelligence: Robotics and Machine Evolution"	David Jefferis	Crabtree Publishing Company	1992
3	"Artificial Intelligence for Robotics"	Fransis X. Govers	Packt Publishing	2018
4	"Handbook of Robotics"	Sicilliano, Khatib	Springer	
5	Introduction to Robotics – Mechanics and Control	John J. Craig		

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/XCPZBD9lbVo
2	https://youtu.be/ARywou8HLQk
3	https://youtu.be/GqxOwxwH1C8
4	https://youtu.be/yCXm5cgG0UA
5	https://youtu.be/bRsjfbqHj8k

SEMESTER S6

ASSISTIVE MEDICAL DEVICES

Course Code	OEBRT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	THEORY

Course Objectives:

1. Identify the elements of formal definitions of AT. Contrast these with information definitions of AT.
2. The major approaches for generating accessibility to tablet and smartphone technologies.
3. Functional manipulative tasks that can be aided by assistive technologies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Definitions of assistive technology, Principles of assistive technology, The Human Activity Assistive Technology Model, Foundational concepts, Application of HAAT, Universal Design **Functional Framework for Assistive Technologies**: Hard and Soft technologies, Allocation of functions, Hard assistive technologies, soft assistive technologies, technology options.	9
2	**Control Interfaces:** Anatomic Sites for Control of AT, Elements of the Human/Technology Interface, Speech, Contexts affecting HTI, Functional movement evaluation, Characteristics of Control Interfaces, CI for direct and indirect selection - Keyboards, Touch screens, Concept keyboards, Eye-Controlled systems, Tracking of body features, Enhancing keyboard control, Pointing interfaces, Switches, Switch arrays, Discrete joysticks, and Chord Keyboards, Selecting a CI, Multiple versus Integrated CI, Evaluation of CI.	9

3	**Accessing the Information Highway:** Cell phone accessibility, Internet accessibility, Raising the floor assistive technology, Cross-platform accessibility - Automatic speech recognition, Brain–Computer Interface, Computer access, Access to phones and tablets. **Assistive Technologies for Seating and Positioning:** Biomechanical principles, seating technologies, Seating assessment, Principles of intervention. **Wheeled Mobility:** Supporting structure, Frame types, Propelling structure, Specialized bases, Smart Wheelchairs	**9**
4	**Assistive Technologies for manipulation:** Low-technology Aids for manipulation, Special-purpose electromechanical aids for manipulation, Electronic aids to daily living, Robotic aids - Principles of Assistive Robots, Robots as personal assistants, Mobile assistive robots **Assistive Technologies for Vision:** Reading aids, Accessing ICT, Mobility and orientation Aids, Assessment - Visual function orientation and mobility **Assistive Technologies for hearing:** Approaches to auditory sensory aids, Aids for auditory impairments, Face- to-Face communication, Alerting devices, Assistive listening devices. Aids for persons with both visual and auditory impairments.	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the framework for assessing the usability of technology and undertake product research and development	**K2**
CO2	Discuss the major approaches to the design of control interfaces that are used with Ats	**K3**
CO3	Apply principles of AT to guide recommendation of access to ICT enabled systems and aids for seating, positioning, and mobility	**K4**
CO4	Describe the mainstream technologies that support individuals with low vision, blindness, and cognitive impairment	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3		2		2		2		3
CO2	3	3	3	3		2		2		2		3
CO3	3	3	3	3		2		2		2		3
CO4	3	3	3	3		2		2		2		3

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Assistive Technologies: Principles and Practice"	Albert M. Cook and Janice M. Polgar	Elsevier	4th Edition, 2015
2	"E-Health, Assistive Technologies and Applications for Assisted Living Challenges and Solutions",	Martina Ziefle and Carsten Rcker	IGI Global snippet	2010
3	"Assistive Technology for People with Disabilities"	Diane Pedrotty Bryant, Brian R Bryant	Pearson	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"KEEP IT SIMPLE A Guide to Assistive Technologies"	Ravonne A. Green and Vera Blair	Libraries Unlimited	2011
2	"Aural Rehabilitation for People with Disabilities",	John M. A. Oyiborhoro	Elsevier, Academic Press	2005
3	"Choosing Assistive Devices A guide for users and professionals"	Helen Pain Lindsay McLellan Sally Gore	Jessica Kingsley Publishers	2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/omjVM1lwkII
2	https://youtu.be/D9yY4XmRYBQ
3	https://youtu.be/3MjIkWxXigM
4	https://youtu.be/92bapdwvBEI
5	https://youtu.be/cmGbALGIS-A

SEMESTER S6

ROBOTICS AND MACHINE VISION LAB

Course Code	PCBRL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRL 507, PCBRT303	Course Type	Practical

Course Objectives:

1. Introduces the student to a Robot Operating System and familiarization of few ROS tools.

Details of Experiment

***Any 10 Experiments mandatory**

Expt.	Experiment
1	Writing a Simple Publisher and Subscriber, Simple Service and Client, Recording and playing back data, Reading messages from a bag file(Python/C++)
2	Getting Started with Turtlesim
3	Familiarisation with Rviz -- Markers: Sending Basic Shapes -- use visualization_msgs/Marker messages to send basic shapes, to send points and lines (C++), Interactive Markers: Writing a Simple Interactive Marker Server, Basic Controls
4	Introduction to tf -- broadcast the state of a robot to tf, get access to frame transformations, Adding a frame, waitFor Transform function, Setting up your robot using tf, publish the state of your robot to tf, using the robot state publisher.
5	Building a Visual Robot Model with URDF from Scratch, Building a Movable Robot Model with URDF, Adding Physical and Collision Properties to a URDF Model.
6	Familiarisation with Gazebo--How to get Gazebo up and running, Creating and Spawning Custom URDF Objects in Simulation, Gazebo ROS API for C-Turtle, Simulate a Spinning Top, Gazebo Plugin - how to create a gazebo plugin, Create a Gazebo Plugin that Talks to ROS
7	Create a Gazebo Custom World (Building Editor, Gazebo 3D Models), Add Sensor plugins like Laser, Kinect, etc. to URDF of mobile robot
8	Create a 3DOF robotic arm from scratch

9	Familiarisation with MoveIt through its RViz plugin, Motion Planning with the Panda or other robot models.
10	Create Moveit package for robotic arm simulation and add controllers, Plan a path for a 3DOF Robotic Arm and execute the same, Move the 3DOF arm to a desired goal point
11	Attach 2DOF gripper as the end effector of 3DOF arm and execute gripping operations, Execute Pick and Place Operation
12	Familiarisation with 2D navigation stack, Basic ROS Navigation, Start robots in simulation.
13	Execute SLAM Mapping (Lidar based) using a differentially driven mobile robot
14	Execute AMCL Navigation in a known environment using a differentially driven mobile robot.
15	Familiarise ROS Serial Arduino for hardware interface.
16	Obstacle avoidance using a differentially driven mobile robot

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

	Course Outcome	**Bloom's Knowledge Level (KL)**
CO1	Understand the applications of ROS in real world complex scenarios	**K2**
CO2	Work with turtlesim, Gazebo, MoveIt and Rviz	**K3**
CO3	Familiarise about the concepts behind navigation	**K1**
CO4	Interface with hardware and analyse the issues in hardware interfacing	**K4**

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	3	2	2	2	3	2			2	2		3
CO2	3	2	2	2	3	2			2	2		3
CO3	3	2	2	2	3	2			2	2		3
CO4	3	2	2	2	3	2			2	2		3

1: Slight (Low),2: Moderate (Medium),3: Substantial (High), : No Correlation

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robot Operating Systems (ROS) for Absolute Beginners	Lentin Joseph	Apress	2018
2	Learning ROS for Robotics Programming	Aaron Martinez, Enrique Fernández	Packt Publishing Ltd,	2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	A Gentle Introduction to ROS	Jason M O'Kane	Create Space	2013
2	Robot Operating System (ROS)	Anis Koubaa	The Complete Reference (Vol.3), Springer,	2018
3	Robot Operating System Cookbook	Kumar Bipin	Packt Publishing	2018
4	A Systematic Approach to learning Robot Programming with ROS	Wyatt Newman	CRC Press,	2017
5	ROS by Example: A do it yourself guide to Robot Operating System	Patrick Gabriel	Lulu	2012

Continuous Assessment (25 Marks)

1. **Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. **Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

BIOMEDICAL & ROBOTIC ENGINEERING

SEMESTER S7

THERAPEUTIC EQUIPMENTS

Course Code	PEBRT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	THEORY

Course Objectives:

1. To understand the working principle of therapeutic devices for cardiac assistance**.**
2. To analyze life-support equipments in clinical field.
3. To describe the current stimulated therapeutic equipments.
4. To Analyze the working of drug delivery systems and describe specialized instruments to view and operate on the internal organs and vessels

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Cardiac pacemakers:** different modes of operation, external and implantable pacemakers, pacemaker standard codes. Need for cardiac pacemaker. **Defibrillator:** AC and DC defibrillator – Need for defibrillator – basic principle. Implantable defibrillator and automated external defibrillator (AED) – functional block diagram. Pacer- cardioverter defibrillator – block diagram, defibrillator analysers. Catheterization - IABP, Stents.	9
2	**Ventilators:** Types of ventilators, Different ventilator operational and therapy modes. Flow sensors and FiO2 sensor. Heart lung machine (HLM) – principle of operation-functional block diagram - types of oxygenators. Extracorporeal membrane oxygenation (ECMO) machine. **Anaesthetic machines:** Block diagram and working. Need of anaesthesia, gas used and their sources, Gas blending and vaporizers, Anaesthesia delivery system, breathing circuit. Computer aided anaesthesia control.	9

3	**Surgical diathermy unit:** Basic principle and working, electrodes and safety aspects in electrosurgical units, Electro-surgical analysers, Neuro drills, Neuro navigation systems, Intra operative nerve monitors, Deep brain simulators. Laser surgery applications in ophthalmology, Short wave diathermy, Ultrasonic therapy unit, Interferential current therapy, Transcutaneous electrical nerve stimulation (TENS) - applications. Transcranial magnetic stimulation- applications. **Dialysis machine**: Block diagram and working.	**9**
4	**Drug delivery systems**: Different approaches for drug delivery, Drug infusion devices - Gravity Drip Systems, Infusion pump, Syringe pump, Implantable pumps, Patient Controlled Analgesia (PCA) Pumps, Elastomeric Infusers. Recent developments in drug infusion systems. **Endoscopy:** Principles, types &applications. Block diagram of a fibre optic endoscope with integral TV cameras. Laparoscopes, Gastro endoscope, Bronchoscope, Stroboscope, Capsule endoscopy, Cryo-surgery techniques and application. Operating microscope, arthroscopy. Modern lithotripter system, laser lithotripsy. Photo therapy unit	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe the working principle of therapeutic devices for cardiac assistance	**K1**
CO2	Analyze life-support equipments in clinical field	**K4**
CO3	Describe the current stimulated therapeutic equipment's	**K2**
CO4	Analyze the working of drug delivery systems and specialized instruments to view and operate on the internal organs and Vessels	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	2	2										
CO3	2	2										
CO4	2	2										
CO5	3	3	1									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of biomedical instrumentation.	Khandpur, Raghbir Singh	McGraw-Hill Education	2004
2	Biomedical engineering fundamentals	Bronzino, Joseph D., and Donald R. Peterson	CRC press	2014
3	Springer handbook of medical technology.	Kramme, Rüdiger, Klaus-Peter Hoffmann, and Robert Steven Pozos, eds.	Springer Science & Business Media,	2011
4	Medical Ventilator System Basics: A Clinical Guide	Lei, Yuan	Oxford University Press	2017
5	Encyclopaedia of medical devices and instrumentation. Vol. 4.	Webster, John G.	Wiley- Interscience	1988

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical instrumentation: application and design	Webster, John G., ed.	John Wiley & Sons,	2009
2	"Automatic ventilation of the lungs."	Mushin, William Woolf.		1980
3	Introduction to Biomedical Equipment Technology	Joseph J. Carr, John M. Brown	Pearson Education (Singapore) Pvt. Ltd.	2001
4	Principles of Applied Biomedical Instrumentation	Geddes & Baker	Wiley, 1989 Biomedical Engineering Handbook, CRC Press	1995

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=lDE7uwhseI4, https://www.youtube.com/watch?v=sNy3EWkBcDM
2	https://www.youtube.com/watch?v=ufaGUPs2fHo, https://www.youtube.com/watch?v=GKsQxt8hlPs
3	https://www.youtube.com/watch?v=-ic0xW-bQzE, https://www.youtube.com/watch?v=0hLgGhyWS3k
4	http://www.digimat.in/nptel/courses/video/102108077/L61.html, https://www.youtube.com/watch?v=sjaEgXmUtH4

SEMESTER S7

BIOMEDICAL INSTRUMENTATION

Course Code	PEBRT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To understand inverse kinematic transformation of position.
2. To understand the basics of industrial manipulator, manipulator kinematics and mechanics of robot motion, forward and inverse kinematic transformation of position.
3. To understand forward and inverse kinematic transformation of velocity, end effectors force transformations.
4. To understand about manipulator dynamics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to human physiological system:** Physiological systems of the body (brief discussion on Heart and cardio vascular system, Anatomy of nervous system, Physiology of respiratory systems) problems encountered in biomedical measurements. Sources of bioelectric potentials – resting and action potentials -propagation of action potentials – bio electric potentials example (ECG, EEG, EMG, ERG, EOG, EGG etc.) **Bio potential electrodes and ECG** Bio potential electrodes – theory – microelectrodes – skin surface electrodes – needle electrodes –biochemical transducers –transducers for biomedical applications. Electro conduction system of the heart. Electro cardiograph – electrodes and leads – Einthoven triangle, ECG read out devices, ECG machine – block diagram.	9

2	**Measurement of blood pressure, blood flow and heart sound** Measurement of blood pressure – direct and indirect measurement– oscillometric measurement – ultrasonic method, measurement of blood flow and cardiac output, plethysmography –photo electricand impedance plethysmographs. Measurement of heart sounds –phonocardiography.	**9**
3	**Measurement of EEG, EMG and Respiratory Parameters** Electro encephalogram –neuronal communication – EEG measurement, recording and analysis. Muscle response– Electromyogram (EMG) – Nerve Conduction velocity measurements- Electromyogram Measurements. Respiratory parameters – Spiro meter, pneumography **Therapeutic Aid** Cardiac pacemakers – internal and external pacemakers, defibrillators. Ventilators,	**9**
4	**Advances in Radiological Imaging** X-rays- principles of generation, uses of X-rays- diagnostic still picture, fluoroscopy, angiography, endoscopy, and diathermy. Basic principle of computed tomography, magnetic resonance imaging system and nuclear medicine system – radiation therapy. Ultrasonic imaging system - introduction and basic principle. **Electrical safety** Electrical safety– physiological effects of electric current –shock hazards from electrical equipment –method of accident prevention, introduction to tele-medicine	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the human anatomy and physiological signal Measurements. Illustrate various techniques used for measurement of Blood flow, blood pressure, and respiration rate and body temperature.	**K2**
CO2	Measurement of blood pressure, blood flow and heart sound	**K2**
CO3	Measurement of EEG, EMG and Respiratory Parameters and therapeutic aid	**K2**
CO4	Advances in Radiological Imaging and Electrical safety	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	3	3										
CO4	3	3										

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical Instrumentation, Application and Design,	J. G. Webster	John Wiley and Sons	
2	Biomedical Instrumentation Measurements	L. Cromwell, F. J. Weibell and L. A. Pfeiffer	Pearson education, Delhi	1990
3	Handbook of Biomedical Instrumentation	R. S. Khandpur	Tata Mc Graw Hill	
4	Introduction to Biomedical Equipment Technology	J. J. Carr and J. M. Brown	Pearson Education	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Joseph Bronzino, Introduction to Biomedical Engg.	John Enderele , Susan Blanchard	Academic Press	
2	Biomedical Instruments, Theory and Design	Welkovitz	Elselvier	
3	Medical Imaging Signals & Systems	Jerry L Prince, Jonathan M Links	Pearson Education	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.youtube.com/watch?v=DEsTkXTtOLA, https://www.youtube.com/watch?v=mK6sPBbChqc
2	https://www.youtube.com/watch?v=DQxiUYXi4UE, https://www.youtube.com/watch?v=3EFq5WbvJTQ
3	https://www.digimat.in/nptel/courses/video/108108125/L45.html, https://www.youtube.com/watch?v=q57ybN7m6V0
4	https://www.digimat.in/nptel/courses/video/102105090/L07.html, https://www.youtube.com/watch?v=TpPXxJ7fPDs

SEMESTER S7

INTRODUCTION TO BIONANOTECHNOLOGY

Course Code	**PEBRT743**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	THEORY

Course Objectives:

1. This course is designed as a broad survey of the field with a wide scope of Bionanoparticles and preparation methods, Analytical tools for biomedical applications in therapeutic, diagnostic, and preventive medicine, and Future goals and concerns of nanobiotechnology.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Introduction to nanotechnology and bionanotechnology, Nanoscale and history of nano, time frame of nanobiotechnology, Bionanotechnology and nanobiotechnology, Biocompatibility and its assessment, Cell-nanomaterial interactions	**9**
2	**Bionanoparticles and preparation methods:** DNA, amyloid fibre, actin filaments, aromatic peptides, enzymes, bacteriophages, minerals, viruses Dendrimers, ferritin etc. Use of microorganism in synthesis of nanoparticles, Green synthesis methods, Recombinant DNA technology, Monoclonal antibodies	**9**
3	**Analytical tools in nanobiotechnology** Tools in Nanobiotechnology: Microscopy and imaging techniques such as SEM, TEM, AFM and STM. Elemental composition by EDS, WDX, ICPMS, SIMS etc. Morphology and crystal structure by XRD and XAS. Fluorescent microscopy and confocal microscopy.	**9**

4	**Applications of nanobiotechnology:** Nano-biotechnology in cancer, nanotherapeutics: gene therapy and immunotherapy. **Future goals and concern of nanobiotechnology** New materials and devices in medicine, biomaterials, energy production etc. Issues with toxicity and environmental impact of nanomaterials.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	60

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Differentiate bionanotechnology and nanobiotechnology.	**K3**
CO2	Describe bionanoparticles and their preparation methods.	**K2**
CO3	Describe the working principle of analytical tools in nanobiotechnology.	**K2**
CO4	Evaluate future goals and concerns of nanobiotechnology	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2			1				2		2
CO2	2	3	2			1				2		2
CO3	2	3	2			1				2		2
CO4	2	3	2			1				2		2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nanobiotechnology: Concepts, Applications and Perspectives	Christof M.Niemeyer (Editor), Chad A. Mirkin (Editor),	Wiley	1st edition, 2004.
2	Nanobiotechnology-II more concepts and applications	Chad A Mirkin and Christof M. Niemeyer	Wiley	1st edition, VCH, 2007.
3	Nanotechnology in Biology and Medicine: Methods, Devices, and Applications	Tuan Vo-Dinh	CRC Press	2nd edition, 2019.
4	Bio-Nanotechnology: A Revolution in Food, Biomedical and Health Sciences	Debasis Bagchi (Editor), Manashi Bagchi (Author), Hiroyoshi Moriyama (Author), Fereidoon Shahidi (Author)	Wiley-Blackwell	1st edition, 2013

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Nanobiomaterials in Antimicrobial Therapy, Applications of Nano-biomaterials	Alexandru Grumezescu (Editor)	William Andrew	1st edition, 2016.
2	Prospects and applications of nanobiotechnology: A medical perspective	Md Fakruddin, Zakir Hossain & Hafsa Afroz	Journal of Nanobiotechnology volume 10, Article number: 31	2012
3	Perspective on Analytical Sciences and Nanotechnology, in Advanced Environmental Analysis: Applications of Nanomaterials	Deepali Sharma, Suvardhan Kanchi,Krishna Bisettya and Venkatasubba Naidu Nuthalapatib		Volume 1, 2016

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/0EWCqCIsFOA?si=3a7Jqm8NIc8WWi1N
2	https://youtu.be/qUEbxTkPIWI?si=m1gMEEMeiNImbbpy
3	https://youtu.be/HJb-6y1O8sI?si=abqvSIaOyOjsh2da

SEMESTER S7

COMPUTATIONAL METHODS TO BIOMEDICAL PROCESS

Course Code	**PEBRT744**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT 205	**Course Type**	Theory

Course Objectives:

1. This course gives an Introduction to practical computational methods for data analysis, modelling and simulation of biomedical systems and instrumentation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Scientific Computing in Biomedical engineering:** Numerical Algorithms and Errors Approximations - Accuracy and precision, definitions of round off and truncation errors-Taylor series, error propagation. Algebraic equations - Formulation and solution of linear algebraic equations, Gauss elimination method Gauss elimination, Gauss-Jordan reduction Iterative Approach for Solution of Linear Systems Jacobi and Gauss-Seidel methods.	9
2	**Non-Linear Models of Biological Systems**: General Form of Nonlinear Equations Examples of Nonlinear Equations in Biomedical Engineering The Method of Successive Substitution. The Method of False Position (Linear Interpolation) The Newton-Raphson Method Newton's Method for Simultaneous Nonlinear Equations. Applications: Friction factor in a catheter, Ventricular pressure measurements	9
3	**Finite Difference Methods:** Interpolation, Differentiation, and Integration Backward, Forward, and Central Differences- Interpolating of Equally Spaced Points Gregory-Newton, Stirling's interpolation Interpolating Polynomials-Interpolation of Equally Spaced Points- Gregory-Newton	9

	interpolation, Interpolation of Unequally Spaced Points Lagrange and spline, Integration Formulas-Newton-Cotes Formulas of Integration **Dynamic Systems**: Examples of Ordinary differential equations in Biomedical Engineering, Solution of Nonlinear Ordinary Differential Equations- Euler, Runge-Kutta methods, Solution of Linear Ordinary Differential Equations-Eigenvalue methods. Applications	
4	Glycolysis pathways of living cells, Dynamics of membrane and nerve cell potentials. Solution of Partial Differential Equations: elliptic, parabolic, hyperbolic equations. Applications: Dynamics of the basilar membrane, Drug diffusion **Computational Modelling of Biosystems**: Numerical Modelling of bioengineering Systems using numerical tools PhysioNet, PhysioBank, and Physio Toolkit, Python packages to simulate neuronal activity. Applications: ECG simulation; EEG simulation; Model of glucose regulation, Diabetes and insulin regulation; Renal clearance	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Analyze the principles related to physiological problems	**K4**
CO2	Solve linear, nonlinear differential equations and partial differential equations using numerical methods.	**K5**
CO3	Analyze Biomedical Engineering databases using numerical packages	**K4**
CO4	Apply numerical methods using modern scientific computing tools in Biomedical Engineering	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	1	3							2
CO2	3	2	1	1	3							2
CO3	3	2	2	1	3							2
CO4	3	2	2	1	3							2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical methods in biomedical engineering.	Dunn, Stanley, Alkis Constantinides, and Prabhas V. Moghe	Elsevier	2005
2	Numerical Methods for Engineers	Steven C. Chapra and Raymond P Canale	5 th edition, TataMcgraw Hill, New Delhi	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational methods in biomedical research.	Khattree, Ravindra, and Dayanand Naik	CRC Press	2007
2	Numerical and statistical methods for bioengineering: applications in MATLAB.	King, Michael R., and Nipa A. Mody	Cambridge University Press	2010
3	Computational Methods in Engineering	S. P. Venkatesan,	Ane Books India, New Delhi	2014

Video Links (NPTEL, SWAYAM…)	
Module No.	Link ID
2	https://youtu.be/vxOMd-1NZH8?si=hWMEiiNjFSlf0lYK
3	https://youtu.be/zSVo_klJ6HM?si=HyC92D0w5JgZWNbk
4	https://youtu.be/ABLwh3auDrk?si=ZMbmKZdDpPlm-6ki

SEMESTER S7

ASSISTIVE MEDICAL DEVICES

Course Code	**PEBRT745**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Artificial organs and Prosthetic engineering	**Course Type**	THEORY

Course Objectives:

1. Identify the elements of formal definitions of AT. Contrast these with information definitions of AT.
2. The major approaches for generating accessibility to tablet and smartphone technologies.
3. Functional manipulative tasks that can be aided by assistive technologies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Definitions of assistive technology, Principles of assistive technology, The Human Activity Assistive Technology Model, Foundational concepts, Application of HAAT, Universal Design **Functional Framework for Assistive Technologies**: Hard and Soft technologies, Allocation of functions, Hard assistive technologies, soft assistive technologies, technology options.	9
2	**Control Interfaces:** Anatomic Sites for Control of AT, Elements of the Human/Technology Interface, Speech, Contexts affecting HTI, Functional movement evaluation, Characteristics of Control Interfaces, CI for direct and indirect selection - Keyboards, Touch screens, Concept keyboards, Eye-Controlled systems, Tracking of body features, Enhancing keyboard control, Pointing interfaces, Switches, Switch arrays, Discrete joysticks, and Chord Keyboards, Selecting a CI, Multiple versus Integrated CI, Evaluation of CI.	9

3	**Accessing the Information Highway:** Cell phone accessibility, Internet accessibility, Raising the floor assistive technology, Cross-platform accessibility - Automatic speech recognition, Brain–Computer Interface, Computer access, Access to phones and tablets. **Assistive Technologies for Seating and Positioning:** Biomechanical principles, seating technologies, Seating assessment, Principles of intervention. **Wheeled Mobility:** Supporting structure, Frame types, Propelling structure, Specialized bases, Smart Wheelchairs	9
4	**Assistive Technologies for manipulation:** Low-technology Aids for manipulation, Special-purpose electromechanical aids for manipulation, Electronic aids to daily living, Robotic aids - Principles of Assistive Robots, Robots as personal assistants, Mobile assistive robots **Assistive Technologies for Vision:** Reading aids, Accessing ICT, Mobility and orientation Aids, Assessment - Visual function orientation and mobility **Assistive Technologies for hearing:** Approaches to auditory sensory aids, Aids for auditory impairments, Face- to-Face communication, Alerting devices, Assistive listening devices. Aids for persons with both visual and auditory impairments.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

<u>Criteria for evaluation:</u>

1. ***Problem Definition (K4 - 4 points)***
    a. *Clearly defines the real-world problem.*
    b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*

2. ***Problem Analysis (K4 - 4 points)***
    a. *Break-down and presents a well-reasoned solution approach.*
    b. *Compare and justify the proposed solutions with evidence and logical reasoning.*
3. ***Evaluate (K5 - 4 points)***
    a. *Thoroughly evaluate the proposed solutions.*
    b. *Compares trade-offs, advantages, and disadvantages.*
    c. *Considers feasibility, scalability, and practical implications.*
4. ***Implementation (K5 - 4 points)***
    a. *Select the most feasible solution by implementing the proposed solutions.*
    b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*
    c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*
5. ***Conclusion (K4- 2 points, K5 – 2 points)***
    a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***
    b. *Reflects critical thinking and informed decision-making.* ***(K5)***

Scoring:

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the framework for assessing the usability of technology and undertake product research and development	**K2**
CO2	Discuss the major approaches to the design of control interfaces that are used with ATs	**K3**
CO3	Apply principles of AT to guide recommendation of access to ICT enabled systems and aids for seating, positioning, and mobility	**K4**
CO4	Describe the mainstream technologies that support individuals with low vision, blindness, and cognitive impairment	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3		2		2		2		3
CO2	3	3	3	3		2		2		2		3
CO3	3	3	3	3		2		2		2		3
CO4	3	3	3	3		2		2		2		3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"Assistive Technologies: Principles and Practice"	Albert M. Cook and Janice M. Polgar	Elsevier	4^{th} Edition, 2015
2	"E-Health, Assistive Technologies and Applications for Assisted Living Challenges and Solutions",	Martina Ziefle and Carsten Rcker	IGI Global snippet	2010
3	"Assistive Technology for People with Disabilities"	Diane Pedrotty Bryant, Brian R Bryant	Pearson	2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	"KEEP IT SIMPLE A Guide to Assistive Technologies"	Ravonne A. Green and Vera Blair	Libraries Unlimited	2011
2	"Aural Rehabilitation for People with Disabilities",	John M. A. Oyiborhoro	Elsevier, Academic Press	2005
3	"Choosing Assistive Devices A guide for users and professionals"	Helen Pain Lindsay McLellan Sally Gore	Jessica Kingsley Publishers	2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/omjVM1lwkII
2	https://youtu.be/D9yY4XmRYBQ
3	https://youtu.be/3MjIkWxXigM
4	https://youtu.be/92bapdwvBEI
5	https://youtu.be/cmGbALGIS-A

SEMESTER S7

ROBOT NAVIGATION

Course Code	PEBRT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT 302	Course Type	THEORY

Course Objectives:

1. Odometry sensors and Colour tracking sensors.
2. Localization-Based Navigation and Programmed Solutions.
3. The navigation function in static deterministic environments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Building blocks of navigation:** Definitions - perception, localization, cognition, motion control **Perception:** Sensors for Perception - Schematic representation of the sensing unit, Obstacle Sensor (Bumper), the odometry sensor, Heading sensors - Gyroscopes, IMUs, Distance sensors - ToF, Phase shift, triangulation, ultrasonic rangefinders, Ground based beacons, GPS, Motion field and optical flow, Color tracking sensors, Feature Extraction - Feature Definition, Target Environment, Environment representation	9
2	**Localization**: General schematic for mobile robot localization, Challenge of Localization - Sensor noise and Sensor aliasing, Effector noise, Error model for odometric position estimation Localization-Based Navigation versus Programmed Solutions, Belief Representation, Map Representation - Continuous Representation, Decomposition Strategies, current challenges in map representation, Probabilistic Map-Based Localization - Markov localization, Kalman filter localization, Application to mobile robots: Kalman filter localization -	9

	schematic, Robot position prediction, Observation, Measurement prediction, Matching, Estimation, Landmark- based navigation.	
3	**Motion in Potential Field and Navigation Function-**Problem Statement, configuration space, free configuration space, Gradient Descent Method of Optimization, Gradient Descent without and with Constraints, Minkowski Sum, Potential Field, Navigation Function - in Static Deterministic Environment, in Static Uncertain Environment, Navigation Function and Potential Fields in Dynamic Environment - Estimation, Prediction, Optimization.	**9**
4	**Satellite Navigation** Introduction to Satellite Navigation, Trilateration, Position Calculation - Multipath Signals, GNSS Accuracy Analysis, Dilution of Precision, Coordinate Systems, UTM Projection, Local Cartesian Coordinates, Velocity Calculation, Urban Navigation - Urban Canyon Navigation, Map Matching, Dead Reckoning – Inertial Sensors, Incorporating GNSS Data with INS - Modified Particle Filter, Estimating Velocity by Combining GNSS and INS, A- GPS, DGPS Systems, RTK Navigation, GNSS Threats	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe sensors for perception in navigation	**K1**
CO2	Compare various methods for robot localization	**K2**
CO3	Explain the techniques of motion in Potential Field and Navigation Function	**K2**
CO4	Use information from satellites for navigation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	3									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh	MIT Press	
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	2020
3	Probabilistic Robots	Thrun, Burgard	MIT Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Planning Algorithms	Steven M. LaValle	Cambridge University Press	
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/fnkrocv4Alg
2	https://youtu.be/d4fC2RguJws
3	https://youtu.be/ii-KBKSODek
4	https://youtu.be/rIOdXt7bD3Q
5	https://youtu.be/ieearzWTCZw

SEMESTER S7

REHABILITATION ENGINEERING

Course Code	PEBRT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To understand the knowledge of engineering in rehabilitation
2. To apply in the design of devices and system

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Rehabilitation Engineering, Measurement and analysis of human movement, disability associated with aging in the work place and their solutions, clinical practice of rehabilitation engineering. Fundamentals of rehabilitation engineering design: design considerations, total quality management in rehabilitation engineering, steel as a structural material, Aluminum for assistive technology design, use of composites for assistive technology design.	9
2	Personal transportation: Introduction; selecting a vehicle; lift mechanisms, wheel chair securement systems, passenger restraint systems; automobile hand-controls; control of secondary functions; Wheel chair safety, standard sand testing: standard tests; normative values; static stability; geometric approach to static stability; stability with road crown and inclination; impact strength tests; fatigue strength tests; finite-element modelling applied to wheel chair design and testing. Augmentative and alternative communication technology, language representation methods and acceleration techniques, Computer applications in Rehabilitation Engineering, telecommunications, and Web Accessibility	9

3	Manual wheelchair design: Introduction; classes of manual wheelchairs; Frame design; materials; wheelchair and rider; Wheels and casters; Components; human factors design considerations. Power wheelchair design, classes of power wheelchairs; motor selection; servo amplifiers; Microprocessor control, shared control; fault-tolerant control; integrated controllers, electromagnetic compatibility; batteries; gear boxes; user interfaces; Postural support and seating. Power wheel chair range testing.	**9**
4	Rehabilitation robotics: Introduction; components and configurations of robots; robot kinematics; robot motion; robot control; robot sensors; human interfaces to robotic systems; recreational devices and vehicles, racing wheel chairs; arm-powered bicycles and tricycles; off-road vehicles; water sports; Adaptive ski equipment; recreational vehicles.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamental knowledge of engineering in rehabilitation	K3
CO2	Understand the parts of wheel chair and auxiliary devices	K2
CO3	Analyze wheel chair safety, standards and testing	K4
CO4	Develop self-learning initiatives and integrate learned knowledge to assess and evaluate the need of the end user	K3

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3			2		2		2		3
CO2	3	3	3			2		2		2		3
CO3	3	2	2			2		2		2		3
CO4	3	3	2			2		2		2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An introduction to rehabilitation engineering	Rory A Cooper, Hisaichi Ohnabe, Douglas A.Hobson	CRC Press	2007
2	The biomedical engineering handbook Biomedical Engineering Fundamentals,	Joseph D. Bronzino	CRC Press	2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Rehabilitation Engineering Applied to Mobility and Manipulation,	Rory A Cooper	CRC	1995
2	Rehabilitation Engineering	Robinson C.J	CRC Press,	1995
3	Rehabilitation Technology	Ballabio E	IOS Press	1993

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/105/105/105105213/

SEMESTER S7

HUMAN-MACHINE INTERFACE

Course Code	**PEBRT753**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. To get knowledge about the state-of-the-art human machine interactive systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to HMI -Natural Communication, types of human-machine interfaces, Human perception and recognition, Psychology of users, attention, thinking, perception of visual, sound and haptic incitements. Multimodality, concept of combined reality, virtual reality, technologies, existing scientific and commercial projects.	9
2	The interaction: Models of interaction, the terms of interaction, interaction framework, the ergonomics of interaction, interaction styles, WIMP interface, Experience, engagement and fun, paradigm of interaction	9
3	Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games-Mobile Information Architecture, Mobile2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies. Designing Web Interfaces–Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies	9
4	Technologies and concepts. Haptic interfaces, Haptic perception and recognition, sensors for sensing of fingers, hands and touching, interactive digital surfaces, manipulation of digital objects, displays with rear projection. Sound interaction. Basics of acoustics. Psychoacoustics. Analysis and synthesis of sound.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of human and computational abilities	**K2**
CO2	Design GUIs for human computer interactions	**K1**
CO3	Understand the fundamental aspects of designing mobile ecosystems	**K2**
CO4	Design the web based interaction systems	**K1**
CO5	Apply appropriate HMI modalities and sensors to design systems that are usable by people	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	2										
CO3	3	2										
CO4	3	3										
CO5	3	3										

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Human Computer Interaction	Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale	Prentice Hall	3rd Edition, 2004.
2	Mobile Design and Development	Brian Fling	O'Reilly Media Inc	First Edition, 2009
3	Designing Web Interfaces	Bill Scottand Theresa Neil	O'Reilly	First Edition, 2009.
4	Wearable Sensors Applications design and implementation	Subhas Chandra Mukhopadhyay and Tarikul Islam	IOP Science	2017
5	Haptics and Haptics interfaces	Katherine Kuchen becker		

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Jonathan Lazar Jinjuan Heidi Feng, Harry Hochheiser	Research Methods in Human Computer Interaction	Wiley	2010.

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.slideshare.net/slideshow/human-machine-interaction-252620230/252620230
2	https://www.youtube.com/watch?v=TALxVQc0W6w

SEMESTER S7

RELIABILITY ENGINEERING

Course Code	PEBRT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Able to design systems to meet the required demands in reliability

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of reliability– key elements; failure analysis – failure density – failure rate – probability of failure - bathtub curve. Basic reliability equations: Reliability in terms of – failure rate – failure density- relation between reliability, failure density and hazard rate - Mean time to failure (MTTF) – Integral equation of MTTF in terms of reliability	9
2	Hazard models – constant hazard model - linearly increasing hazard model - Weibull model. Expressions for reliability, failure density, and probability of failure of these models – problems	9
3	System reliability - reliability block diagrams (RBD), Components connected in series – components connected in parallel – mixed configuration – problems. Fault Tree Analysis (FTA) Symbols used in FTA, Fault Tree Construction -Estimation of system reliability from Fault tree problems.	9
4	Reliability improvement methods –Redundancy –unit redundancies –element redundancies -problems. Simplification of design –parts derating –operating environment; Cost of reliability –factors to be considered for optimizing the reliability cost.	9

	Reliability allocation -basic concepts. Reliability allocation for a series system - Proportional to uniform failure rate method of reliability allocation (Other methods need not be introduced)	

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the fundamental concepts of reliability	**K2**
CO2	To compute the reliability of products and systems	**K5**
CO3	To apply the techniques of reliability improvement and allocation in engineering design	**K3**
CO4	To analyze the failure data of engineering systems	**K4**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2									2
CO2	3	3	2									2
CO3	3	3	2									2
CO4	3	3	2									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Reliability Engineering	L. S. Srinath	Affiliated East-West Press Ltd	Fourth Edition, ,1985

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Reliability Engineering	E. Balaguruswamy	Tata McGraw Hill Publishing Co	1984
2	Reliability & Maintainability Engineering	Charles E. Ebling	Waveland Press, Inc.	Third Edition,2019
3	Reliability Engineering Theory and Practice	Alessandro Birolini	Springer	Third Edition, 2013
4	Introduction to Reliability Engineering	Lewis E	John Wiley & Sons	Second Edition,1995
5	Reliability Engineering	Singiresu S. Rao	Pearson India	First edition,2016

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/fvJg28Y4eDQ?si=ZDOI1c7M3rlH7skV
2	https://youtu.be/iVHXvZcfhrA?si=6ko7yV9kzVdffVoW
3	https://youtu.be/_c-iZ2BAXPw?si=EXQasd5Sr5FM-uQY
4	https://youtu.be/hH7I7qXbn1Q?si=cyF95vueAiz_SOct

SEMESTER S7

AI & MACHINE LEARNING FOR ROBOTS

Course Code	PEBRT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Robotic applications are able to mimic some of the critical operations that a human being is capable of. This was possible mainly due to the integration of Artificial Intelligence into robotic application
2. Artificial Intelligence can be applied to a wide range of engineering application and is a topic of study by itself
3. Provides an introduction to the areas of AI that can be used for robotic application which includes computer vision, path planning, object recognition etc.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview Application areas of AI -natural language processing, vision and speech processing, robotics, expert systems -basic overview Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Stochastic Gradient Descent, Challenges Motivating Deep Learning	9
2	Deep Feed forward Networks- Example: Learning XOR, Gradient-Based Learning, Hidden Units. Architecture Design, Back-Propagation and Other Differentiation Algorithms, Sequence Modelling: Recurrent and Recursive Nets -Need for sequence models, basic RNN architecture and types	9

3	Machine vision - Introduction, Computer vision - Introduction, Image formation, Basic image processing operations - edge detection, texture, optical flow, segmentation. challenges in image detection, Image features optimization. Case study- application of AI in ball Tracking in football game, crop monitoring using drones, traffic sign detection, pedestrian detection	**9**
4	Robotics - Robotic perception, Localization and mapping, Machine learning in robot perception, Application domains Case study- Use of AI in typical pick and place task, localization of a differential drive robot	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	**15**	**10**	**10**	**40**

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. ***Problem Definition (K4 - 4 points)***
    a. *Clearly defines the real-world problem.*
    b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*
2. ***Problem Analysis (K4 - 4 points)***
    a. *Break-down and presents a well-reasoned solution approach.*
    b. *Compare and justify the proposed solutions with evidence and logical reasoning.*
3. ***Evaluate (K5 - 4 points)***
    a. *Thoroughly evaluate the proposed solutions.*
    b. *Compares trade-offs, advantages, and disadvantages.*
    c. *Considers feasibility, scalability, and practical implications.*

4. ***Implementation (K5 - 4 points)***
    a. *Select the most feasible solution by implementing the proposed solutions.*
    b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*
    c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*
5. ***Conclusion (K4- 2 points, K5 – 2 points)***
    a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***
    b. *Reflects critical thinking and informed decision-making.* ***(K5)***

<u>Scoring:</u>

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the different learning techniques used in Machine learning	**K1**
CO2	Recognize the need for multilayer neural network for solving complex tasks	**K3**
CO3	Understand the fundamental concepts of Image processing and its application in computer vision	**K2**
CO4	Explain the different ways of perception of the environment by a robot and its use in path planning	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	2	2									2
CO3	3	2	2									2
CO4	3	2	2									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Deep Learning	Ian Goodfellow, YoshuaBengio, Aaron Courville	MIT Press	2016
2	Artificial Intelligence - A Modern Approach	Stuart J. Russell and Peter Norvig	Third Edition	2016
3	Pattern Recognition and Machine Learning	Bishop, C. ,M	Springer	2006
4	Robot vision	Berthold Klaus, Paul Horn	The MIT Press	1987
5	Computer Vision: Algorithms and Applications	Richard Szeliski		2010
6	A survey of deep learning techniques for autonomous driving."	Grigorescu, Sorin, et al.	Journal of Field Robotics	2020

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to AI Robotics	Robin R. Murphy	The MIT Press	
2	Artificial Intelligence and Machine Learning,	Chandra S.S.V, AnandHareendran S.	PHI	
3	Computer Vision – Models, Learning and Inference	Simon J. D. Prince	Cambridge University Press	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/2XXVjVqU2Nk?si=_6wUekUjQGCH0wwi
2	https://youtu.be/aPfkYu_qiF4?si=UayleJy59rOgDHn1
3	https://youtu.be/3LaVxEX3F0o?si=AhCKhPEqxGd0EcRM
4	https://youtu.be/_M-nDb0MIa4?si=lBS2QFxX-UJsexUr

SEMESTER S7

ADVANCED MEDICAL IMAGE PROCESSING TECHNIQUES

Course Code	**OEBRT721**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	**Course Type**	Theory

Course Objectives:

1. Summarize the types of CT used. Explain their need and principle
2. Mention the types of image reconstruction techniques used in MRI.
3. The applications of transforms in biomedical engineering. The different spatial domain smoothing techniques
4. an image observation model and explains the reasons for degradations in the observed image

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**X- Ray Imaging**: Basic principles of image formation, Block diagram of an X-Ray machine, Digital radiography - Basic principles **X-Ray Computed Tomography**: Basic principles, Contrast scale, Different generations of CT scanners, Basic principles of image reconstruction. **Ultrasonic Imaging**: Physical principles, Transducer parameters, Different modes - A-mode, M-mode (echocardiograph), B-mode, Principles of Doppler ultrasonic imaging. **Magnetic Resonance Imaging**: Principles of MRI, T1 weighted images, T2 weighted images, Proton density weighted images, Applications of MRI **Nuclear Medicine Imaging Modalities**: Emission Computed Tomography- SPECT & PET	9
2	**Image Perception:** MTF of the visual system, Monochrome vision model, Color vision model - HSI model, Image sampling and quantization. **Image Transforms:** Two dimensional orthogonal and unitary transforms, Properties of unitary transforms- DCT, KL Transform	9

3	**Image Enhancement in Spatial Domain**: Grey level transformations, Histogram processing - Histogram equalization and modification, Smoothing and sharpening spatial filters, Zooming and interpolation. **Image Enhancement in Frequency Domain:** Filtering- low pass, high pass, band pass and band stop filters, Homomorphic filter	9
4	**Image Restoration**: Inverse filtering, Wiener filtering. **Image Segmentation:** Detection of discontinuities- Point, line, edge, canny edge detector. Edge linking - Hough Transform. Segmentation based on thresholding - Basic global thresholding, variable thresholding. Region based segmentation- region growing, region splitting and merging	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify major processes involved in formation of medical images.	**K1**
CO2	Understand transforms used in image processing.	**K2**
CO3	Understand the different methods of image processing both in the time and transform domains.	**K2**
CO4	Describe fundamental methods for image enhancement and segmentation	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2							1		1
CO2	3	2	3							1		1
CO3	3	3	2							1		1
CO4	3	2	2							1		1

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Webb's The Physics of Medical Imaging	Flower M. A	CRC Press	2012
2	Digital Image Processing	Gonzalez Rafel C, Woods Richard E,	Taylor & Francis	2018
3	Fundamentals of Digital Image Processing	Jain Anil K	Prentice Hall of India	1989

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical Imaging Analysis	Atam P Dhawan	Wiley Inter science publication	2003
2	Ultrasound Physics & Instrumentation	D L Hykes, W R Hedrick & D E Starchman	Churchill Livingstone, Melbourne	1985
3	Ultrasonic Bioinstrumentation	Douglas A Christensen	John Wiley, New York	1988
4	Digital Image Processing	Pratt William K	John Wiley and Sons	2001
5	Biomedical Image Processing	Thomas M. Deserno	Springer-Verlag Berlin Heidelberg	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/117/105/117105135/
2	https://onlinecourses.nptel.ac.in/noc22_ee116/preview

SEMESTER S7

ROBOTIC REHABILITATION

Course Code	OEBRT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To understand the knowledge of engineering in rehabilitation
2. To apply in the design of devices and system

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Rehabilitation Engineering, Measurement and analysis of human movement, disability associated with aging in the work place and their solutions, clinical practice of rehabilitation engineering. Fundamentals of rehabilitation engineering design: design considerations, total quality management in rehabilitation engineering, steel as a structural material, Aluminum for assistive technology design, use of composites for assistive technology design.	9
2	Personal transportation: Introduction; selecting a vehicle; lift mechanisms, wheel chair securement systems, passenger restraint systems; automobile hand-controls; control of secondary functions; Wheel chair safety, standard sand testing: standard tests; normative values; static stability; geometric approach to static stability; stability with road crown and inclination; impact strength tests; fatigue strength tests; finite-element modeling applied to wheel chair design and testing. Augmentative and alternative communication technology, language representation methods and acceleration techniques, Computer applications in Rehabilitation Engineering, telecommunications, and Web Accessibility	9

3	Manual wheelchair design: Introduction; classes of manual wheelchairs; Frame design; materials; wheelchair and rider; Wheels and casters; Components; human factors design considerations. Power wheelchair design, classes of power wheelchairs; motor selection; servo amplifiers; Microprocessor control, shared control; fault-tolerant control; integrated controllers, electromagnetic compatibility; batteries; gear boxes; user interfaces; Postural support and seating. Power wheel chair range testing.	**9**
4	Rehabilitation robotics: Introduction; components and configurations of robots; robot kinematics; robot motion; robot control; robot sensors; human interfaces to robotic systems; recreational devices and vehicles, racing wheel chairs; arm-powered bicycles and tricycles; off-road vehicles; water sports; Adaptive ski equipment; recreational vehicles.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamental knowledge of engineering in rehabilitation	**K3**
CO2	Understand the parts of wheel chair and auxiliary devices	**K2**
CO3	Analyze wheel chair safety, standards and testing	**K4**
CO4	Develop self-learning initiatives and integrate learned knowledge to assess and evaluate the need of the end user	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3			2		2		2		3
CO2	3	3	3			2		2		2		3
CO3	3	2	2			2		2		2		3
CO4	3	3	2			2		2		2		3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	An introduction to rehabilitation engineering	Rory A Cooper, Hisaichi Ohnabe, Douglas A.Hobson	CRC Press	2007
2	The biomedical engineering handbook Biomedical Engineering Fundamentals,	Joseph D. Bronzino	CRC Press	2006

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Rehabilitation Engineering Applied to Mobility and Manipulation,	Rory A Cooper	CRC	1995
2	Rehabilitation Engineering	Robinson C.J	CRC Press,	1995
3	Rehabilitation Technology	Ballabio E	IOS Press	1993

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://www.my-mooc.com/en/categorie/robotics
2	https://onlinecourses.nptel.ac.in/noc20_me03/preview

SEMESTER S7

ROBOT NAVIGATION

Course Code	OEBRT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT 302	Course Type	THEORY

Course Objectives:

1. Odometry sensors and Colour tracking sensors.
2. Localization-Based Navigation and Programmed Solutions.
3. The navigation function in static deterministic environments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Building blocks of navigation:** Definitions - perception, localization, cognition, motion control **Perception:** Sensors for Perception - Schematic representation of the sensing unit, Obstacle Sensor (Bumper), the odometry sensor, Heading sensors - Gyroscopes, IMUs, Distance sensors - ToF, Phase shift, triangulation, ultrasonic rangefinders, Ground based beacons, GPS, Motion field and optical flow, Color tracking sensors, Feature Extraction - Feature Definition, Target Environment, Environment representation	9
2	**Localization**: General schematic for mobile robot localization, Challenge of Localization - Sensor noise and Sensor aliasing, Effector noise, Error model for odometric position estimation Localization-Based Navigation versus Programmed Solutions, Belief Representation, Map Representation - Continuous Representation, Decomposition Strategies, current challenges in map representation, Probabilistic Map-Based Localization - Markov localization, Kalman filter localization, Application to mobile robots: Kalman filter localization -	9

	schematic, Robot position prediction, Observation, Measurement prediction, Matching, Estimation, Landmark- based navigation.	
3	**Motion in Potential Field and Navigation Function-**Problem Statement, configuration space, free configuration space, Gradient Descent Method of Optimization, Gradient Descent without and with Constraints, Minkowski Sum, Potential Field, Navigation Function - in Static Deterministic Environment, in Static Uncertain Environment, Navigation Function and Potential Fields in Dynamic Environment - Estimation, Prediction, Optimization.	**9**
4	**Satellite Navigation** Introduction to Satellite Navigation, Trilateration, Position Calculation - Multipath Signals, GNSS Accuracy Analysis, Dilution of Precision, Coordinate Systems, UTM Projection, Local Cartesian Coordinates, Velocity Calculation, Urban Navigation - Urban Canyon Navigation, Map Matching, Dead Reckoning – Inertial Sensors, Incorporating GNSS Data with INS - Modified Particle Filter, Estimating Velocity by Combining GNSS and INS, A- GPS, DGPS Systems, RTK Navigation, GNSS Threats	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe sensors for perception in navigation	**K1**
CO2	Compare various methods for robot localization	**K2**
CO3	Explain the techniques of motion in Potential Field and Navigation Function	**K2**
CO4	Use information from satellites for navigation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	3									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh	MIT Press	
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	2020
3	Probabilistic Robots	Thrun, Burgard	MIT Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Planning Algorithms	Steven M. LaValle	Cambridge University Press	
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/fnkrocv4Alg
2	https://youtu.be/d4fC2RguJws
3	https://youtu.be/ii-KBKSODek
4	https://youtu.be/rIOdXt7bD3Q
5	https://youtu.be/ieearzWTCZw

SEMESTER S7

MEDICAL ROBOTICS

Course Code	**OEBRT724**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. With the increase in automation, robots are used in all walks of life.
2. It was observed by many researches that the ability of a robot can be increased several folds if it was capable of movement.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Components of robotic system**: History of medical robots, Classification of medical robots, Present status and future trends. Basic components of robotic system. Basic terminology – Accuracy, Repeatability, Resolution, Degree of freedom. Mechanisms and transmission, End effectors, Grippers different methods of gripping, Mechanical grippers-Slider crank mechanism, Rotary actuators, Cam type gripper, Magnetic grippers, Vacuum grippers, Air operated grippers; Specifications of robot.; **Drive systems-** Pneumatic Drives-Hydraulic Drives-Mechanical Drives-Electrical Drives, DC Servo Motors, Stepper Motors, AC Servo Motors-Salient Features and Applications	**9**
2	**Sensors in robots-** Touch sensors, Tactile sensor, Proximity and range sensors, Robotic vision sensor, Force sensor, Light sensors, Pressure sensors. **Robot kinematics:** Forward Kinematics, Inverse Kinematics and Difference; Forward Kinematics and Reverse Kinematics of manipulators with Two, Three Degrees of Freedom (in 2 Dimension), Four Degrees of freedom (in 3 Dimension) Jacobians, Velocity and Forces-Manipulator Dynamics, Trajectory Generator.	**9**

3	**Robot programming Languages**-VAL Programming-Motion Commands, Sensor Commands , End Effect or commands and simple Programs **Robots in health care**: Tele operated Robot-Assisted Minimally Invasive Surgery, Robot design concepts Robot assisted Image-Guided Interventions: CT, MR, Ultrasound, Other applications: Robotic catheters for heart electrophysiology, Humanoids, Micro/nano robots for biomedicine-Delivery, surgery, sensing, and detoxification	9
4	**Case Studies**: Cardiac, abdominal, and urologic procedures with tele-operated robots (3 procedures each), Neurosurgery and Orthopedic surgery with robots. Other types of healthcare robots: Physically assistive robots, socially assistive robots, Rehabilitation robotics, Design methodologies-Technological choices - Security. Professional and medical ethics; Robo ethics- social and ethical implications	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental computational issues involved in mobile robotics and issues related to locomotion	**K2**
CO2	Translate the working principle of different visual and non-visual sensors to select the appropriate ones for a particular application	**K3**
CO3	Explain the techniques used for representing and reasoning about space	**K2**
CO4	Classify the different software architecture in the development of robotic applications	**K1**
CO5	State the techniques used for pose maintenance and localization techniques used in robotics	**K1**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2		2									2
CO3	3		2		2							2
CO4	3	2	3	2	2							2
CO5	3	2	3		2							2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computational Principles of Mobile Robotics	Gregory Dudek and Michael Jenkin	Cambridge University Press	
2	Introduction to Autonomous Mobile Robots	Scaramuzza	MITPress,USA	2011.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier, USA	2014
2	Sensors for mobile robot	H R Everett	CRCPress	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc21_me44/preview
2	https://www.fun-mooc.fr/en/cours/mobile-robots-and-autonomous-vehicles/

SEMESTER 8

BIOMEDICAL & ROBOTIC ENGINEERING

SEMESTER S8

BIOSTATISTICS

Course Code	PEBRT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To collect, analyze, and form statistical conclusions from case studies using any of the modern statistical tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Descriptive statistics:** Measures of location, Arithmetic Mean, Median, Mode, Geometric mean and harmonic mean, measures of spread, Variance and standard deviation, coefficient of variation, properties of Arithmetic Mean and Variance, Grouped data, Graphic Methods. **Probability:** Notations, Laws of probability, Bayes' rule and screening tests, Bayesian Inference, ROC Curves. Random variables. **Discrete Probability distributions:** Probability mass function, expected value, variance and cumulative distribution. Binomial distribution, Poisson distribution.	9
2	**Continuous Probability distribution:** Normal distribution, properties of standard normal distribution. **Estimation:** Relationship between Population and Sample, Random number trials, clinical trials, Estimation of Mean, Variance, Binomial distribution and Poisson distribution. One sided confidence Intervals	9
3	**Hypothesis testing:** One sample inference: One sample test for the mean of a normal distribution, one- and two-sided alternatives. Categorical data: Fisher's Exact Test, Chi- Square Goodness-of-Fit Test.	9

	Nonparametric methods: The Sign Test, The Wilcoxon Signed-Rank Test.	
4	Regression and Correlation Methods: Fitting regression lines- method of least squares, Inference about parameters from regression lines, interval estimation for linear regression, correlation coefficient, multiple regression. Multi-sample inference: Introduction to ANOVA	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of statistical analysis. Analyse the concepts of probability and distributions.	**K2**
CO2	Examine the data and to identify distribution forms relating to the variables.	**K1**
CO3	Analyse applications of correlation and regression in biostatistics.	**K1**
CO4	Apply hypothesis testing via some statistical distributions in biological problems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3	3	2	2					1		2
CO4	3	3	3	2	2					1		2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of Biostatistics	Bernard Rosner	Duxbury Press	5th edition, 2010
2	Introduction to Biostatistics and Research Methods	PSS Sunder Rao and J Richard	PHI Learning	2012, 5th Ed.

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Fundamentals of biostatistics	Khan, Irfan A., and Atiya Khanum	Ukaaz	2004
2	Introduction to the Practice of Statistics	Moore and McCabe	W.H. Freeman and Company	2009
3	Probability and Statistics	N.G. Das	N.G. Das	2016
4	Principles of Biostatistics	Marcello Pagano	Chapman and Hall/CRC	2018, 2^{nd} Ed

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/nou23_biot05/preview

SEMESTER S8

AI FOR MEDICAL IMAGE ANALYSIS

Course Code	PEBRT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To study and implement AI for medical Image analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**AI in the field of medicine**: Programs for screening, Businesses in the field of Medical Science, Application of AI in present day, Myths and realities concerning AI, Availablity of the information for review, generation of models, prospects of AI in the field of Medicine.	9
2	**AI in Health Diagnostics:** Advances in AI applications for Medical Imaging- Detection of Diabetic Retinopathy in retinal Fundus Images, Dermatological Classifications of skin cancer, Data issues, Moving Computational Advances into Clinical Practice, Coronary Artery Diseases, Non-invasive Diagnostics, AI in medical applications, Evolution of standards for AI in Medical Applications, Devices and Appps for Data Collection and Analysis, Improving Information Management, User Experience, and Cognitive Support	9
3	**Large Scale Health Data:** Environmental Data – the missing data stream, Capturing data on toxin Exposure. **Disease detection using AI:** Feature Extraction, Transfer Learning, Prediction and classification, Experimental data and setup, performance Evaluation metrics, Experimental Results.	9

	Breast Cancer detection from mammograms using AI. Breast Tumor detection in ultrasound images using AI	
4	AI based skin cancer detection – Machine learning techniques. Brain stroke detection from computed tomography images using AI, AI for COVID -19 detection from computed tomography scans, AI based retinal disease classification using optical disease classification using optical coherence tomography images, Ai based Alzheimer's disease detection.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	How AI is implemented in the field of medicine	**K2**
CO2	AI in Health Diagnostics.	**K1**
CO3	Data collection using AI	**K1**
CO4	Disease detection using AI	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3	3	2	2					1		2
CO4	3	3	3	2	2					1		2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The role of Artificial Intelligence Revolutionising the Medical Field	Prf. Ganesh Vasudeo Manerkar, Dr. Anshad A S, Dr. Nishi S Das, Dr. Atul P Kulkarni, Mr. Kaladi Govindaraju	Cosmas	
2	Applications of Artificial Intelligence in Medical Imaging	Abdilhamit Subasi	Academic Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Advanced Machine Vision Paradigms for Medical Image Analysis	Tapan K. Gandhi, Siddhartha Bhattacharyya, Sourav	Academy Press	2020
2	Artificial Intelligence in Medical Imaging: Opportunities	Erik R. Ranschaert, Sergey Morozov, Paul R. Algra	Springer	2019

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc22_bt34/preview
2	https://www.classcentral.com/course/youtube-computer-science-artificial-intelligence-47565

SEMESTER S8

MEDICAL DEVICE REGULATIONS & STANDARDS

Course Code	**PEBRT863**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	-	**Course Type**	Theory

Course Objectives:

1. To understand the need for regulatory control in medical device development
2. To describe the global medical device regulatory landscape
3. To describe the structure of country specific medical device regulatory organizations and approval processes
4. To understand the requirements for quality systems and risk management of medical devices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Overview of Medical Devices** Medical device definitions, need for medical devices, Need for regulatory control, Classification of medical devices (Indian, American and European), Overview of medical device design - Medical product life cycle, Medical device development process, Medical device design process; Addressing regulation and compliance need; Medical device design control process; Testing – Verification and Validation; Risk management procedures for medical devices. Stake holders in regulatory framework development.	**9**
2	**Overview of Regulatory Landscape** Evolution of medical device regulation, General structure – benefits of regulation, Medical Device Nomenclature (GMDN / EMDN & Alternate Options), Global Harmonization – GHTF & IMDRF, RAPS. **HTA overview**– Clinical investigation of Medical Devices - Clinical investigation plan for Medical Devices; Good clinical practice for clinical	**9**

	investigation of medical devices (ISO 14155:2011); DHF Overview, GMP, Adverse event reporting of medical device. Medical Device Approval: Regulatory Bodies & Notified bodies, leading organizations. **HO Medical Devices:** Global efforts, Ensuring availability, affordability and accessibility (Technical specifications and regulatory efforts). Overview: List of basic documents of National or State level importance for regulatory efforts	
3	**Basic Regulatory Frame work** Patient safety and risk management, Sterile / Non-Sterile devices & packaging, Basics of classification of In vitro diagnostics & Medical Devices (active / non active). Regulatory approval process for Medical Devices: Comparison of US FDA, CE and Indian Medical device Rule; Device risk classification, Registration, Clinical trials, Premarket Notification, Pre-Market Approval (PMA), Investigational Device Exemption (IDE) and In-vitro Diagnostics.	**9**
4	**Advanced Regulatory Framework** Quality System Requirements: Quality management of Medical Devices - International Organization for Standardization, ISO 9001 - QMP Overview & Process Approach, ISO 13485 (deeper understanding), ICMED 13485, ISO 14971 - Risk Management of Medical Devices, Unique Device Identification (UDI), Labelling requirements and Post market surveillance**. Quality: Reliability, Product testing and Technical Standards Reliability**: HFE, MTBF, FMEA, FTA, Black box testing and white box testing, Environmental Stress Screening / Testing, Accelerated Life Testing. **Product/Materials testing**: Biocompatibility, overview of ASTM, ISO Technical standards, Indian National standards other international standards.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Understand the need for regulatory control in medical device development	**K2**
CO2	Study the global medical device regulatory landscape	**K4**
CO3	Explain the structure of country specific medical device regulatory organizations and approval processes	**K1**
CO4	Understand the requirements for quality systems and risk management of medical devices	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3				3		3		2		2
CO2	2	3				3		3		2		2
CO3	2	3				3		3		2		2
CO4	2	3				3		3		2		2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Medical Device Quality Assurance and Regulatory Compliance	Richard C. Fries	CRC Press LLC	2019
2	Reliable design of medical devices	Richard C. Fries	CRC Press	2012
3	Medical Device Design Innovation from Concept to Market	Peter Ogrodnik	Elsevier	2013
4	Regulation of medical devices A step-by-step guide	WHO Regional Publications	, Eastern Mediterranean Series, World Health Organization	2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Handbook of medical device design	Richard C. Fries	CRC Press	2019
2	Handbook of Medical Device Regulatory Affairs in Asia”	Jack Wong, Raymond Tong,	Jenny Stanford Publishing	Second Edition, 201
3	Medical device software verification, validation and compliance	Vogel, David A	Artech House	2011

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://archive.nptel.ac.in/courses/127/106/127106136/
2	https://archive.nptel.ac.in/courses/127/106/127106136/
3	https://archive.nptel.ac.in/courses/127/106/127106136/
4	https://archive.nptel.ac.in/courses/127/106/127106136/

SEMESTER S8

BIOMECHATRONICS

Course Code	PEBRT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. Understand types of sensors used in biomedical applications.
2. To be familiar with various equipment in bio-medical applications and the techniques of diagnosis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Bio potential and Bio electrodes Cell structure – electrode – electrolyte interface- electrode potential- resting and action potential – electrodes for their measurement- Propagation of nerve impulses, Refractory period- Electrode electrolyte interface, Half-cell potential, Polarisable and Non-polarisable electrodes - Skin electrode interface – Bio- electrodes: Surface- Micro-. Needle electrodes - Equivalent circuits of electrodes	9
2	Electrical Activity of the heart Cardiac muscle, Action potentials in cardiac muscle, SA node, Origin and propagation of rhythmical excitation & contraction, refractoriness, regular and ectopic pace makers, Electrocardiogram - Einthoven triangle, Standard 12-lead configurations - ECG Machine, Arrhythmias.	9
3	Electrical Activity of the brain Electrical activity of brain – Sleep stages, Brain waves, waveforms & measurements, Evoked potentials, 10-20 electrodes placement system for EEG - EEG machine.	9

4	Measurement of Physiological Parameters Electrocardiograph measurements – blood pressure measurement: by ultrasonic method – plethysmography – blood flow measurement by electromagnetic flow meter - cardiac output measurement by dilution method Biomedical Equipment's Heart lung machine – artificial ventilator –– Basic ideas of CT scanner – MRI and ultrasonic scanner – cardiac pacemaker – DC – defibrillator patient safety - electrical shock hazards- Centralized patient monitoring system	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand types of sensors	**K2**
CO2	Study the electrical activity of heart	**K1**
CO3	Study the electrical activity of brain	**K1**
CO4	Explore the various medical equipment and measurement of Physiological Parameters	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	3	2									
CO3	3	3	2									
CO4	3	3	2									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Textbook of Medical Physiology	Arthur C. Guyton	Prism Books (Pvt) Ltd &W.B.Saunders Company	12th edition, 2012
2	Handbook of Medical Instrumentation	Khandpur R S	Tata McGraw Hill	2004
3	Bio Medical Instrumentation	Arumugam M	Anuradha agencies	2002

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Electronics in Medicine and Biomedical Instrumentation	Nandini K. Jog	PHI	Second Edition 2013
2	Instrument Transducer – An Introduction to their performance and design	Hermann K P. Neubert	Clarendon Press	1975
3	Principles of applied Biomedical Instrumentation	L.A. Geddes, L.E. Baker	Pearson	1989

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.swayam2.ac.in/nou23_bt05/preview
2	https://onlinecourses.swayam2.ac.in/nou23_bt05/preview
3	https://onlinecourses.swayam2.ac.in/nou23_bt05/preview
4	https://onlinecourses.swayam2.ac.in/nou23_bt05/preview

SEMESTER S8

ADVANCED BIOMEDICAL SIGNAL PROCESSING & APPLICATIONS

Course Code	**PEBRT865**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	5/3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PBBRT504 Biomedical Signal Processing	**Course Type**	Theory

Course Objectives:

1. To understand interpretative models in biological signal processing.
2. To apply Wavelet Analysis-The Short-Time Fourier Transform, Time-Frequency Resolution, Multiresolution Analysis, Wavelet Transform
3. Understand Geometrical Scaling and Self-Similarity
4. To Understand Nonlinear Models of Signals-Nonlinear Signals and Systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Interpretative Models in Biological Signal Processing-Mathematical Instruments for Signal Processing- Descriptive Methods, The Black-Box Models, Interpretative Models, Examples-Mathematical Models and Signals in Intensive Care Units, Mathematical Models and Cardiovascular Variability Signals, Mathematical Models and EEG Signals during Epilepsy, Mathematical Models, Electrophysiology, andFunctional Neuroimaging.	9
2	Wavelet Analysis-The Short-Time Fourier Transform, Time-Frequency Resolution, Multiresolution Analysis, Wavelet Transform, Generalization of the Short-Time Fourier Transform, Wavelet Transform and Discrete Filter Banks, Matching Pursuit, Applications to Biomedical Signals- Analysis of ECG, Analysis of Spectral Variability of Heart Rate, Analysis of a Signal from a Laser Doppler	9

3	Geometrical Scaling and Self-Similarity, Measures of Dimension, Self-Similarity and Functions of Time, Theoretical Signals Having Statistical Similarity, Measures of Statistical Similarity for Real Signals, Generation of Synthetic Fractal Signals, Fractional Differencing Models, Example-measurements of local blood flow in the tissues of the heart using a fractal model.	**9**
4	Nonlinear Models of Signals-Nonlinear Signals and Systems: Basic Concepts, Poincare Sections and Return Maps, Chaos, Measures of Nonlinear Signals and Systems, Characteristic Multipliers and Lyapunov Exponents, Estimating the Dimension of Real Data, Tests of Null Hypotheses Based on Surrogate Data, Applications-Bifurcations in a Model of a Cardiac pacemaker Cell.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Assignment: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. ***Problem Definition (K4 - 4 points)***
   a. *Clearly defines the real-world problem.*
   b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*
2. ***Problem Analysis (K4 - 4 points)***
   a. *Break-down and presents a well-reasoned solution approach.*
   b. *Compare and justify the proposed solutions with evidence and logical reasoning.*
3. ***Evaluate (K5 - 4 points)***
   a. *Thoroughly evaluate the proposed solutions.*

b. *Compares trade-offs, advantages, and disadvantages.*
c. *Considers feasibility, scalability, and practical implications.*

4. ***Implementation (K5 - 4 points)***
    a. *Select the most feasible solution by implementing the proposed solutions.*
    b. *Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.*
    c. *Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).*
5. ***Conclusion (K4- 2 points, K5 – 2 points)***
    a. *Summarizes findings and insights. State which solution is most appropriate for the problem.* ***(K4)***
    b. *Reflects critical thinking and informed decision-making.* ***(K5)***

Scoring:

1. ***Accomplished (4 points)****: Exceptional analysis, clear implementation, and depth of understanding.*
2. ***Competent (3 points)****: Solid performance with minor areas for improvement.*
3. ***Developing (2 points)****: Adequate effort but lacks depth or clarity.*
4. ***Minimal (1 point)****: Incomplete or significantly flawed.*

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	**Part B**	**Total**
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	What are the steps involved in the identification of a NARMA model?	**K2**
CO2	Illustrate band-pass interpretation of the STFT	**K2**
CO3	Compare the Boxing count dimension and Similarity dimension	**K2**
CO4	Define Poincare section, Classify different ICA methods	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		2					2		3
CO2	3	3	3		2					2		3
CO3	3	3	3		2					2		3
CO4	3	3	3		2					2		3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Advanced Methods of Biomedical Signal Processing	Sergio Cerutti, Carlo Marchesi	IEEE	2011
2	Biomedical Signal Processing and Signal Modeling	Eugene N. Bruce	Wiley	2000

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Biomedical Signal Processing Advances in Theory	Ganesh Naik	Springer	2020 Edition
2	Digital Signal Processing for Medical Imaging Using Matlab	Gopi, E. S.	Springer	2012
3	Signals and systems in biomedical engineering: signal processing and physiological systems modelling	Devasahayam, S. R	Springer	2012

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	http://tjsec.digimat.in/nptel/courses/video/108105101/L49.html, https://www.digimat.in/nptel/courses/video/108106151/L01.html
2	https://terna.digimat.in/nptel/courses/video/103106114/L23.html, https://www.youtube.com/watch?v=dTU9ribI9k4
3	Geometrical Scaling and Self-Similarity, Geometrical Scaling and Self-Similarity
4	http://www.digimat.in/nptel/courses/video/108102113/L04.html

SEMESTER S8

HUMAN MACHINE INTERFACE

Course Code	OEBRT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	THEORY

Course Objectives:

1. Understand the fundamental aspects of designing mobile ecosystems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to HMI -Natural Communication, types of human-machine interfaces, Human perception and recognition, Psychology of users, attention, thinking, perception of visual, sound and haptic incitements. Multimodality, concept of combined reality, virtual reality, technologies, existing scientific and commercial projects.	9
2	The interaction: Models of interaction, the terms of interaction, interaction framework, the ergonomics of interaction, interaction styles, WIMP interface, Experience, engagement and fun, paradigm of interaction	9
3	Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies. Designing Web Interfaces – Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies	9

4	Technologies and concepts. Haptic interfaces, Haptic perception and recognition, sensors for sensing of fingers, hands and touching, interactive digital surfaces, manipulation of digital objects, displays with rear projection. Sound interaction. Basics of acoustics. Psychoacoustics. Analysis and synthesis of sound.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of human and computational abilities and limitations.	**K1**
CO2	Design GUIs for human computer interactions.	**K2**
CO3	Understand the fundamental aspects of designing mobile ecosystems.	**K2**
CO4	Apply appropriate HMI modalities and sensors to design systems that are usable by people.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2									
CO2	2	3	2									
CO3	2	3	2									
CO4	2	3	2									

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Human Computer Interaction	Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale	Pearson Prentice Hall, 2004.	3RD
2	Mobile Design and Development	Brian Fling	O'Reilly Media Inc	2009
3	Designing Web Interfaces	Bill Scott and Theresa Neil	O'Reilly	2009
4	Wearable Sensors Applications design and implementation	Subhas Chandra Mukhopadhyay and Tarikul Islam	IOP Science	2017
5	Haptics and Haptic Interfaces	Katherine Kuchenbecker		

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Research Methods in Human Computer Interaction	Jonathan Lazar Jinjuan Heidi Feng, Harry Hochheiser	John Wiley & Sons	2010

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://doi.org/10.1007/978- 3-642-41610-1_19-1

SEMESTER 8

BIOMEDICAL TRANSPORT PHENOMENA

Course Code	OEBRT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	THEORY

Course Objectives:

1. Solve transport equations using methods of advanced mathematics.
2. Localization-Based Navigation and Programmed Solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to biomedical transport phenomena- role of diffusion, convection in physiological systems- Fluid kinematics: control volumes, velocity field, flow rate, acceleration, stream lines, stream tubes and streak lines- conservation relations and boundary conditions. Fluid statics: surface tension, membrane and cortical tension -Newtonian and non-Newtonian fluids - Laminar and turbulent flow	9
2	Cardiovascular transport system: review of physiological process- Rheology of flow of blood: large vessels, small tubes, capillaries- regulation of blood flow- Flow in branching vessels-flow in specific arteries- Trans vascular transport: Introduction, pathways for trans endothelial transport- Rates of trans vascular transport: osmotic pressure- starling's law of filtration- Kedem-Katchalsky equation	9
3	Respiratory transport system: review of physiological process - Oxygen-haemoglobin dissociation curve – Oxygen levels in blood- The Hill equation - Dynamics of oxygenation of blood in lung capillaries - Oxygen and carbon dioxide transport in blood oxygenators, bio artificial organs, and tissue engineered constructs - Oxygen transport in perfusion bioreactors - Oxygen	9

	transport in the Krogh tissue cylinder. Transport in kidneys: review of physiological process- Mechanism of transmembrane transport: Direct diffusion, facilitated transport, Active transport - Quantitative analysis of Glomerular filtration: Hydraulic conductivity of Glomerular Capillaries, Solute transport across Glomerular Capillaries - Quantitative analysis of Tubular Reabsorption: Mass balance equations, Fluxes of passive diffusion & Convection, Mathematical modeling of carrier mediated transport.	
4	Drug transport in solid tumors: introduction - Drug delivery in cancer treatment - Quantitative analysis of transvascular transport- Factors that affect drug administration: The Oie-Tozer equation - A model for intravenous injection of drug – Application to controlled release of drugs by osmotic pumps – Controlled release of drugs from transdermal patches – Controlled release of drug from implantable devices.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic engineering principles of momentum, mass, and heat transfer in integrated form through an array of examples and analysis from biological systems.	**K1**
CO2	Formulate differential equations of change for momentum, heat and mass transfer problems	**K2**
CO3	Solve transport equations using methods of advanced mathematics.	**K2**
CO4	Apply biotransport fundamentals to the design and interpretation of biomedical systems and experiments.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2							2		3
CO2	2	3	2							2		3
CO3	2	3	2							2		3
CO4	2	3	2							2		3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Transport Phenomena in Biological Systems	G.A. Truskey, F. Yuan, D.F. Katz	Pearson Prentice Hall, 2009.	2
2	Basic Transport Phenomena in Biomedical Engineering	Ronald L Fournier, Ph.D., PE	CRC Press	4TH
3	Fields, Forces, and Flows in Biological Systems	A.J. Grodzinsky, Fields	Garland Science	2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Transport Phenomena	R.B. Bird, W.E. Stewart, E.N. Lightfoot	John Wiley & Sons	2002
2	Physicochemical Hydrodynamics:An Introduction	R.F. Probstein	Wiley Interscience	2003

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://online-learning.tudelft.nl/courses/the-basics-of-transport-phenomena/
2	https://www.mooc-list.com/tags/transport-phenomena

SEMESTER S8

ROBOT MOTION PLANNING

Course Code	OEBRT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCBRT 302	Course Type	THEORY

Course Objectives:

1. Odometry sensors and Colour tracking sensors.
2. Localization-Based Navigation and Programmed Solutions.
3. The navigation function in static deterministic environments

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Building blocks of navigation:** Definitions - perception, localization, cognition, motion control **Perception:** Sensors for Perception - Schematic representation of the sensing unit, Obstacle Sensor (Bumper), the odometry sensor, Heading sensors - Gyroscopes, IMUs, Distance sensors - ToF, Phase shift, triangulation, ultrasonic rangefinders, Ground based beacons, GPS, Motion field and optical flow, Color tracking sensors, Feature Extraction - Feature Definition, Target Environment, Environment representation	9
2	**Localization**: General schematic for mobile robot localization, Challenge of Localization - Sensor noise and Sensor aliasing, Effector noise, Error model for odometric position estimation Localization-Based Navigation versus Programmed Solutions, Belief Representation, Map Representation - Continuous Representation, Decomposition Strategies, current challenges in map representation, Probabilistic Map-Based Localization - Markov localization, Kalman filter localization, Application to mobile robots: Kalman filter localization -	9

	schematic, Robot position prediction, Observation, Measurement prediction, Matching, Estimation, Landmark- based navigation.	
3	**Motion in Potential Field and Navigation Function-**Problem Statement, configuration space, free configuration space, Gradient Descent Method of Optimization, Gradient Descent without and with Constraints, Minkowski Sum, Potential Field, Navigation Function - in Static Deterministic Environment, in Static Uncertain Environment, Navigation Function and Potential Fields in Dynamic Environment - Estimation, Prediction, Optimization.	**9**
4	**Satellite Navigation** Introduction to Satellite Navigation, Trilateration, Position Calculation - Multipath Signals, GNSS Accuracy Analysis, Dilution of Precision, Coordinate Systems, UTM Projection, Local Cartesian Coordinates, Velocity Calculation, Urban Navigation - Urban Canyon Navigation, Map Matching, Dead Reckoning – Inertial Sensors, Incorporating GNSS Data with INS - Modified Particle Filter, Estimating Velocity by Combining GNSS and INS, A- GPS, DGPS Systems, RTK Navigation, GNSS Threats	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Describe sensors for perception in navigation	**K1**
CO2	Compare various methods for robot localization	**K2**
CO3	Explain the techniques of motion in Potential Field and Navigation Function	**K2**
CO4	Use information from satellites for navigation	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	3									2

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh	MIT Press	
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	2020
3	Probabilistic Robots	Thrun, Burgard	MIT Press	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Planning Algorithms	Steven M. LaValle	Cambridge University Press	
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://youtu.be/fnkrocv4Alg
2	https://youtu.be/d4fC2RguJws
3	https://youtu.be/ii-KBKSODek
4	https://youtu.be/rIOdXt7bD3Q
5	https://youtu.be/ieearzWTCZw

SEMESTER S8

BIOSENSORS AND TRANSDUCERS

Course Code	**OEBRT834**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	**Course Type**	Theory

Course Objectives:

1. To provide an exposure for the student to learn about various sensors and transducers, enabling them to understand their principles and applications.
2. To facilitate the selection process by explaining the underlying theory followed by appropriate case studies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Requirement of sensors in robots used in industry, agriculture, medical field, transportation, military, space and undersea exploration, human robot interactions, robot control, robot navigation, tele-operational robot etc. **Proprioceptive or Internal sensors** Position sensors- encoders- linear, rotary, incremental linear encoder, absolute linear encoder, Incremental rotary encoder, absolute rotary encoder; potentiometers; LVDTs; velocity sensors-optical encoders, tacho generator, Hall effect sensor, acceleration sensors, Heading sensors- Compass, Gyroscope sensor, IMU, GPS, real time differential GPS; Force sensors-strain gauge based and Piezo electric based, Torque sensors; **Note- Block schematic representations, Interpretation of typical manufacturer's datasheet and Numerical problems of the main sensors are to be covered.**	**9**

2	**Exteroceptive or External sensors-** contact type, noncontact type; Tactile, proximity-detection of physical contact or closeness, contact switches, bumpers , inductive proximity, capacitive proximity; semiconductor displacement sensor; Range sensors- IR, sonar, laser range finder, optical triangulation (1D), structured light(2D),performance comparison range sensors; motion/ speed sensors-speed relative to fixed or moving objects, Doppler radar, Doppler sound; Block schematic representations; Numerical problems; **Note- Block schematic representations, Interpretation of typical manufacturer's datasheet and Numerical problems of the main sensors are to be covered.**	**9**
3	**Introduction to transducers**; Requirement of transducers in robots, medicine etc; Differences between Sensors and transducers; Transducer performance characteristics based on static and dynamic properties; Classification of transducers based on physical effect, physical quantity and source of energy- Active vs Passive, Principle of transduction, Analog and Digital transducer, Primary and Secondary transducer; Transducer and Inverse Transducer	**9**
4	Position transducers, Displacement transducer– LVDT's, Captive Armatures, Unguided Armatures, Force-Extended Armatures; Velocity Transducers - LVT; Accelerometer- using potentiometer, Strain gage, Piezoelectric; Light transducers; Force transducers; Piezoelectric transducer; **Actuators for robots-** classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.	**9**

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse and select the most appropriate sensors and transducers for a robotic application	**K1**
CO2	Explain fundamental principle of working of sensors and transducers for robots	**K2**
CO3	Interpret typical manufacturer's datasheet of sensors and transducers and use them for selection in typical applications	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									3
CO2	3	2	2									3
CO3	3	2	2									3

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics Engineering: An Integrated Approach	Richard D. Klafter	Prentice Hall Inc.	
2	Sensors and Transducers	D. Patranabis	PHI Learning Private Limited	

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Sensors and Actuators: Control System Instrumentation	Clarence W. de Silva	CRC Press	2007
2	Introduction to Robotics	SK Saha	Mc Graw Hill Education	
3	Mechatronics	W. Bolton	Pearson Education Limited	
4	Automation, Production Systems and Computer Integrated Manufacturing	Groover M.P,	Prentice Hall Ltd.	1997
5	A first course on electric drives	Pillai S. K	Wieley Eastern Ltd, New Delhi	
6	Journal of sensors, Special issue- Sensors for Robotics	Aiguo Song , Guangming Song, Daniel a Constantinescu, Lei Wang, and Quanjun Song		Volume 2013
7	Mechatronics: Integrated mechanical electronic systems	K.P.Ramachandran, G.KVijayaraghavan	Wiley India	
8	Linear Electric Actuators	I.Boldea		
9	Piezoelectric Actuators (Electrical Engineering Developments)	Joshua E. Segel		2012
10	Sensors and Transducers Used in Robots", Basics of Robotics,	Morecki, Adam and Knapczyk	Springer	1999, pp 275—304
11	Robot sensors and transducers	Ruocco, S	Springer Science \&Business Media,	2013
12	Instrumentation, Measurement and Analysis	Nakra &Chaudhary		2016
13	HANDBOOKOFFORCETRANSDUCERS	Hard cover and Stefanescu		
14	Transducer sand Instrumentation	DVS Murthy		

Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc24_ee45/preview
2	https://www.youtube.com/watch?v=B-IoN9fl2yc